

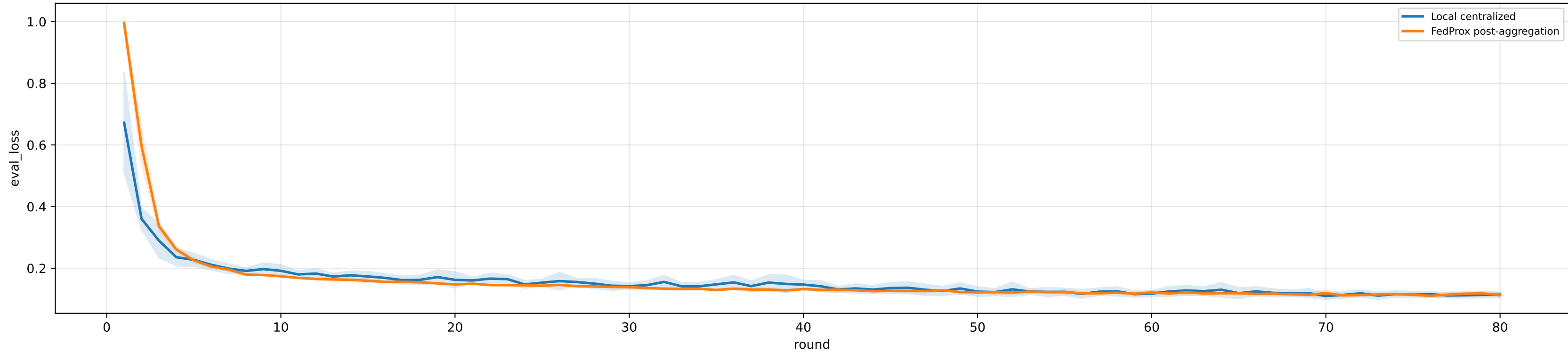
Controlled task-specialization level: Low | FedProx diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

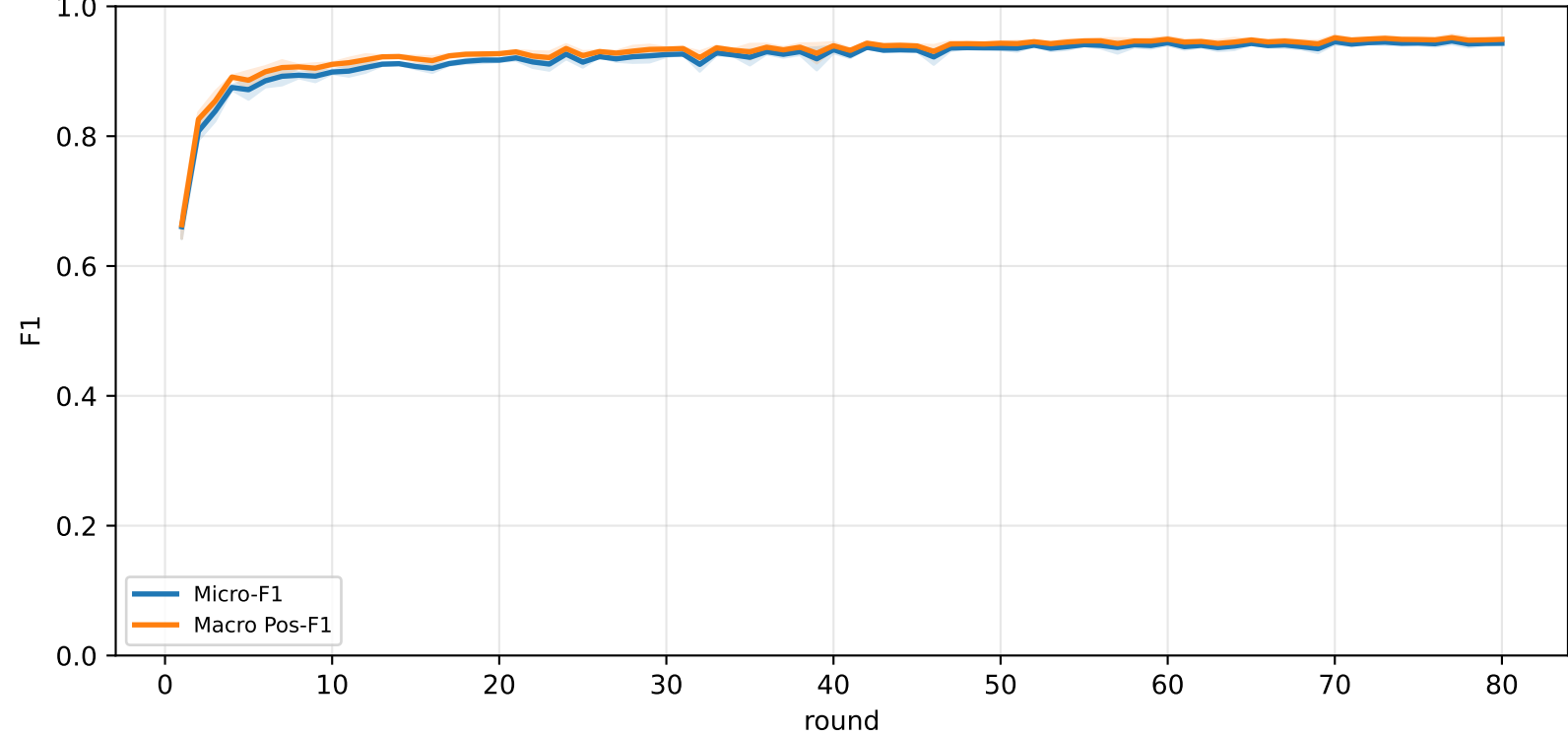
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.1% ± 0.1%	94.7% ± 0.1%	99.0% ± 0.2%	98.3% ± 0.3%	96.2% ± 0.3%	93.3% ± 0.4%	86.9% ± 0.2%
FedProx	93.6% ± 0.2%	94.2% ± 0.2%	98.9% ± 0.2%	98.3% ± 0.1%	96.2% ± 0.1%	92.7% ± 0.6%	85.0% ± 0.5%

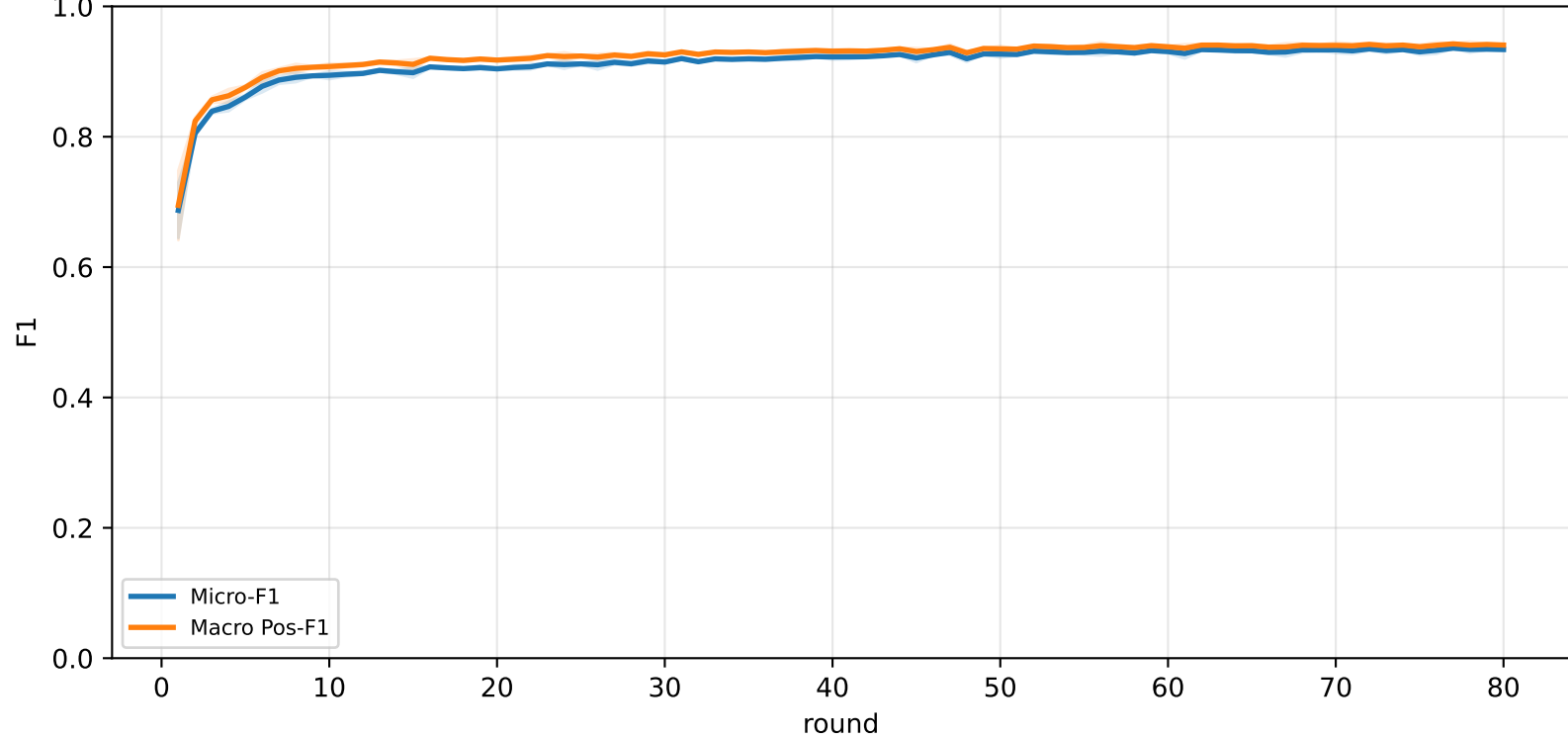
Validation loss comparison



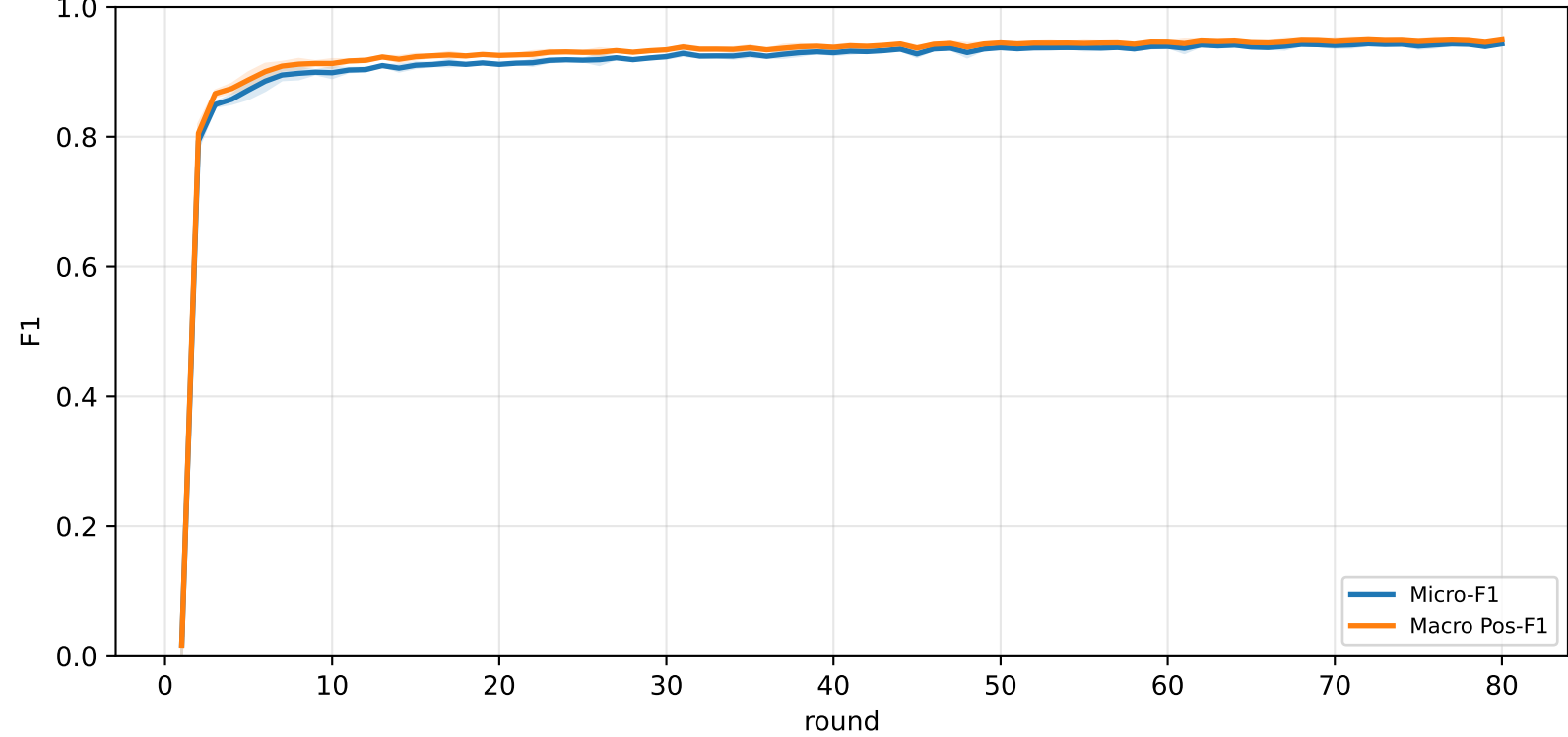
Pooled local centralized micro / macro F1



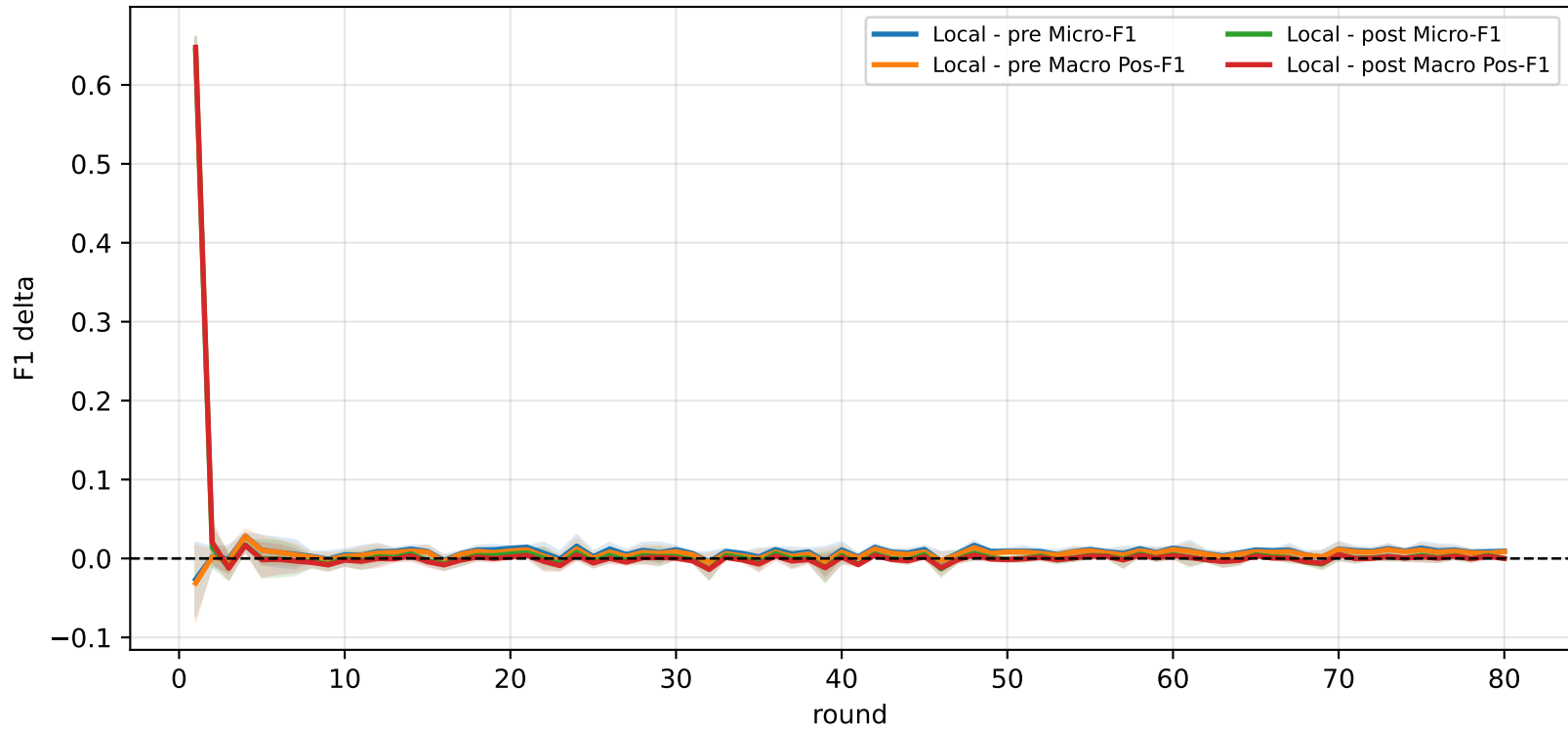
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)

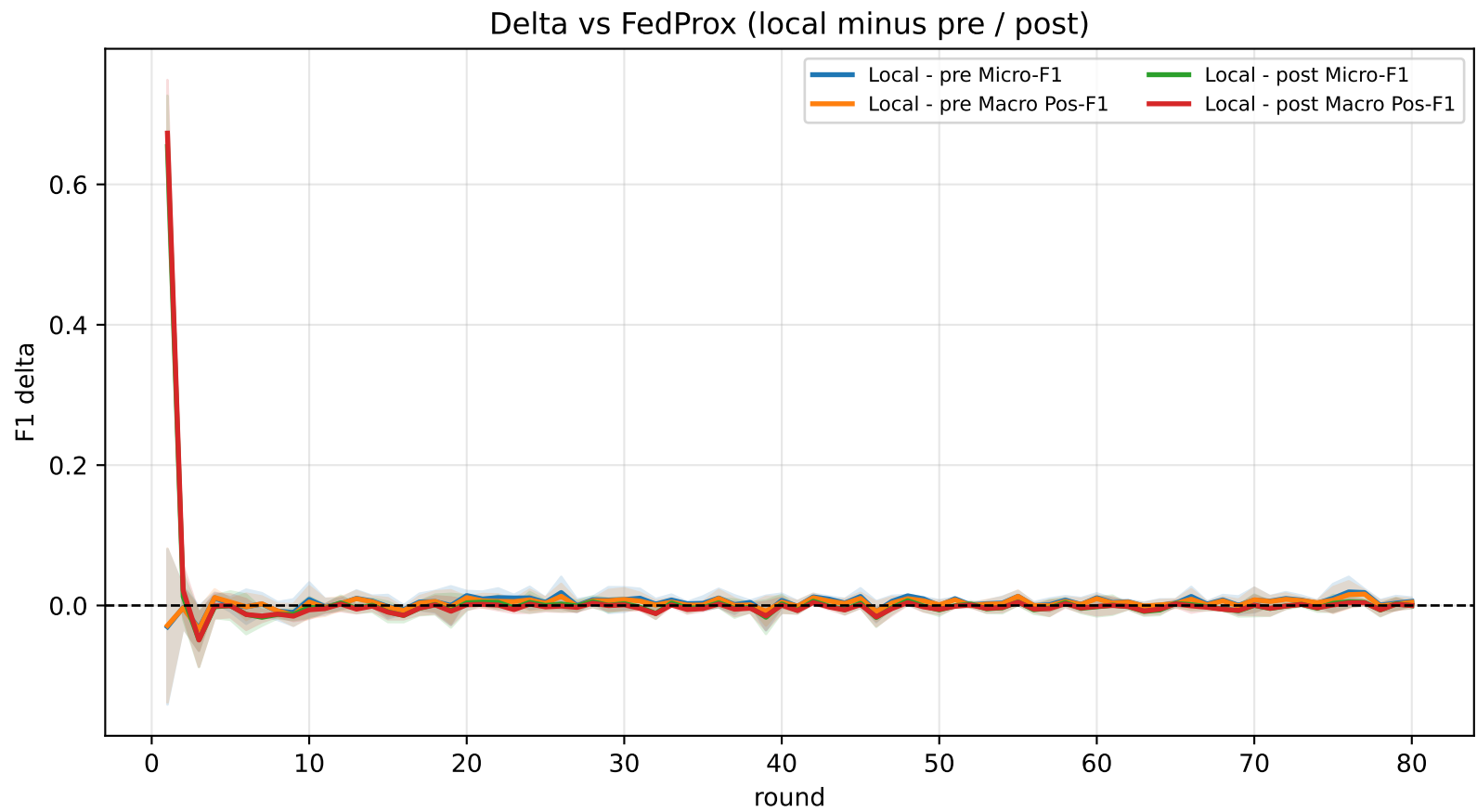
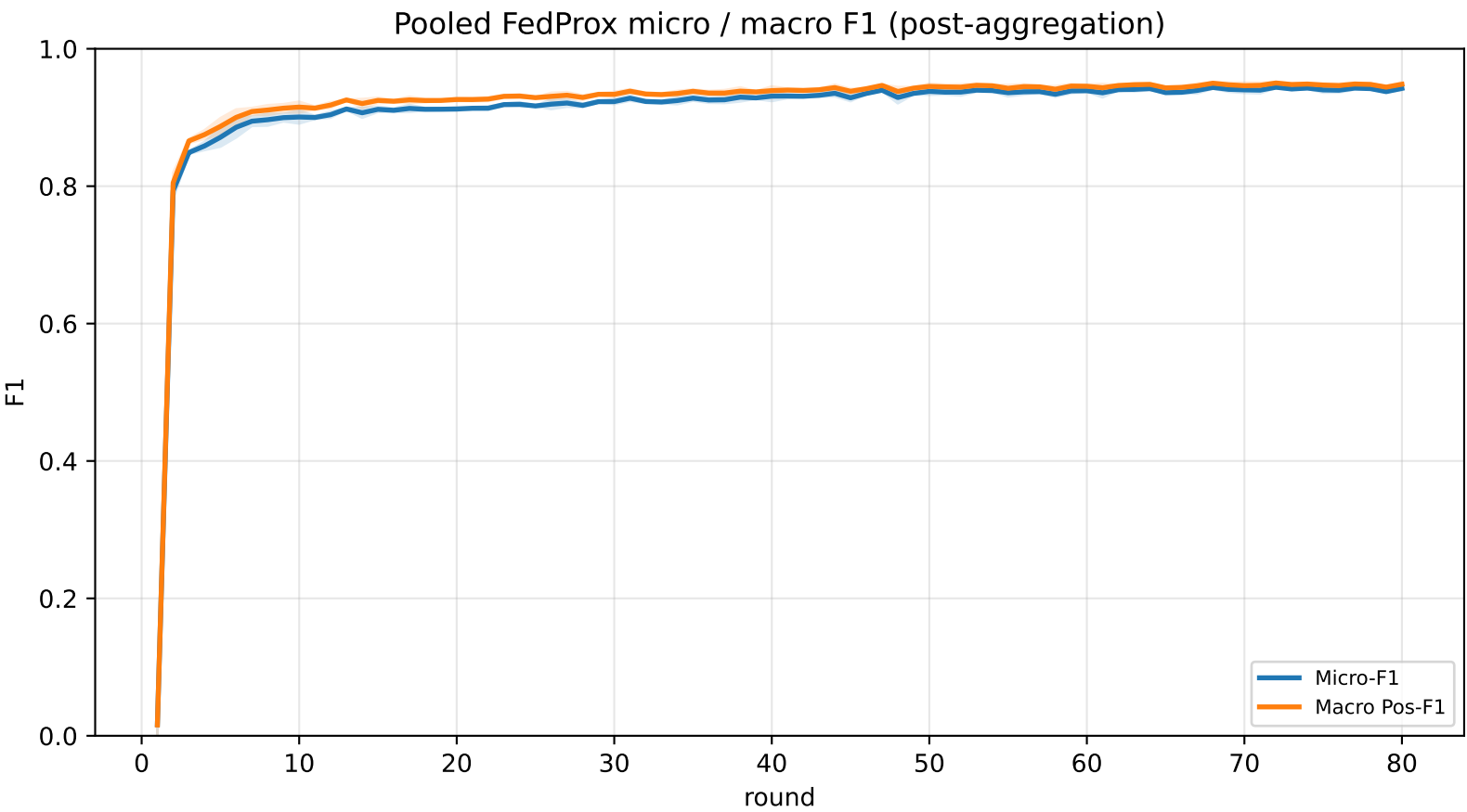
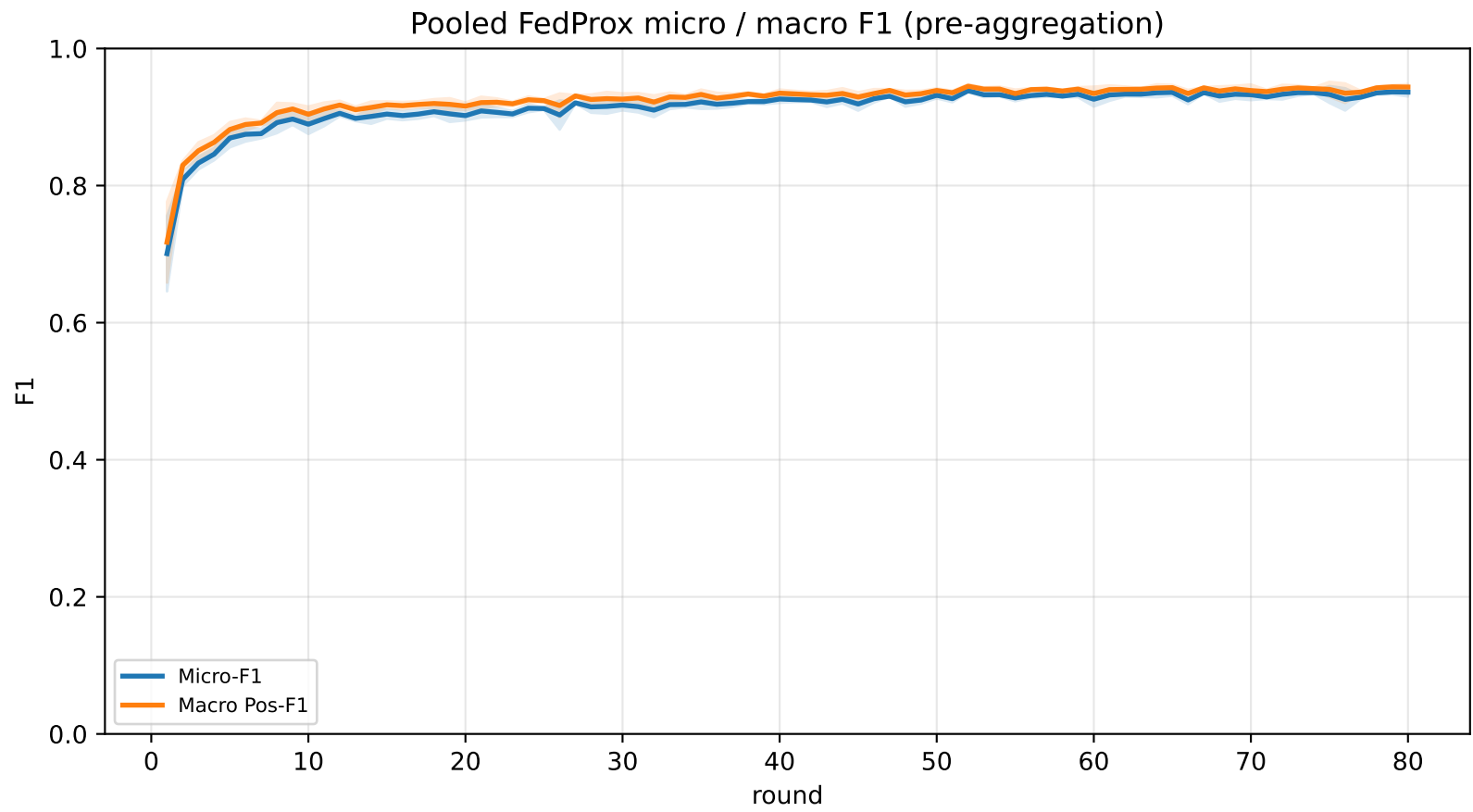
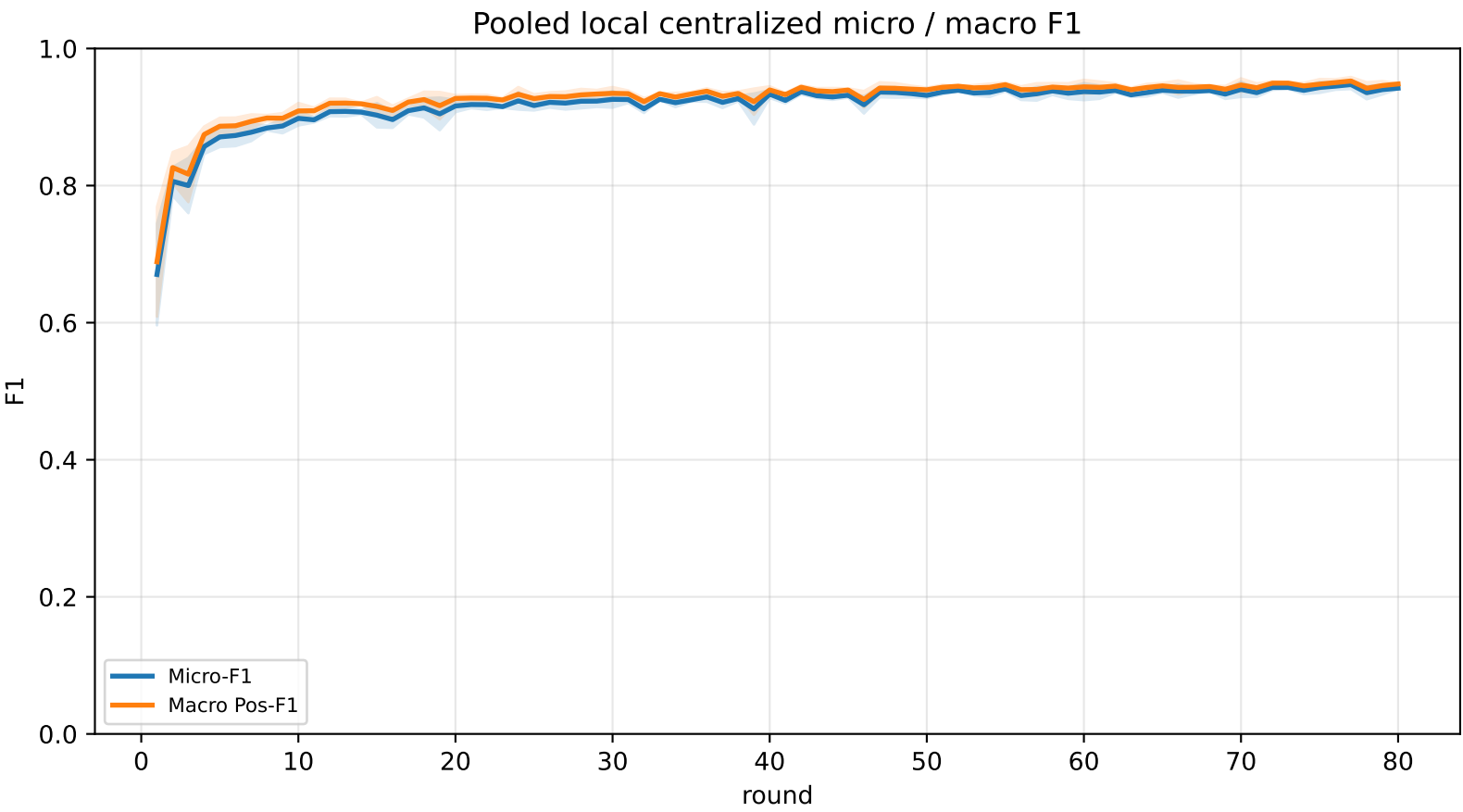
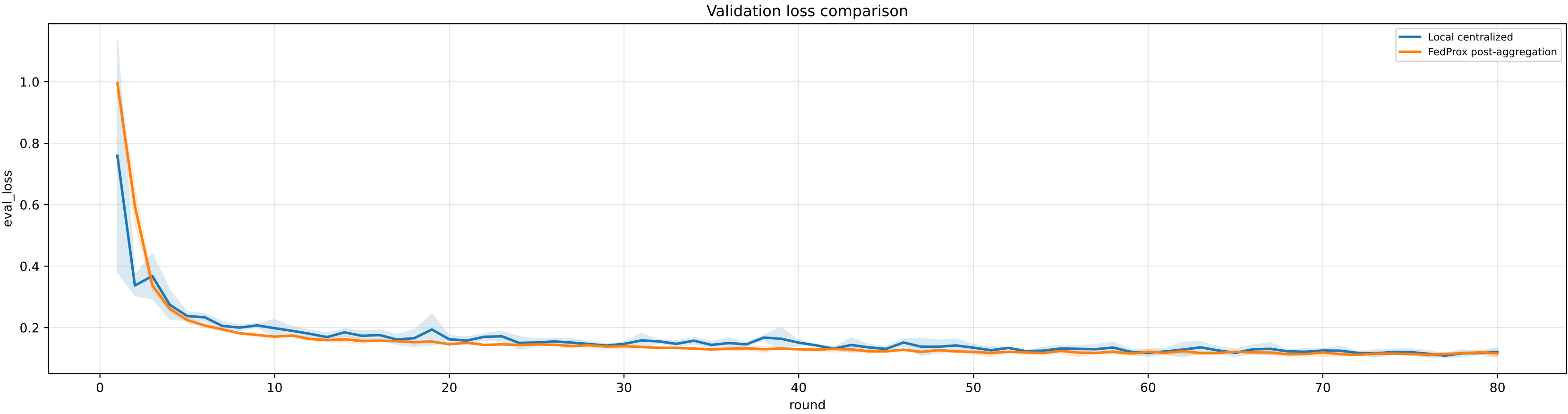


Delta vs FedProx (local minus pre / post)



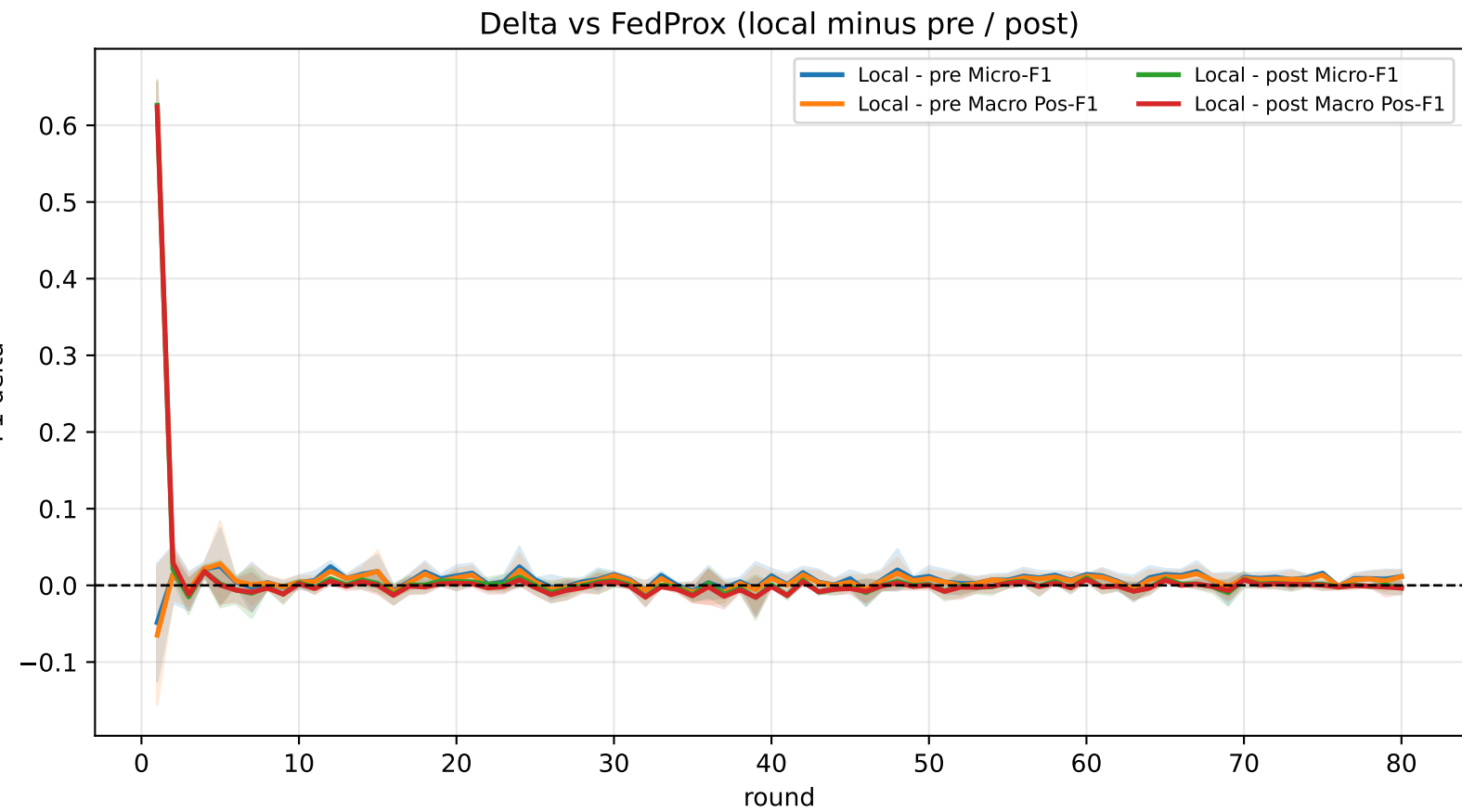
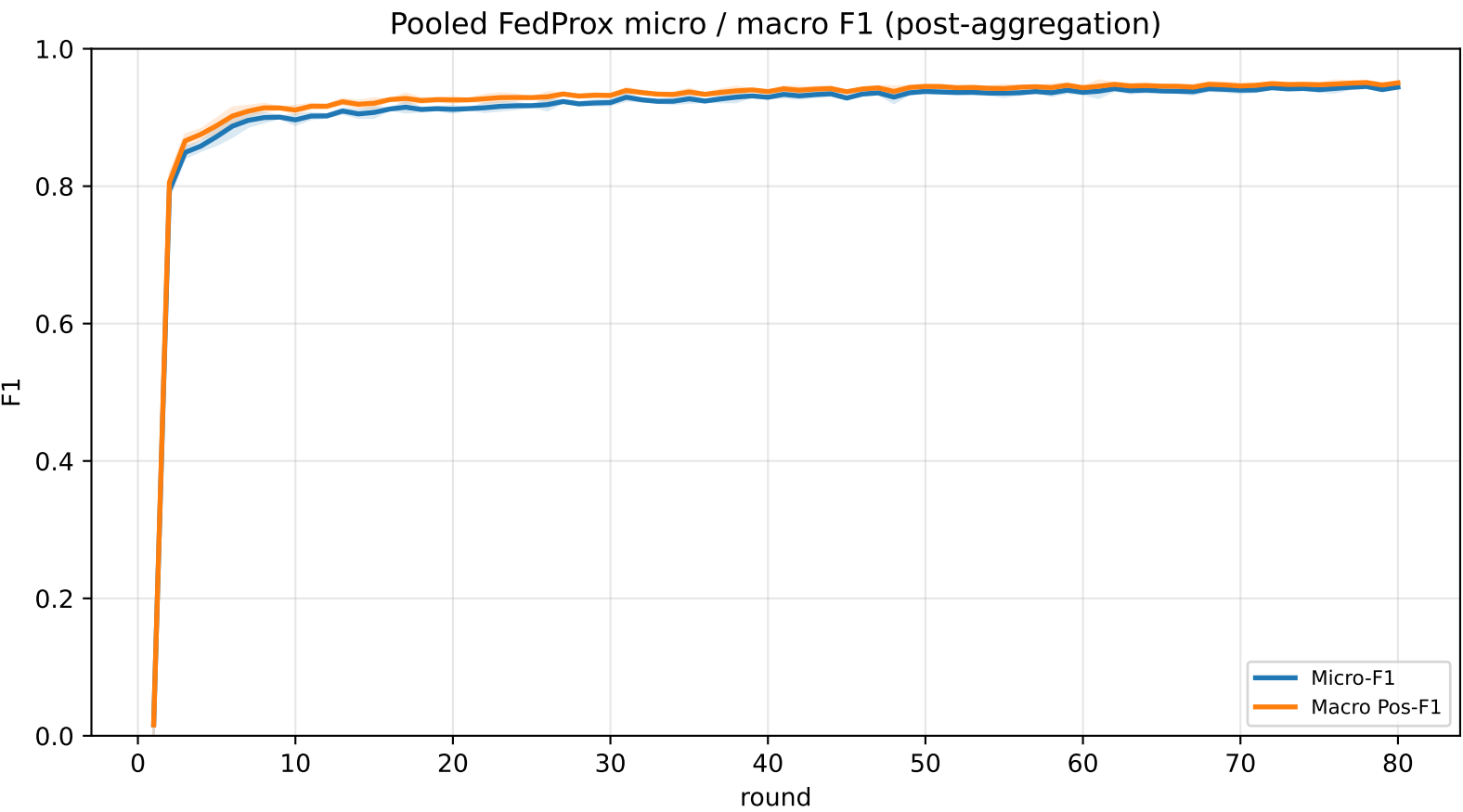
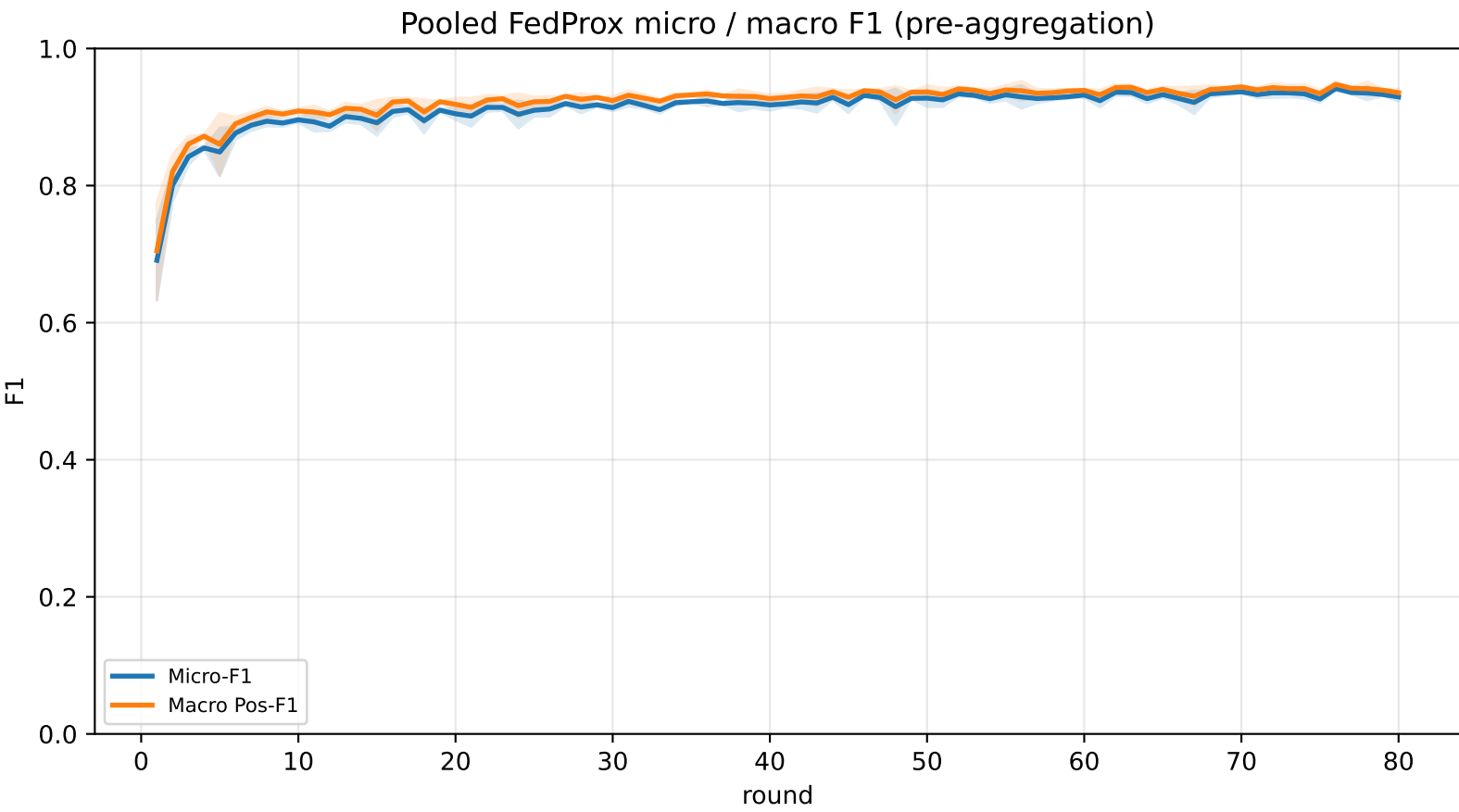
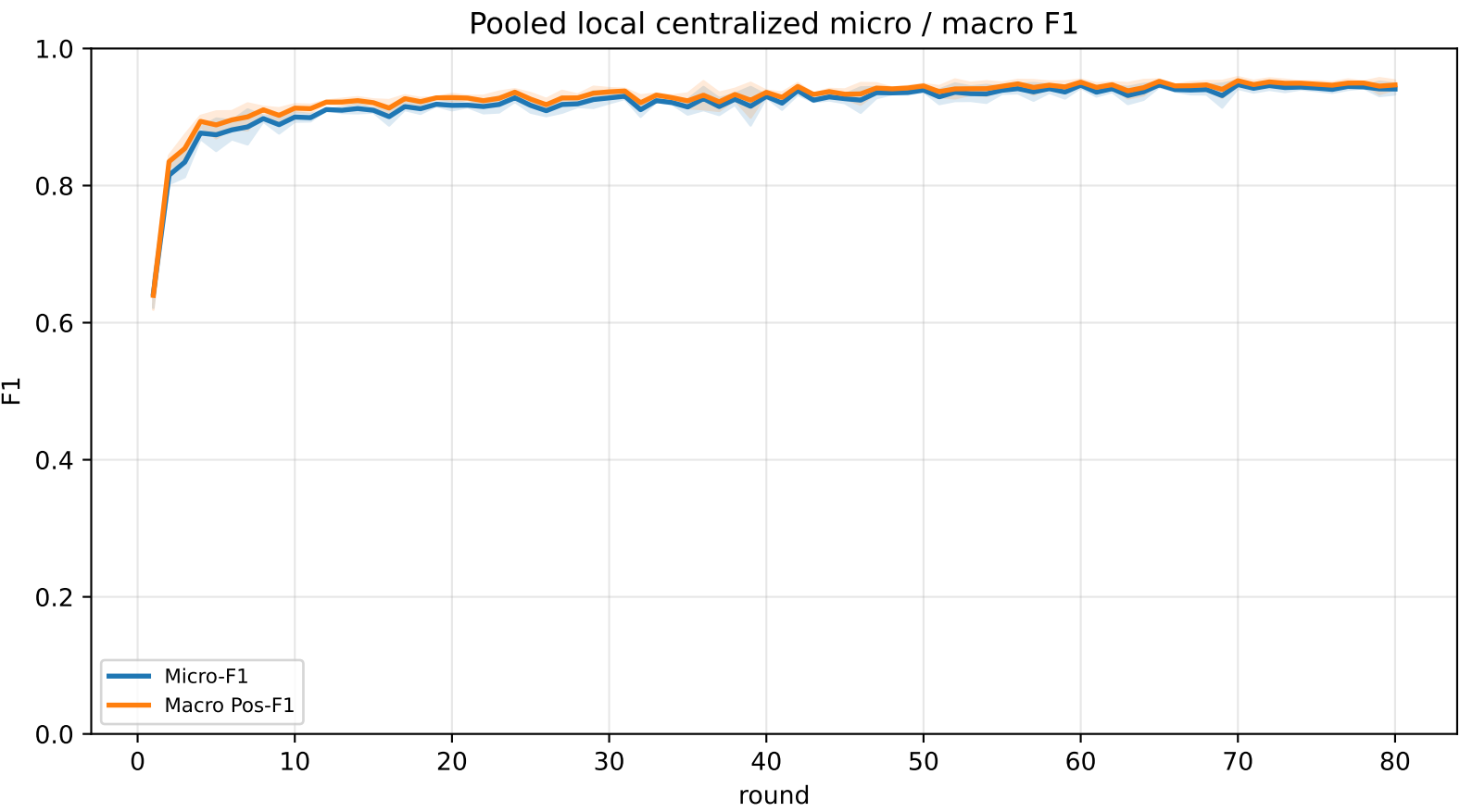
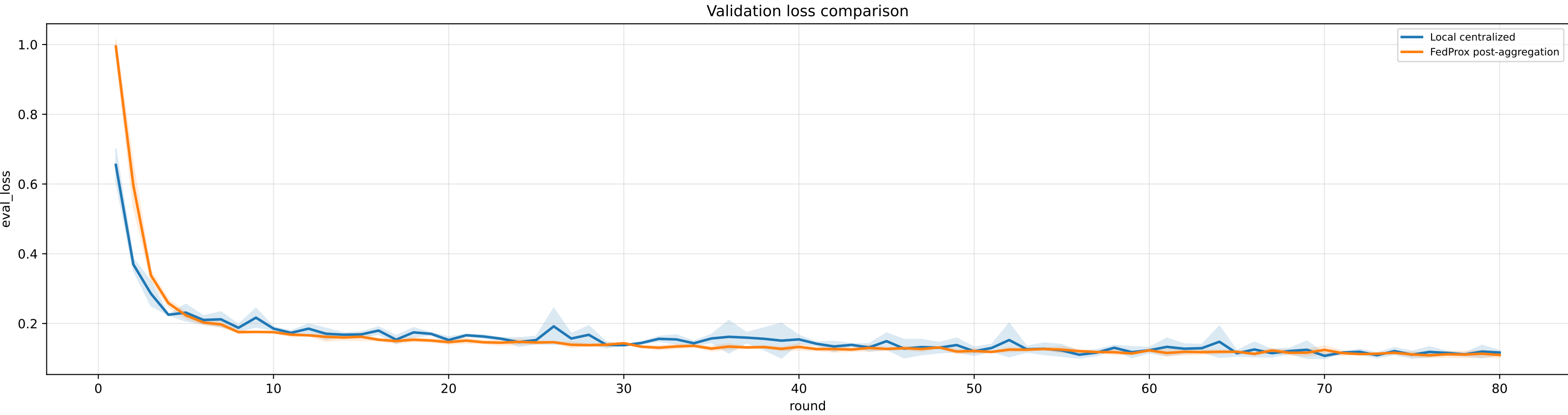
family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.1% ± 0.4%	94.7% ± 0.3%	99.0% ± 0.3%	98.4% ± 0.1%	96.1% ± 0.9%	92.7% ± 0.7%	87.4% ± 1.3%
FedProx	93.6% ± 0.4%	94.3% ± 0.3%	99.1% ± 0.4%	98.3% ± 0.3%	96.3% ± 0.2%	92.8% ± 0.6%	84.9% ± 1.1%



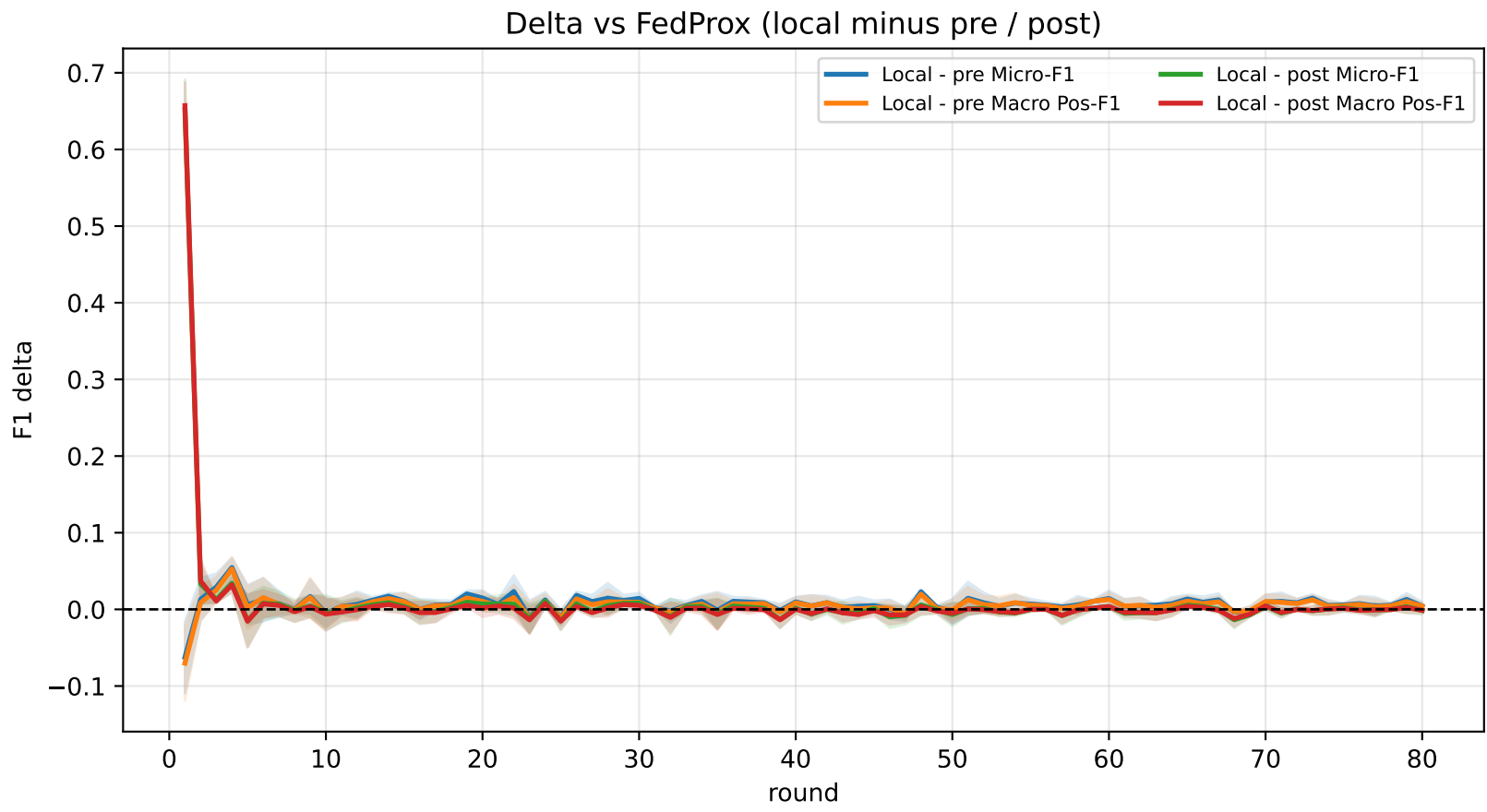
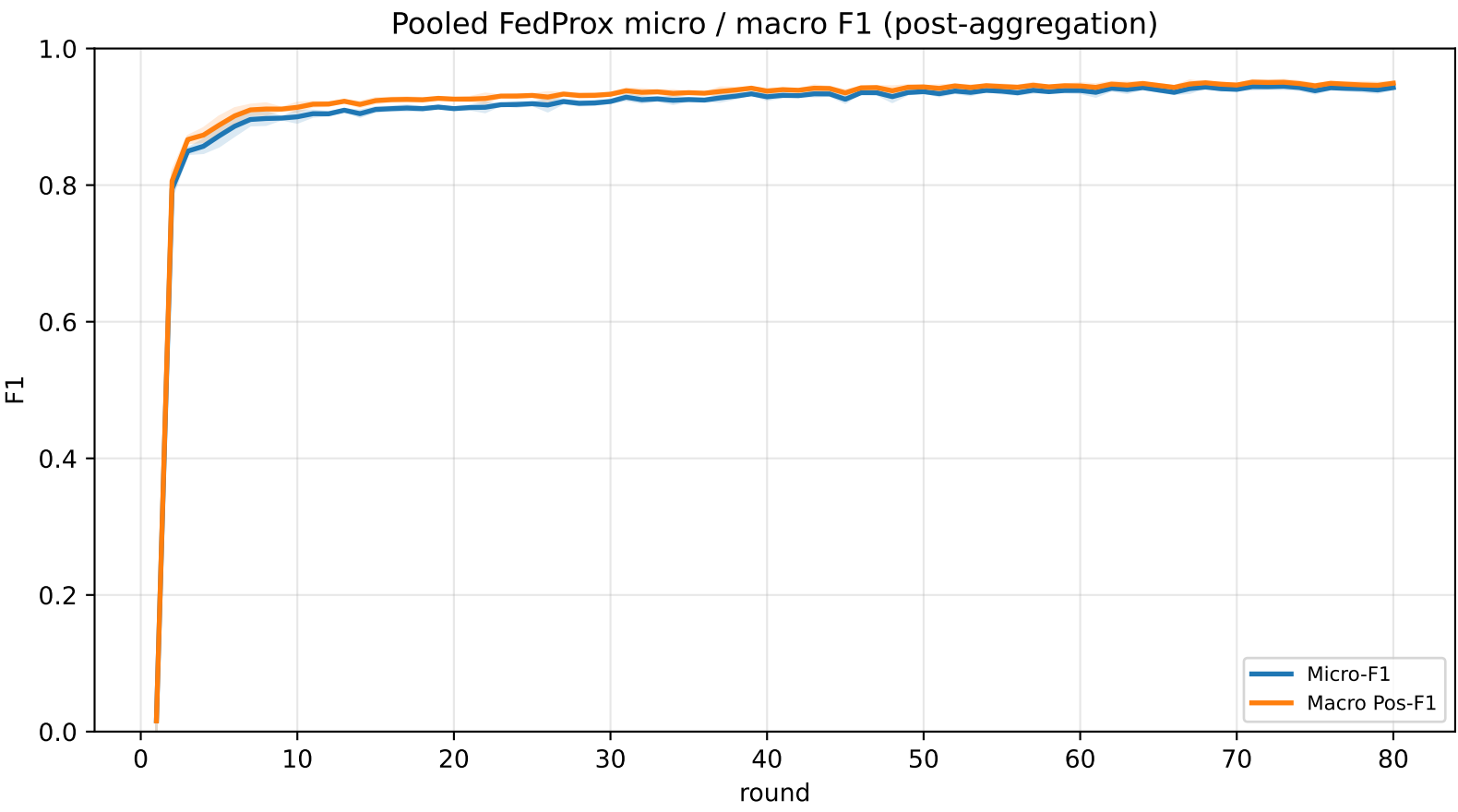
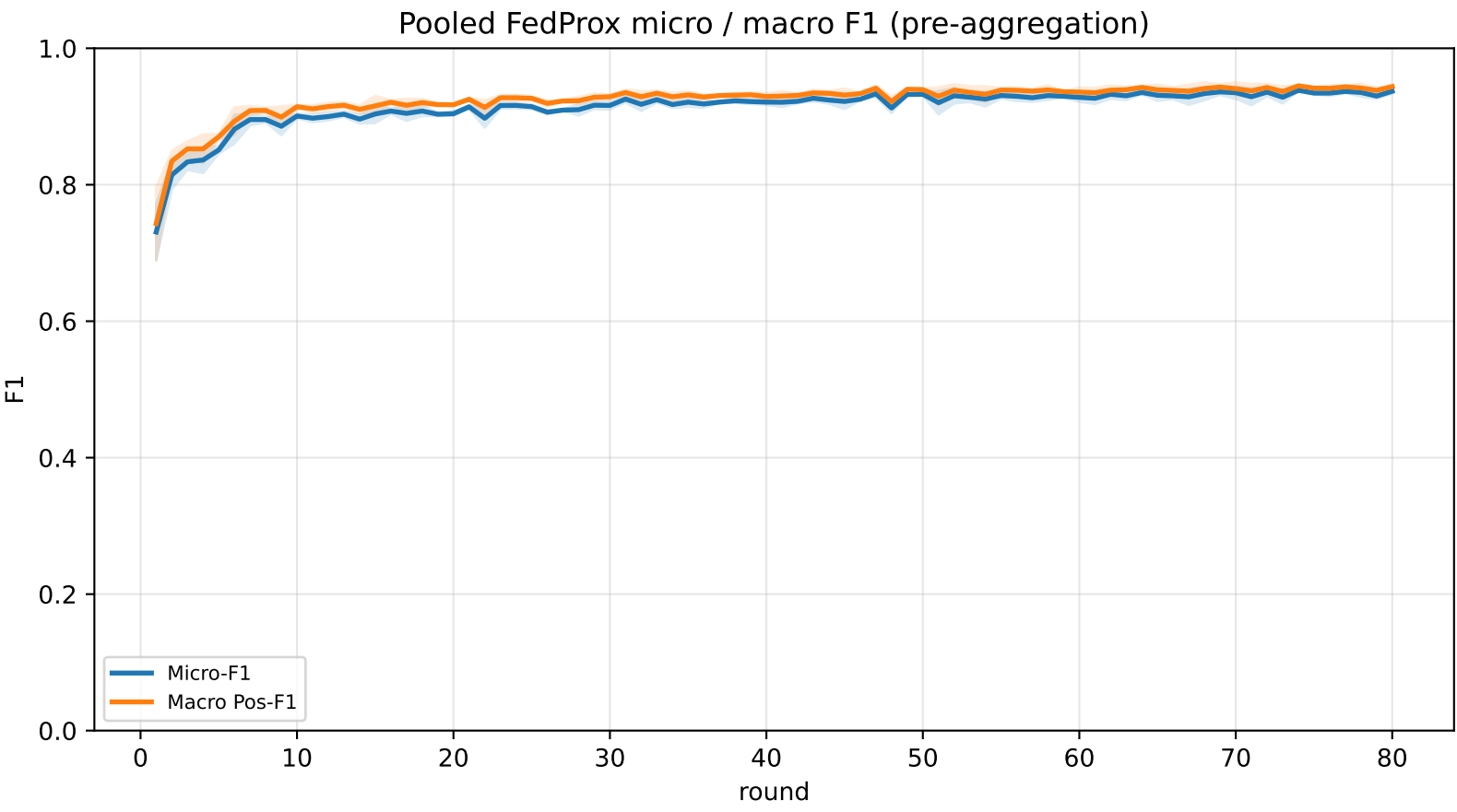
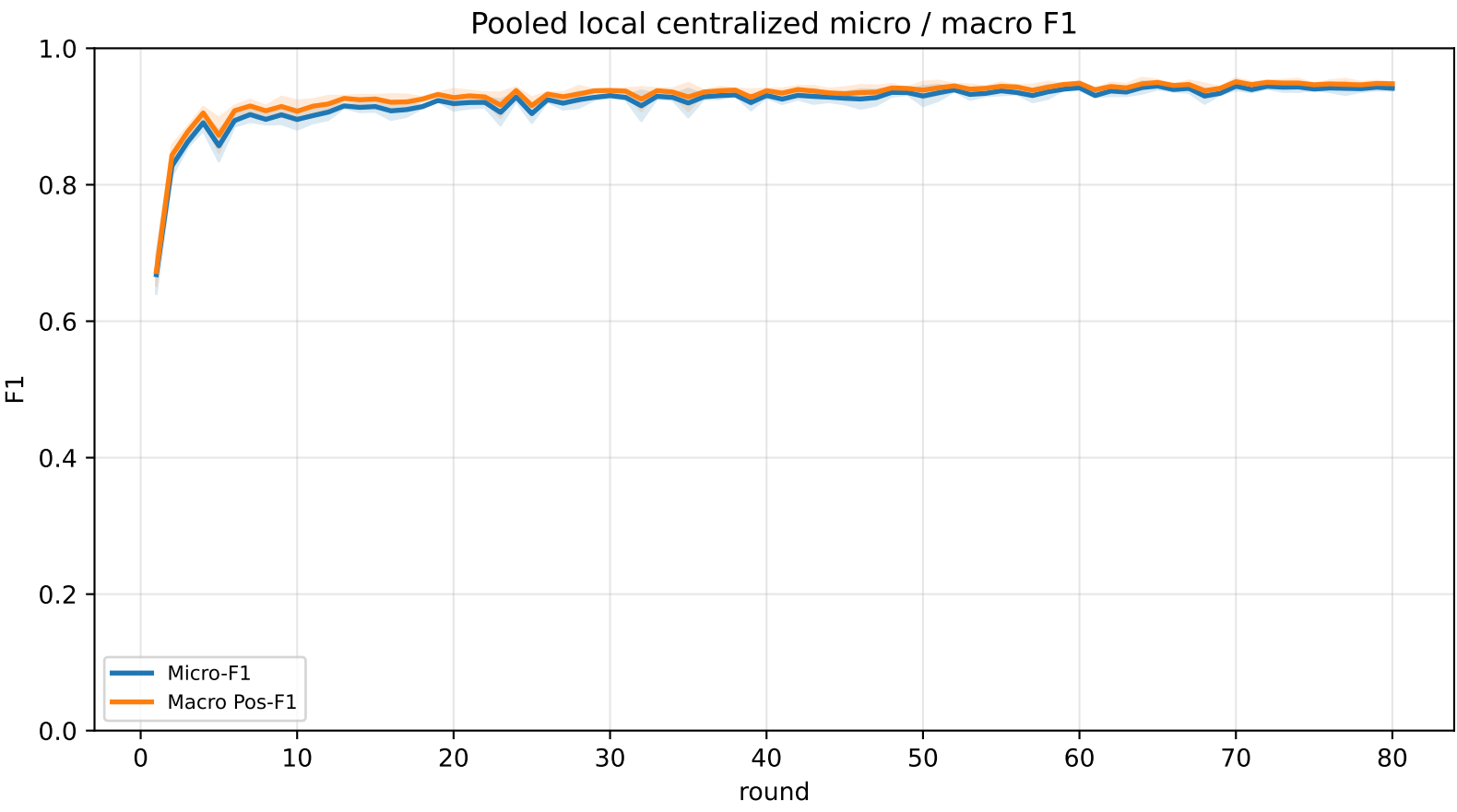
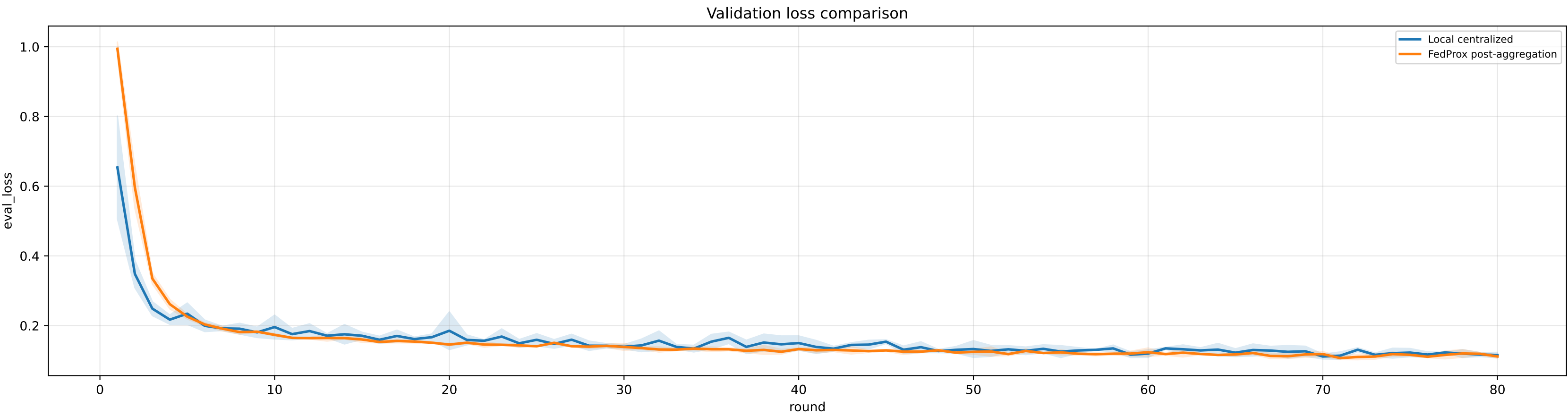
family=specialized_cycle3 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.1% ± 0.1%	94.7% ± 0.1%	99.0% ± 0.2%	98.2% ± 0.3%	95.9% ± 0.1%	93.2% ± 0.3%	86.9% ± 0.5%
FedProx	93.4% ± 0.2%	94.1% ± 0.2%	98.8% ± 0.3%	97.8% ± 0.0%	96.1% ± 0.5%	92.8% ± 0.6%	84.6% ± 0.7%



family=specialized_cycle4 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.6% ± 0.7%	94.3% ± 0.6%	98.9% ± 0.2%	98.1% ± 0.6%	96.4% ± 0.4%	92.7% ± 0.6%	85.4% ± 2.5%
FedProx	93.5% ± 0.2%	94.2% ± 0.2%	99.1% ± 0.1%	98.3% ± 0.1%	96.3% ± 0.3%	92.3% ± 0.6%	85.1% ± 0.6%

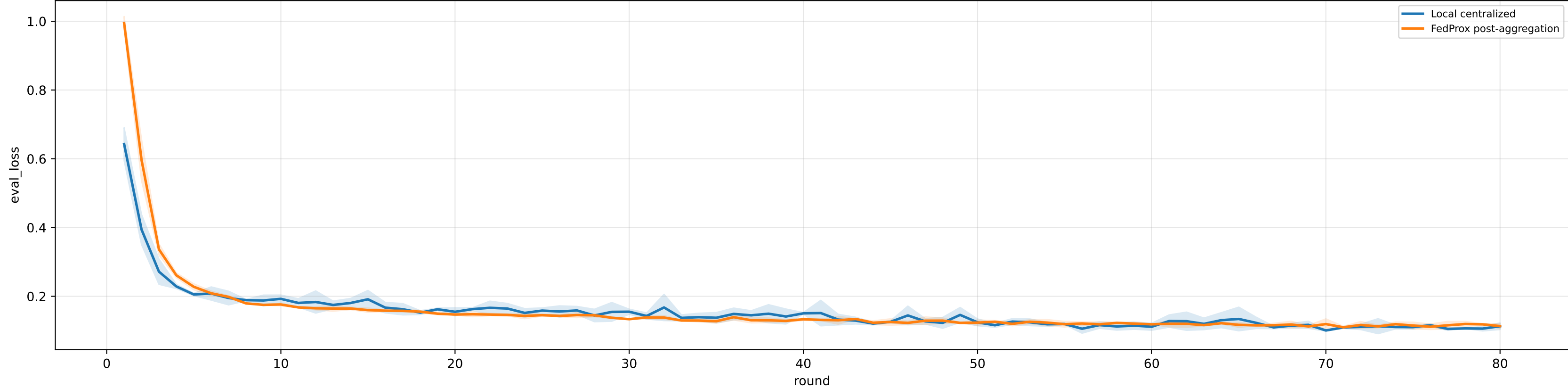


family=specialized_cycle5 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 20.8%±0.0% / true 26.0% |
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 |
family_centroid_jsd_mean=0.003

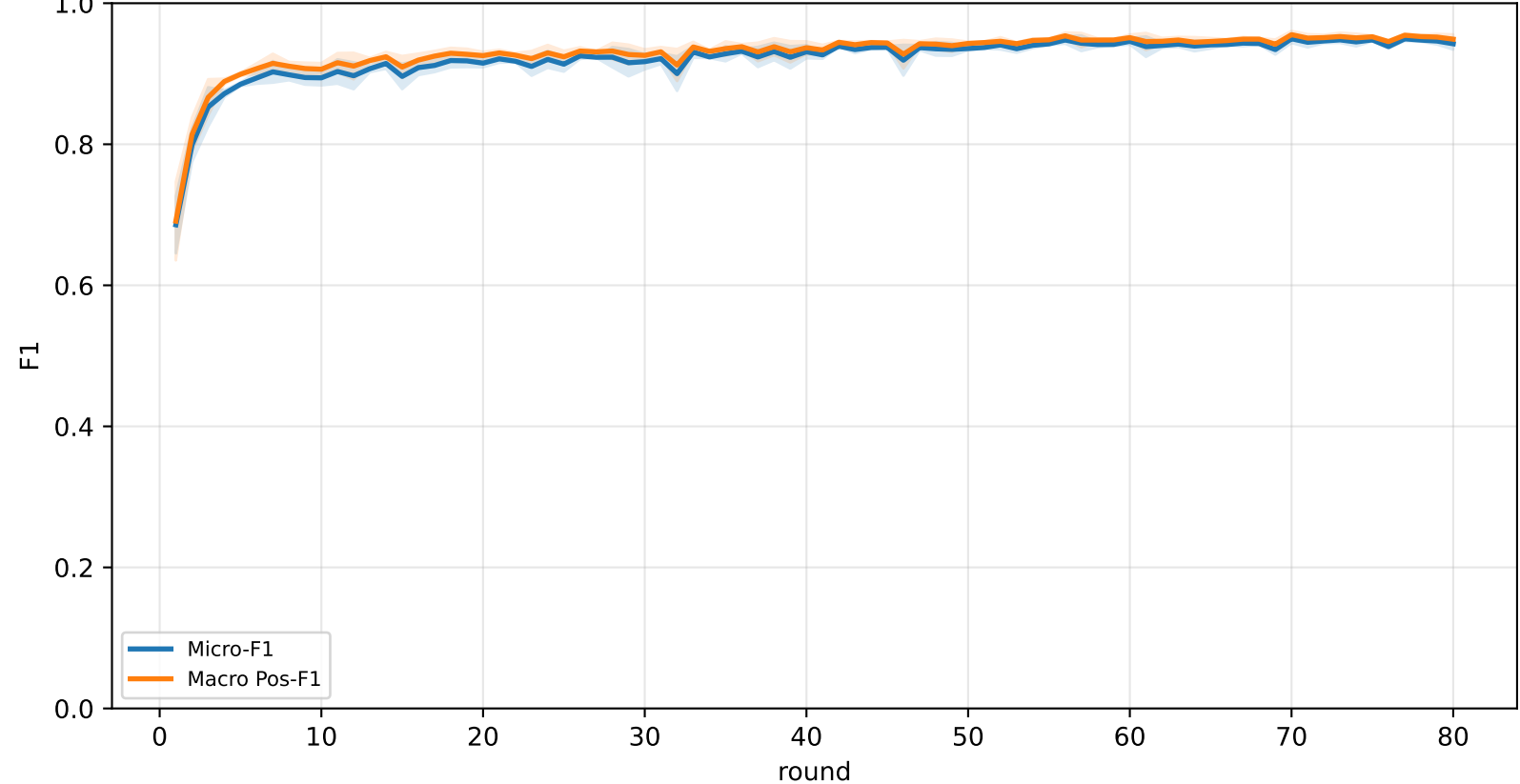
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.4% ± 0.4%	95.0% ± 0.3%	99.0% ± 0.2%	98.5% ± 0.3%	96.6% ± 0.2%	93.9% ± 1.0%	87.1% ± 0.8%
FedProx	93.7% ± 0.3%	94.4% ± 0.2%	98.8% ± 0.3%	98.8% ± 0.1%	96.2% ± 0.3%	92.8% ± 0.8%	85.3% ± 0.4%

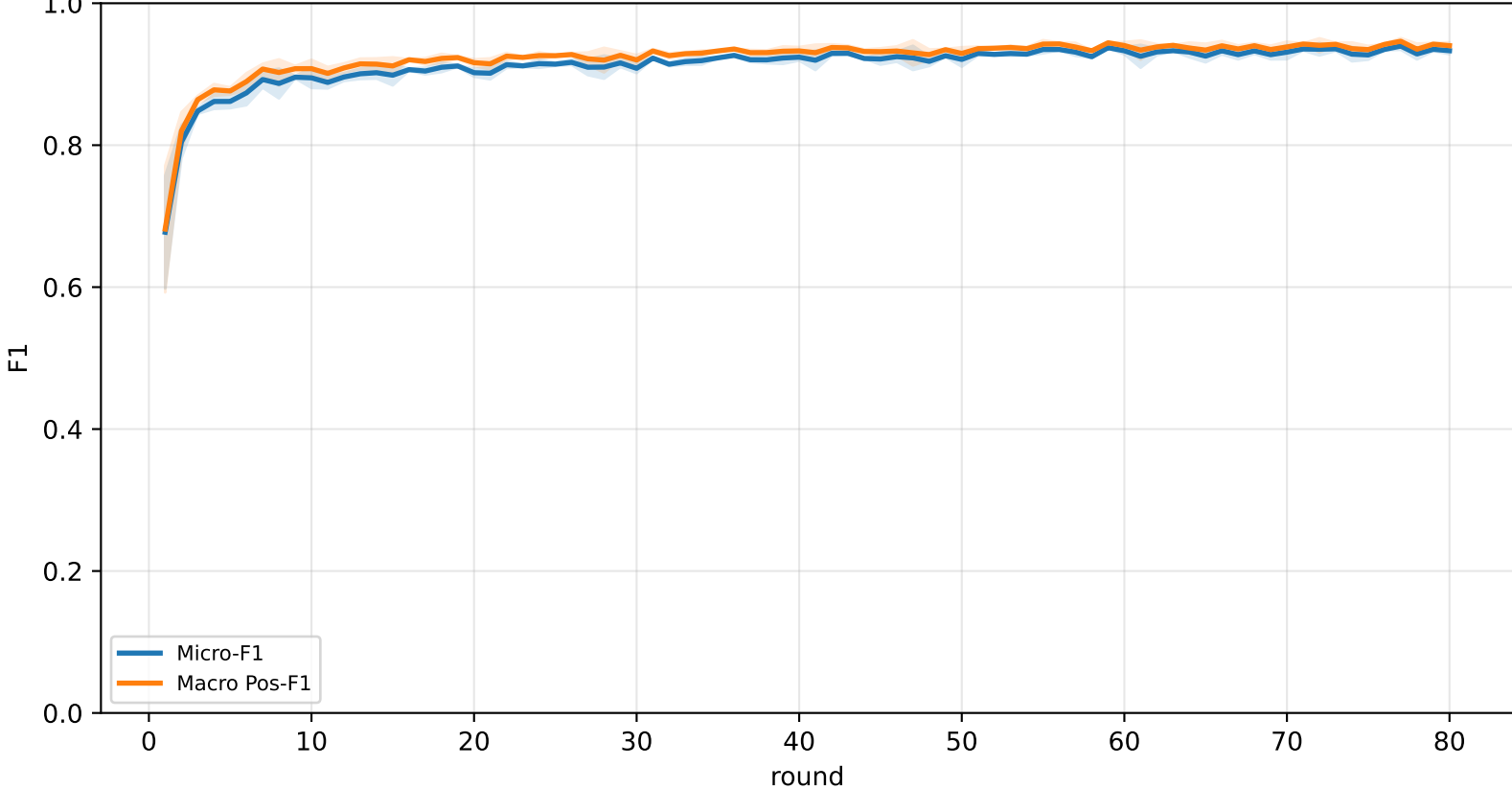
Validation loss comparison



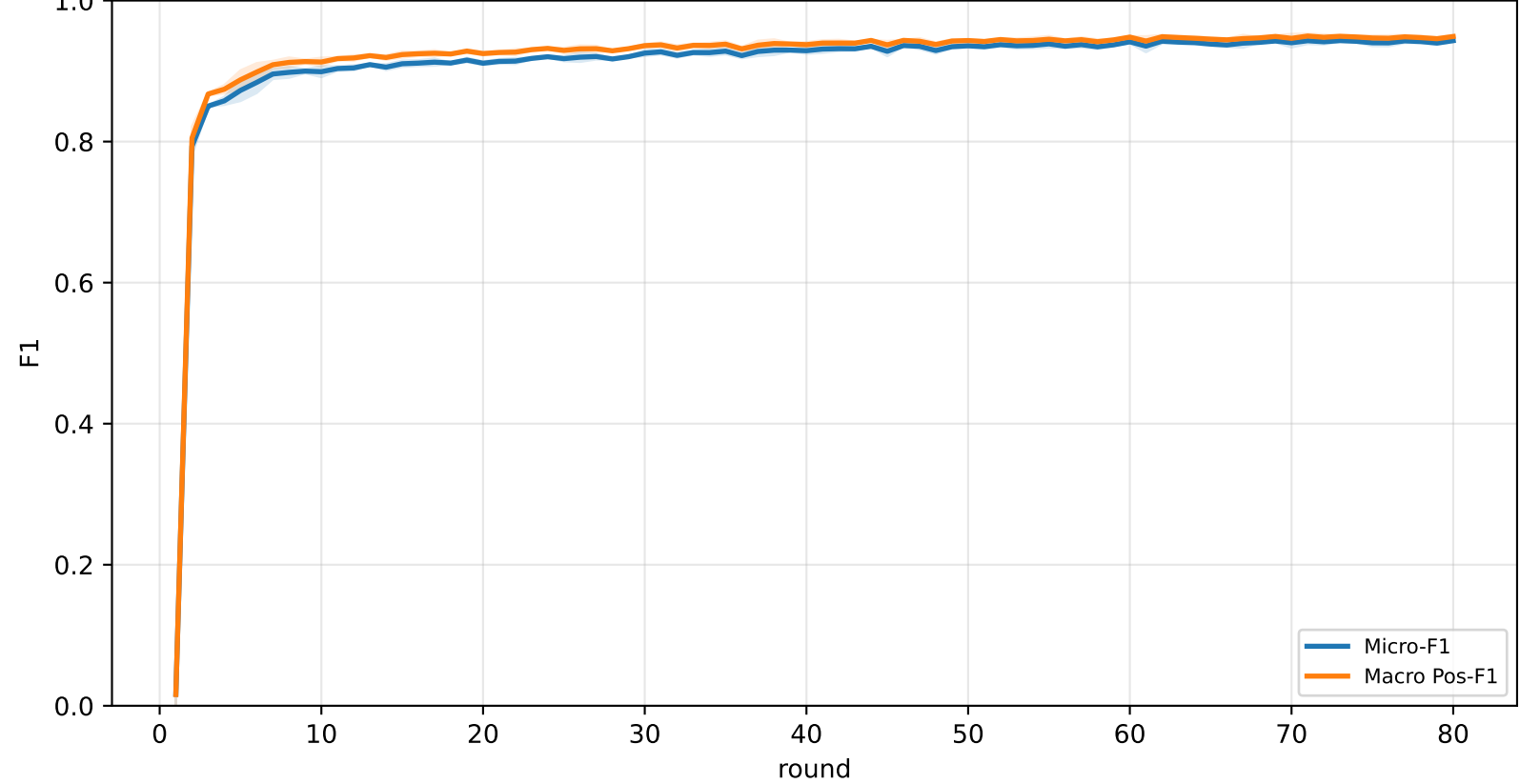
Pooled local centralized micro / macro F1



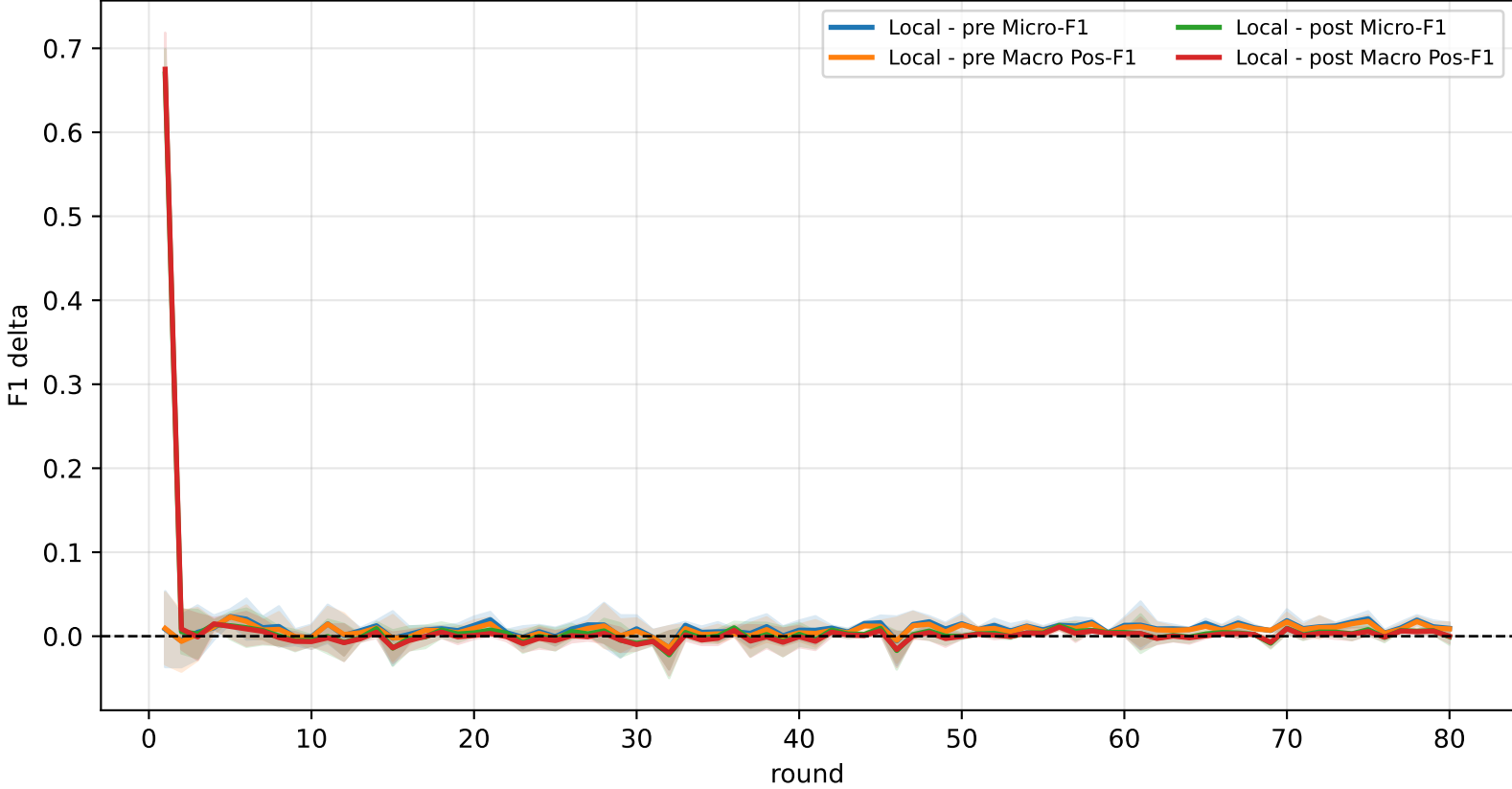
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)

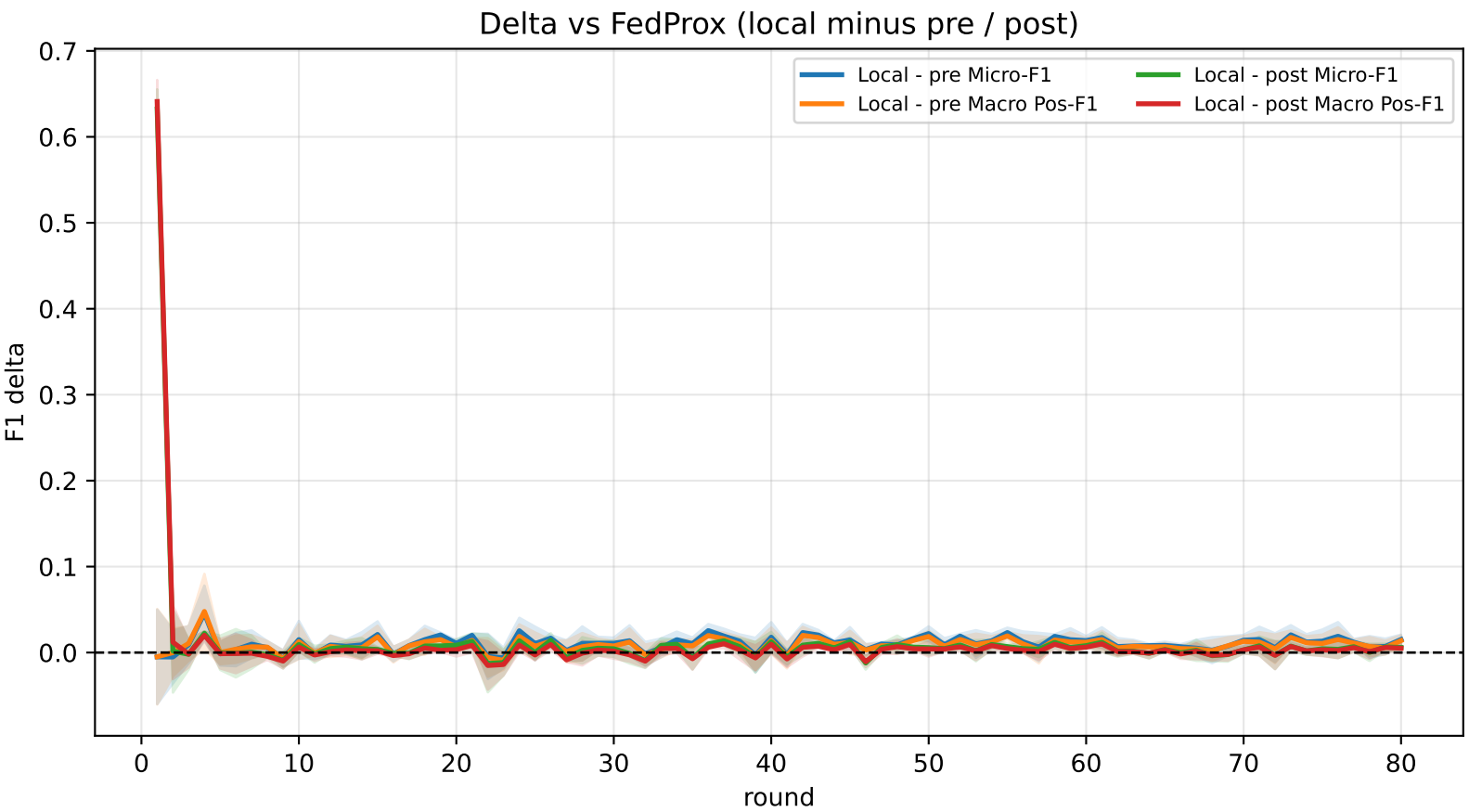
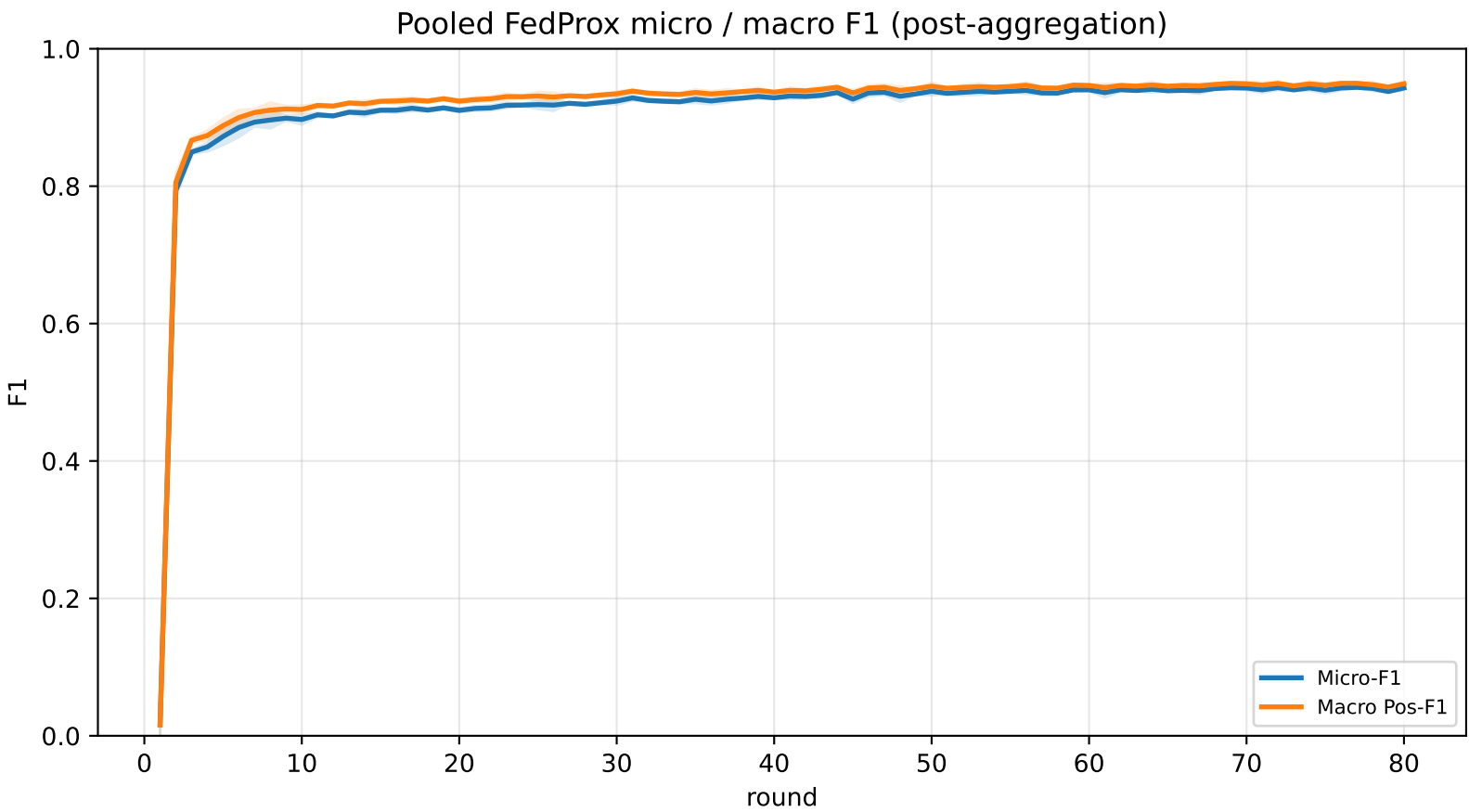
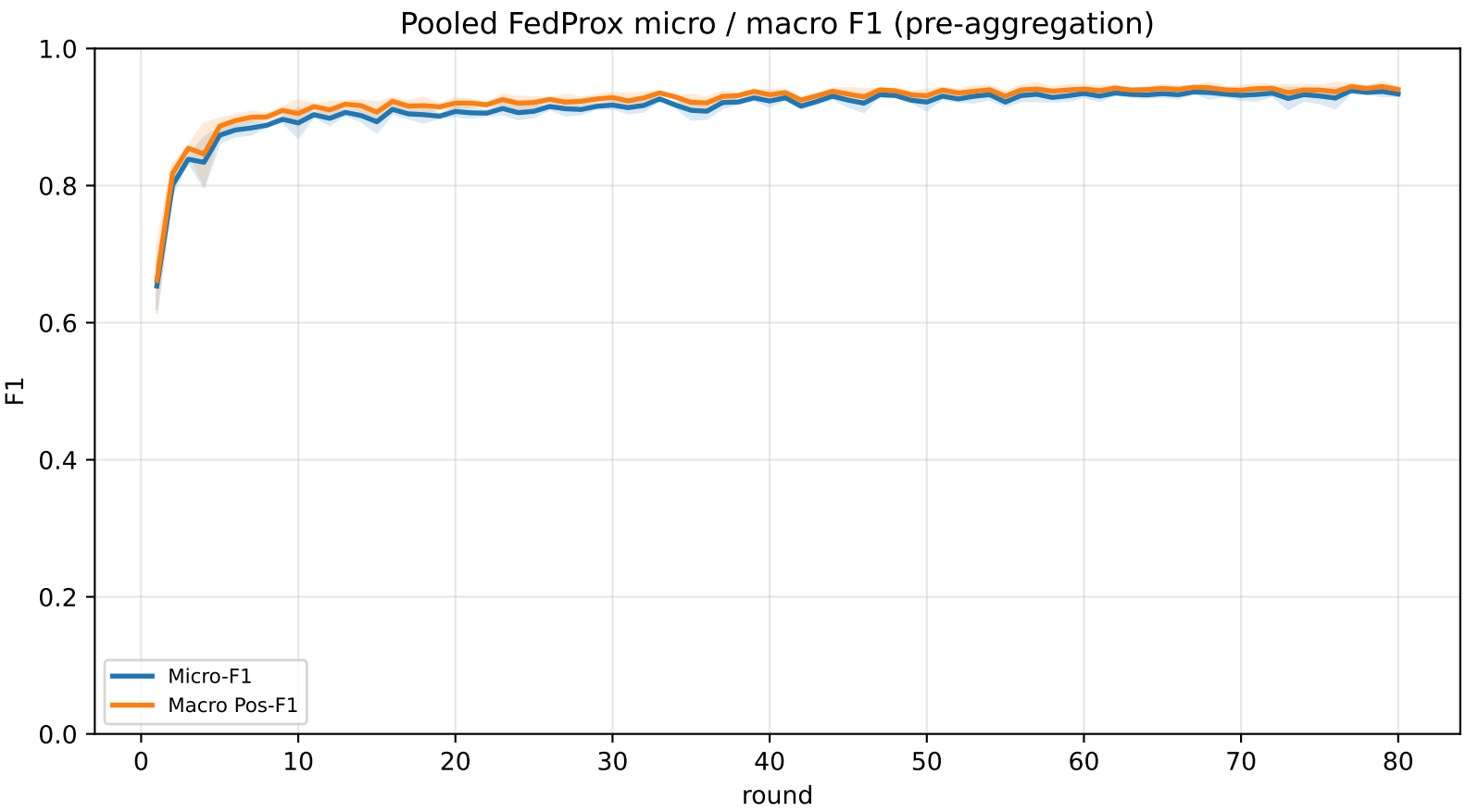
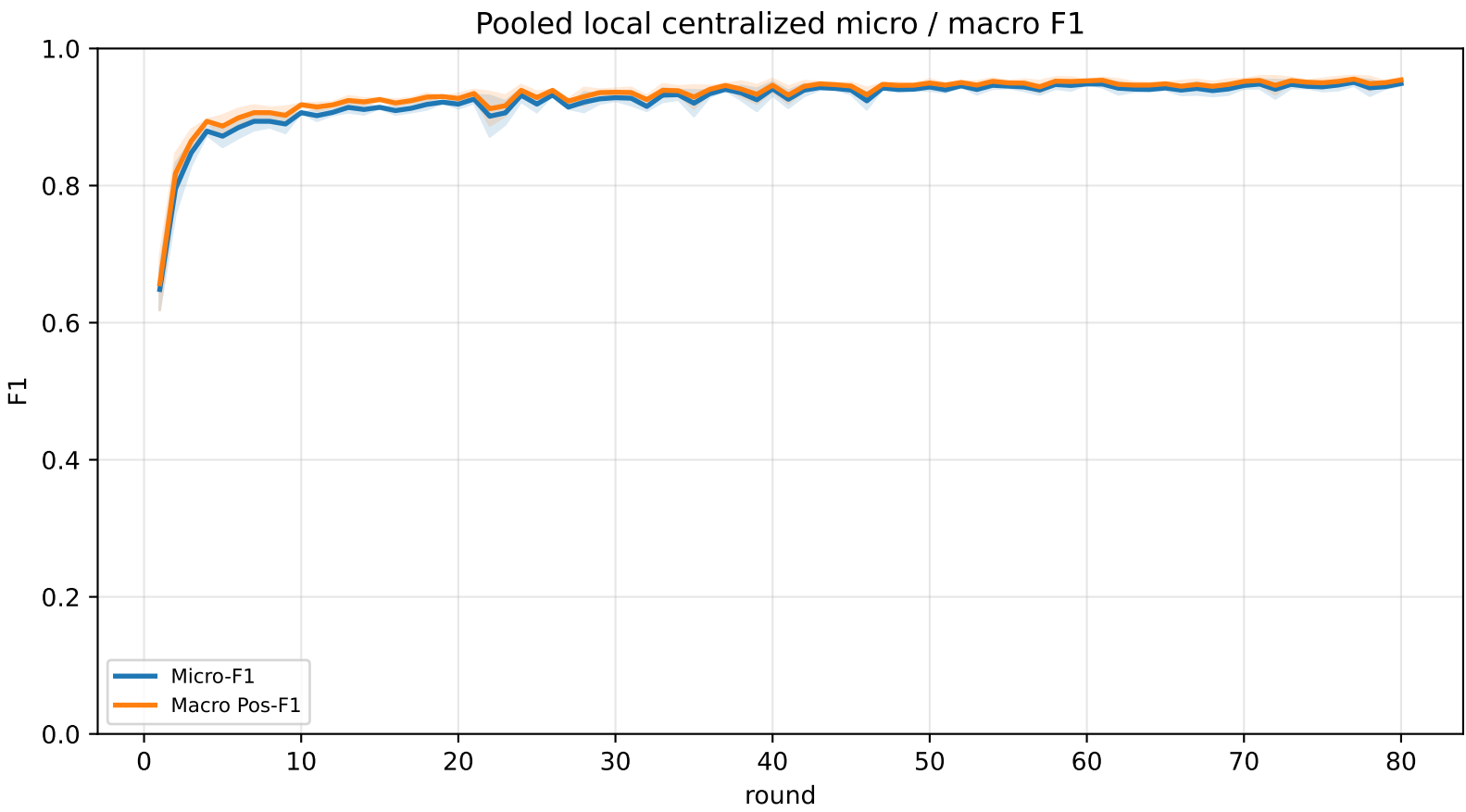
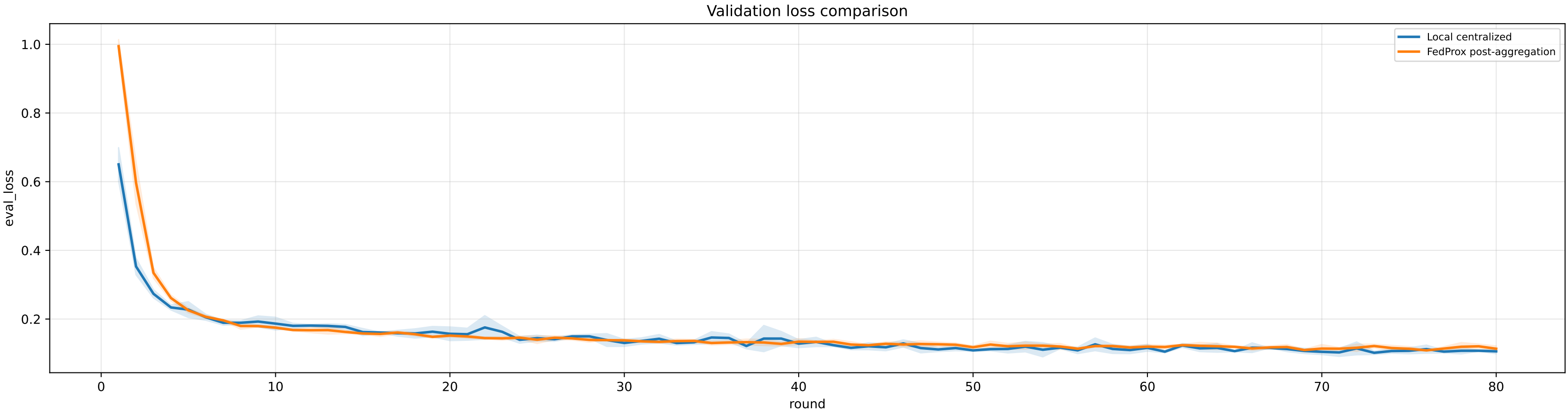


Delta vs FedProx (local minus pre / post)



family=specialized_cycle6 | n_clients=1 | cycle2=visible 15.5%±0.0% / true 19.4% | cycle3=visible 18.9%±0.0% / true 23.6% | cycle4=visible 20.5%±0.0% / true 25.6% | cycle5=visible 20.5%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.3% ± 0.5%	94.9% ± 0.4%	98.9% ± 0.1%	98.2% ± 0.5%	96.0% ± 0.4%	93.8% ± 1.0%	87.6% ± 0.5%
FedProx	93.6% ± 0.2%	94.3% ± 0.2%	98.9% ± 0.4%	98.3% ± 0.4%	96.0% ± 0.5%	92.9% ± 0.4%	85.2% ± 0.3%



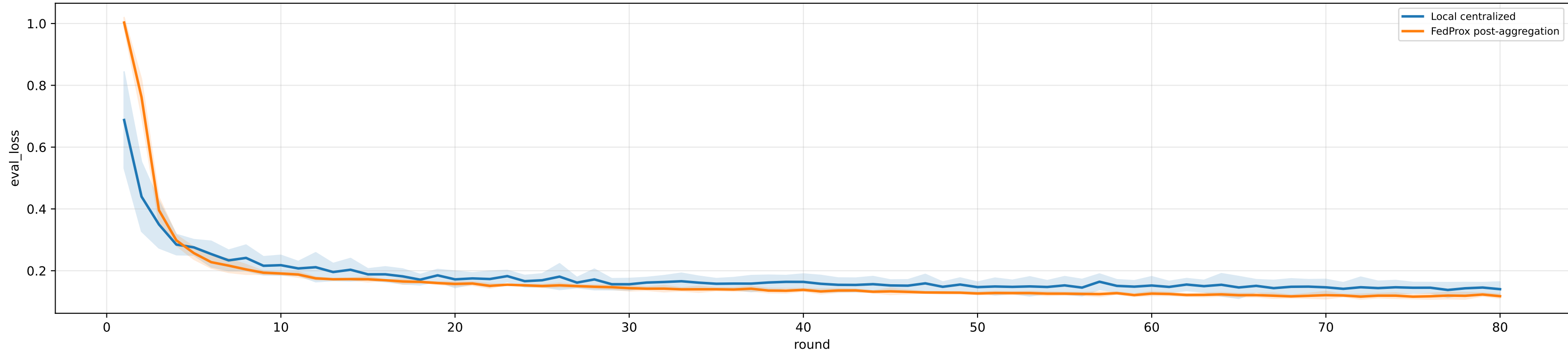
Controlled task-specialization level: Mid | FedProx diagnostics / aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

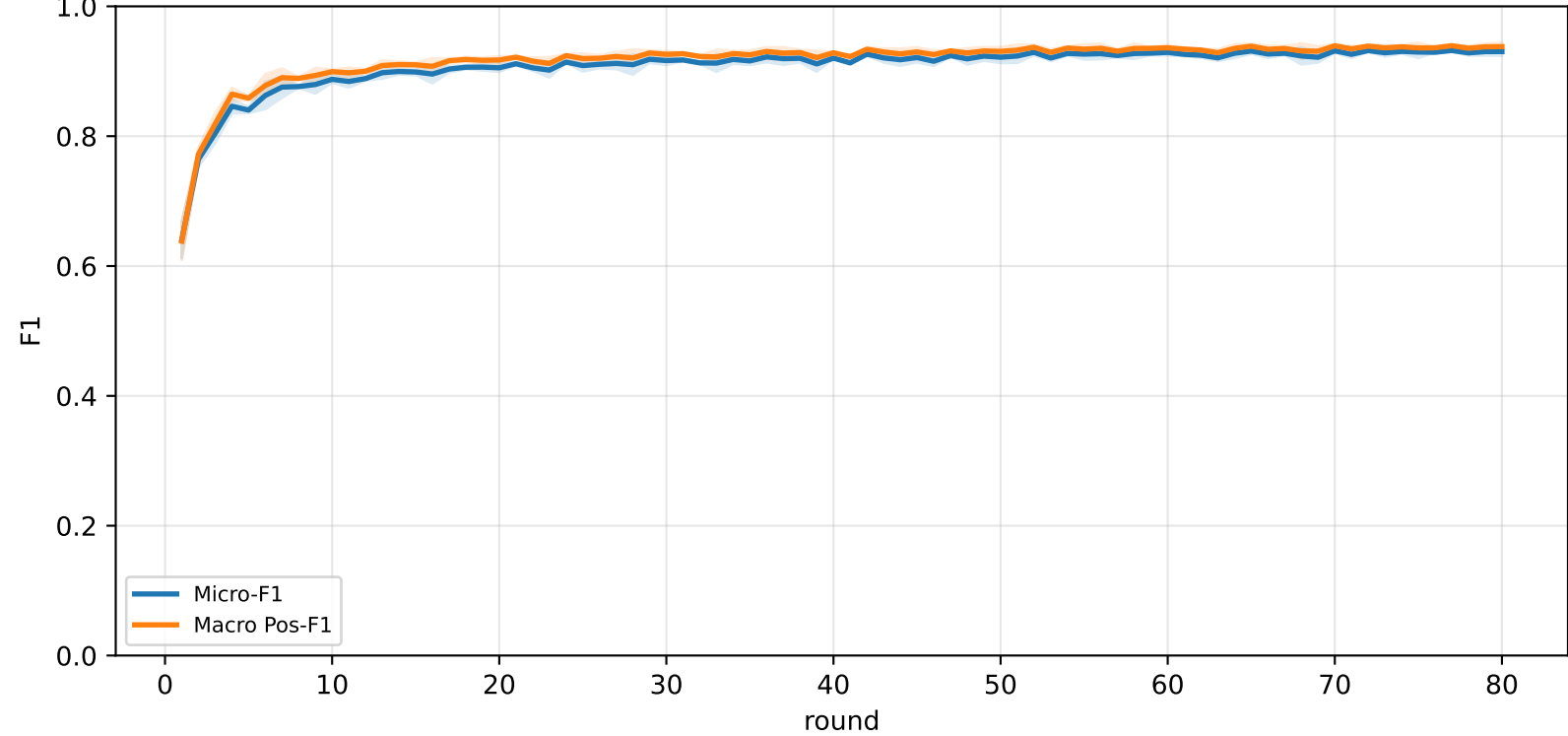
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.7% ± 0.3%	93.5% ± 0.2%	98.9% ± 0.3%	98.0% ± 0.1%	95.7% ± 0.7%	91.4% ± 0.5%	83.5% ± 0.9%
FedProx	93.1% ± 0.2%	93.9% ± 0.2%	98.9% ± 0.2%	98.3% ± 0.2%	96.0% ± 0.3%	92.4% ± 0.5%	83.9% ± 0.8%

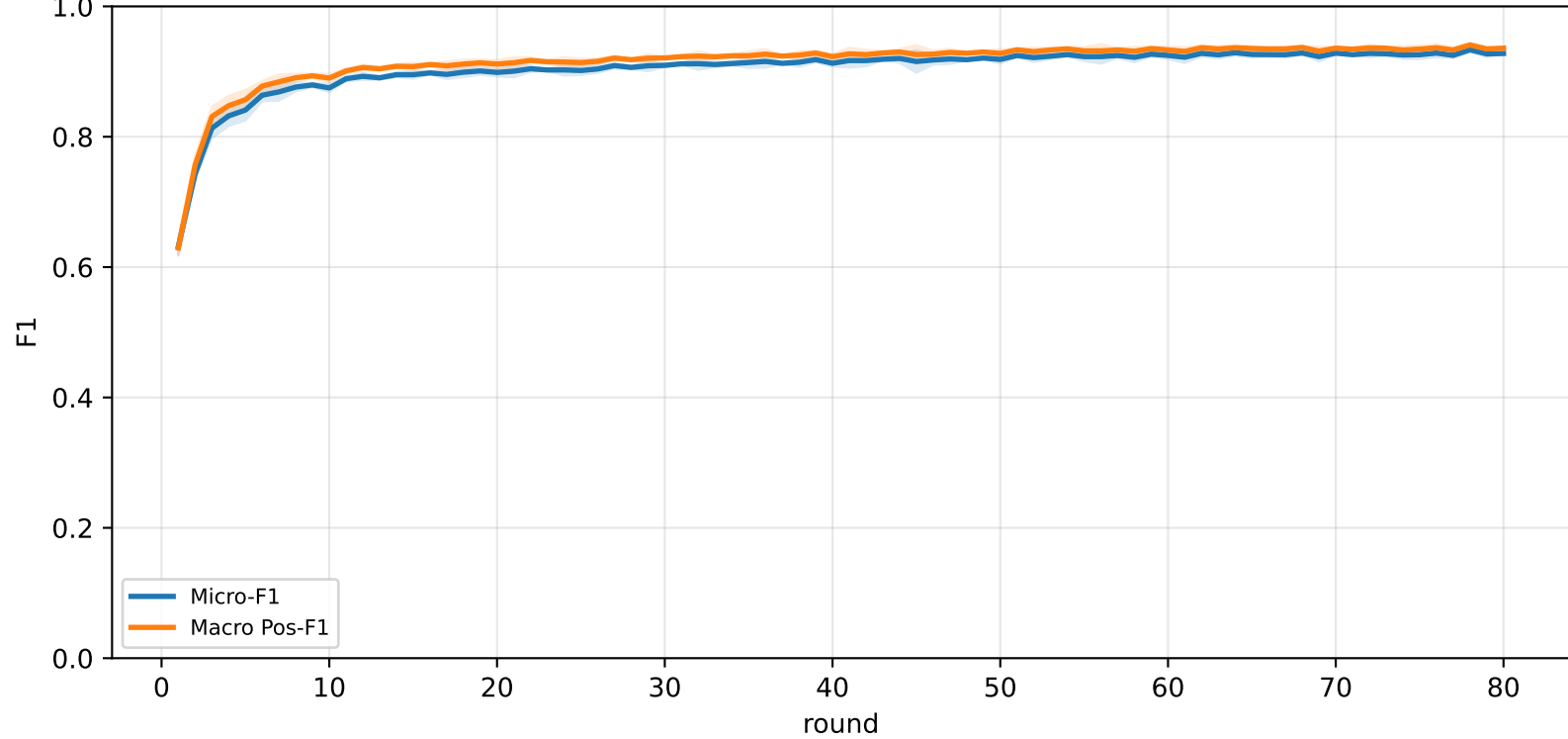
Validation loss comparison



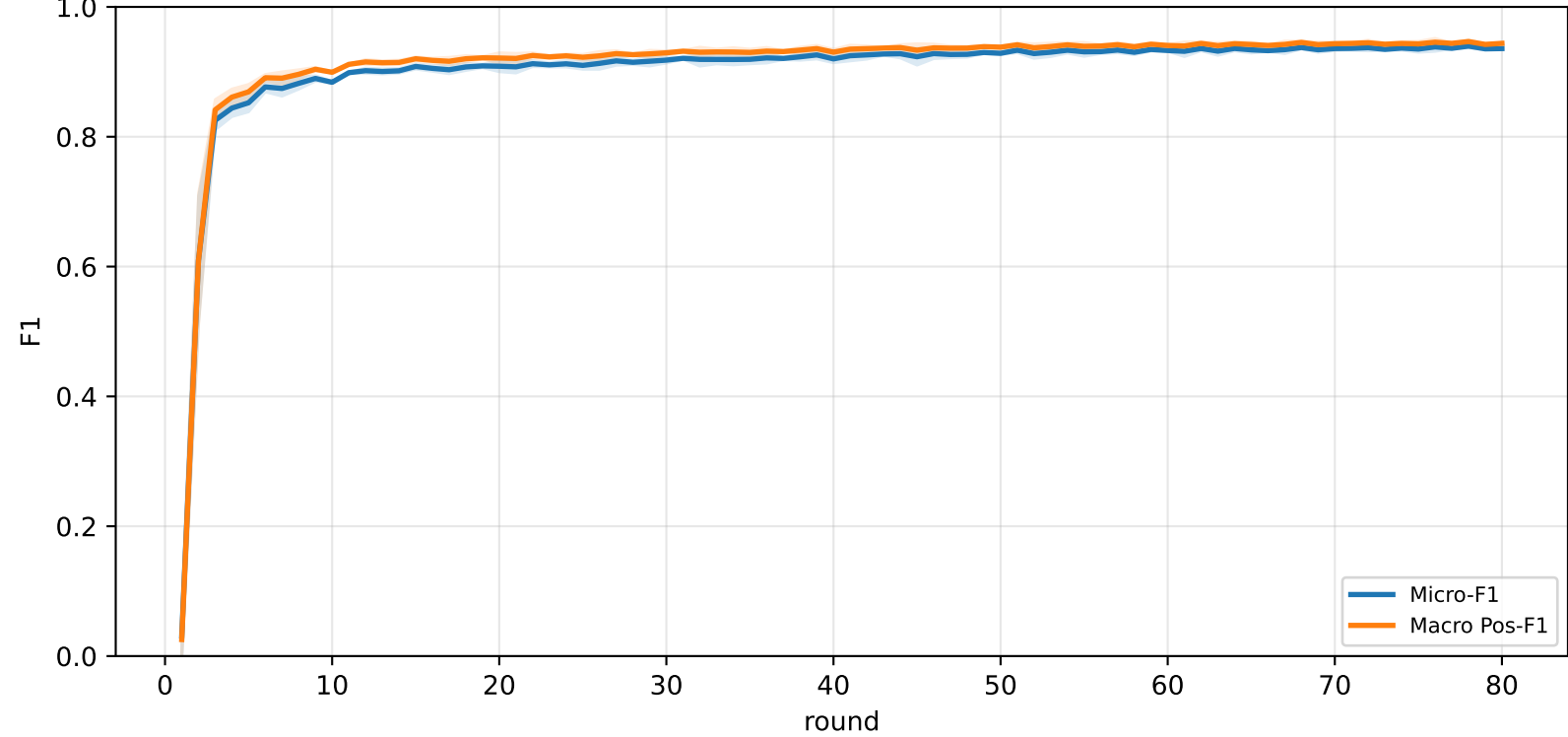
Pooled local centralized micro / macro F1



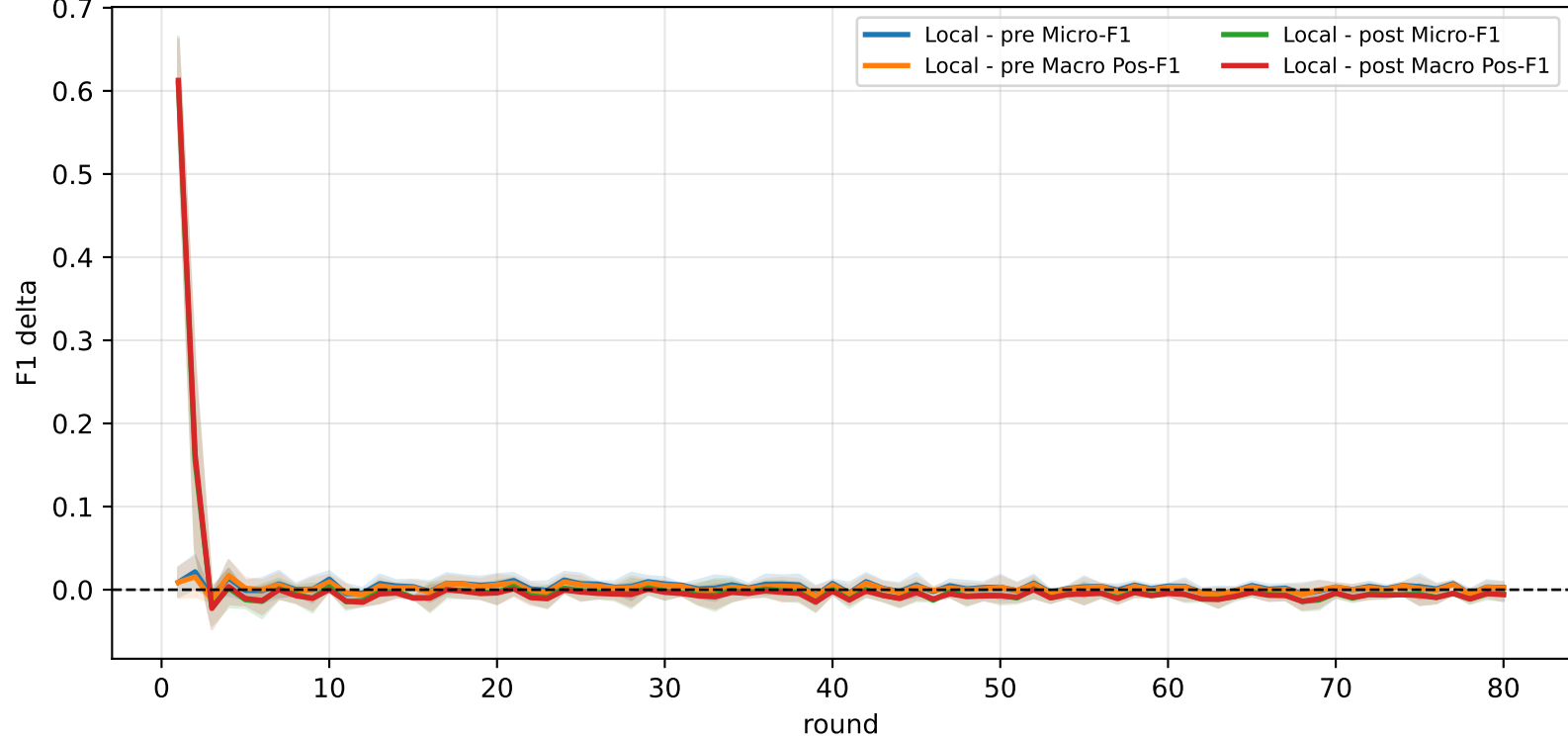
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)

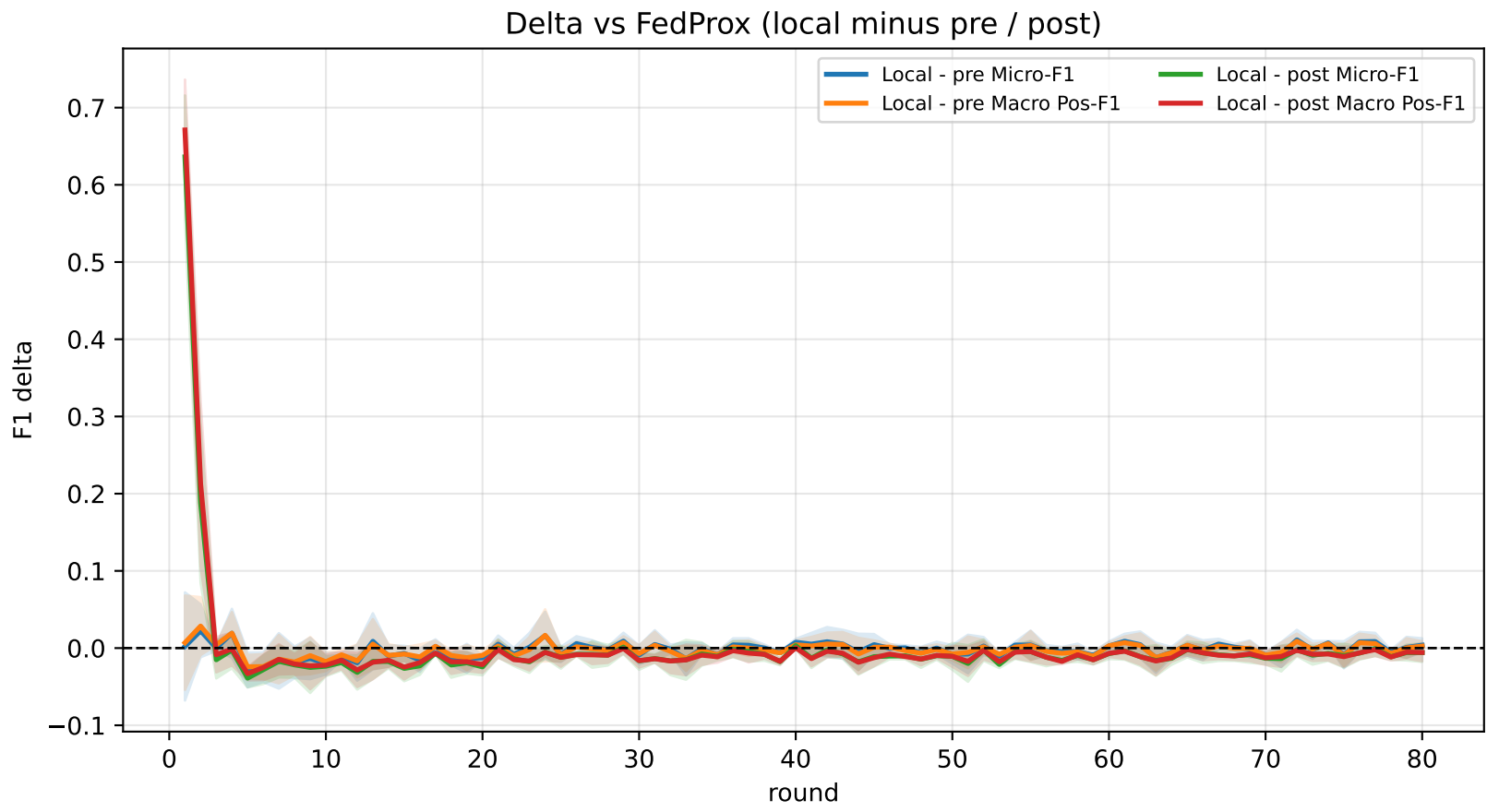
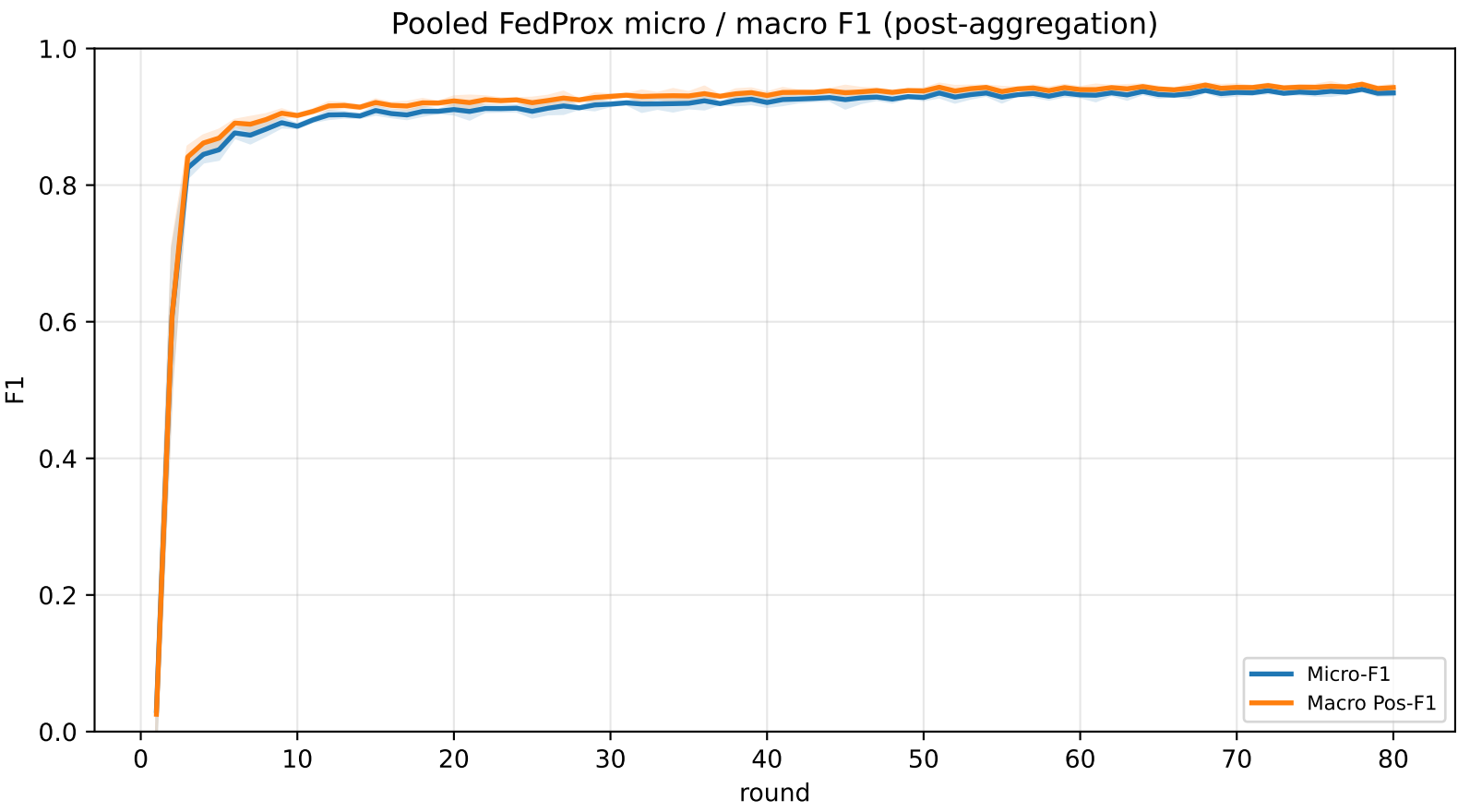
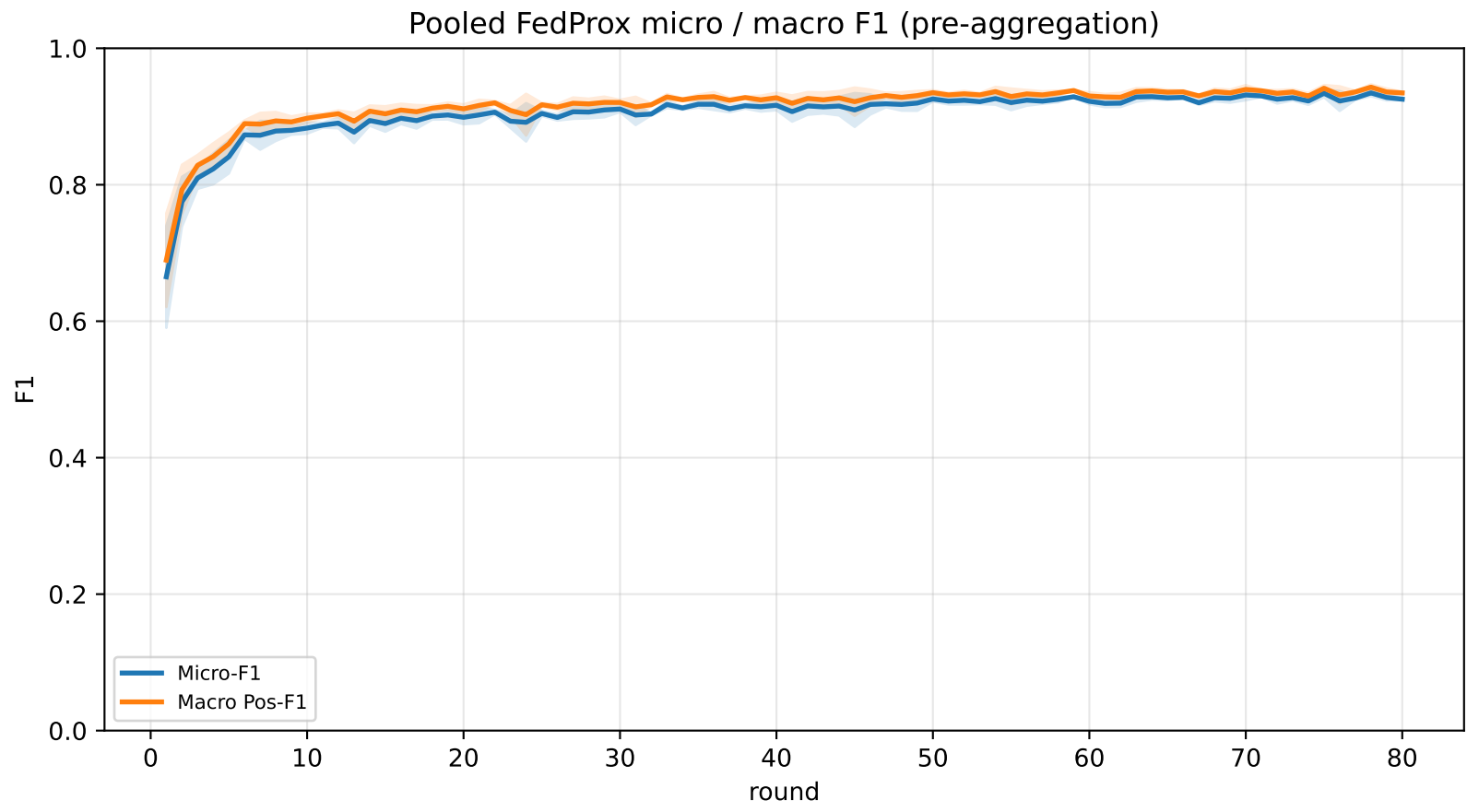
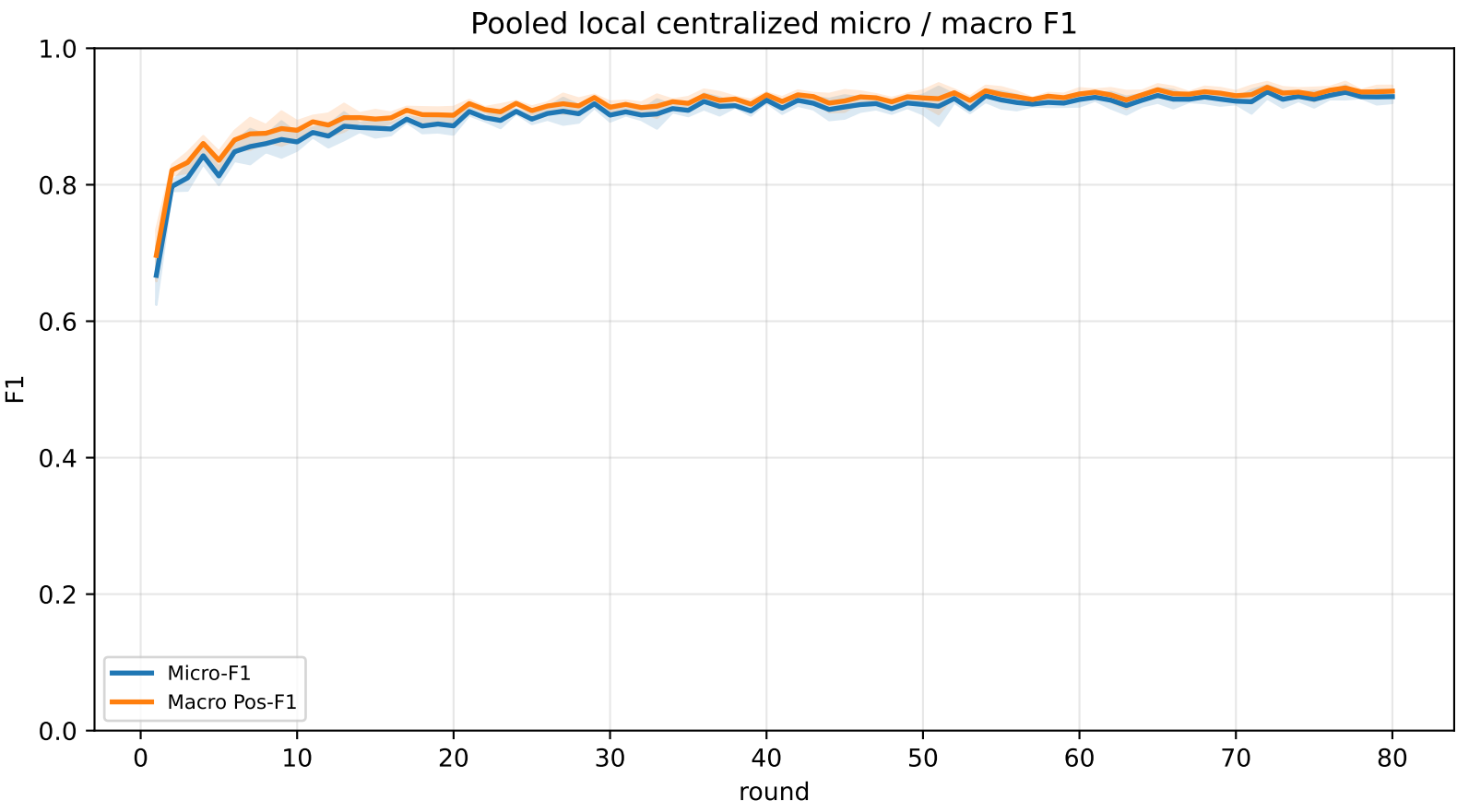
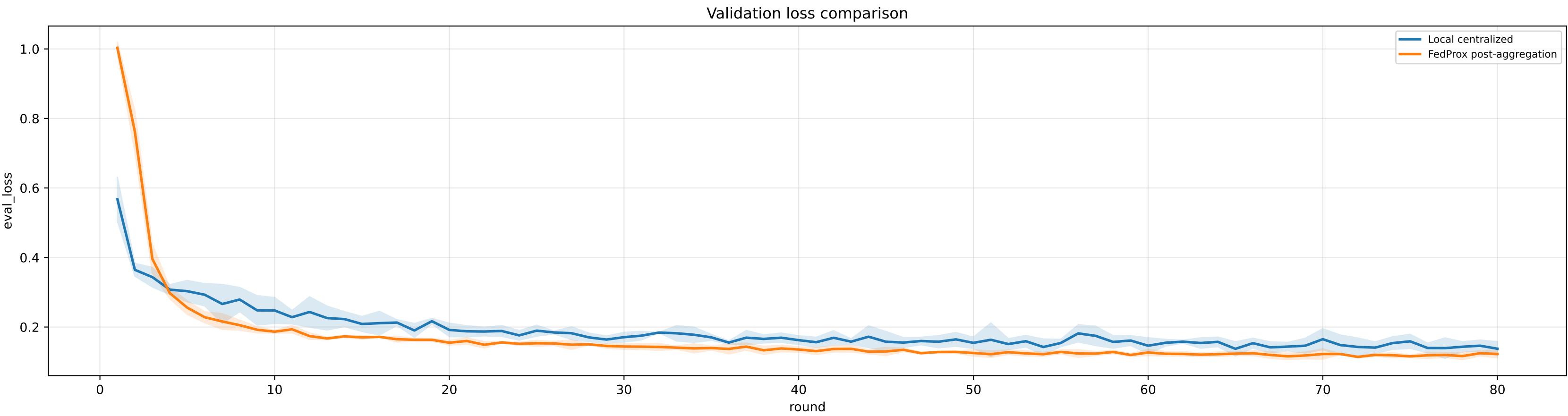


Delta vs FedProx (local minus pre / post)



family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.7% ± 1.0%	93.5% ± 0.8%	99.1% ± 0.2%	98.1% ± 0.1%	95.2% ± 0.9%	91.2% ± 1.4%	84.0% ± 3.0%
FedProx	93.3% ± 0.3%	94.1% ± 0.3%	99.1% ± 0.4%	98.4% ± 0.3%	96.2% ± 0.3%	92.5% ± 0.7%	84.2% ± 1.1%

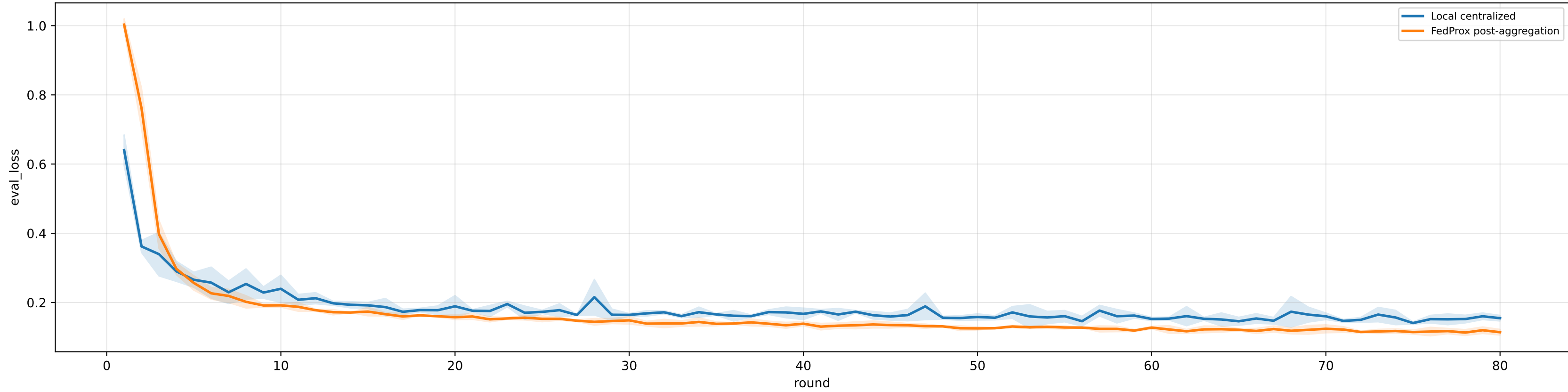


family=specialized_cycle3 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

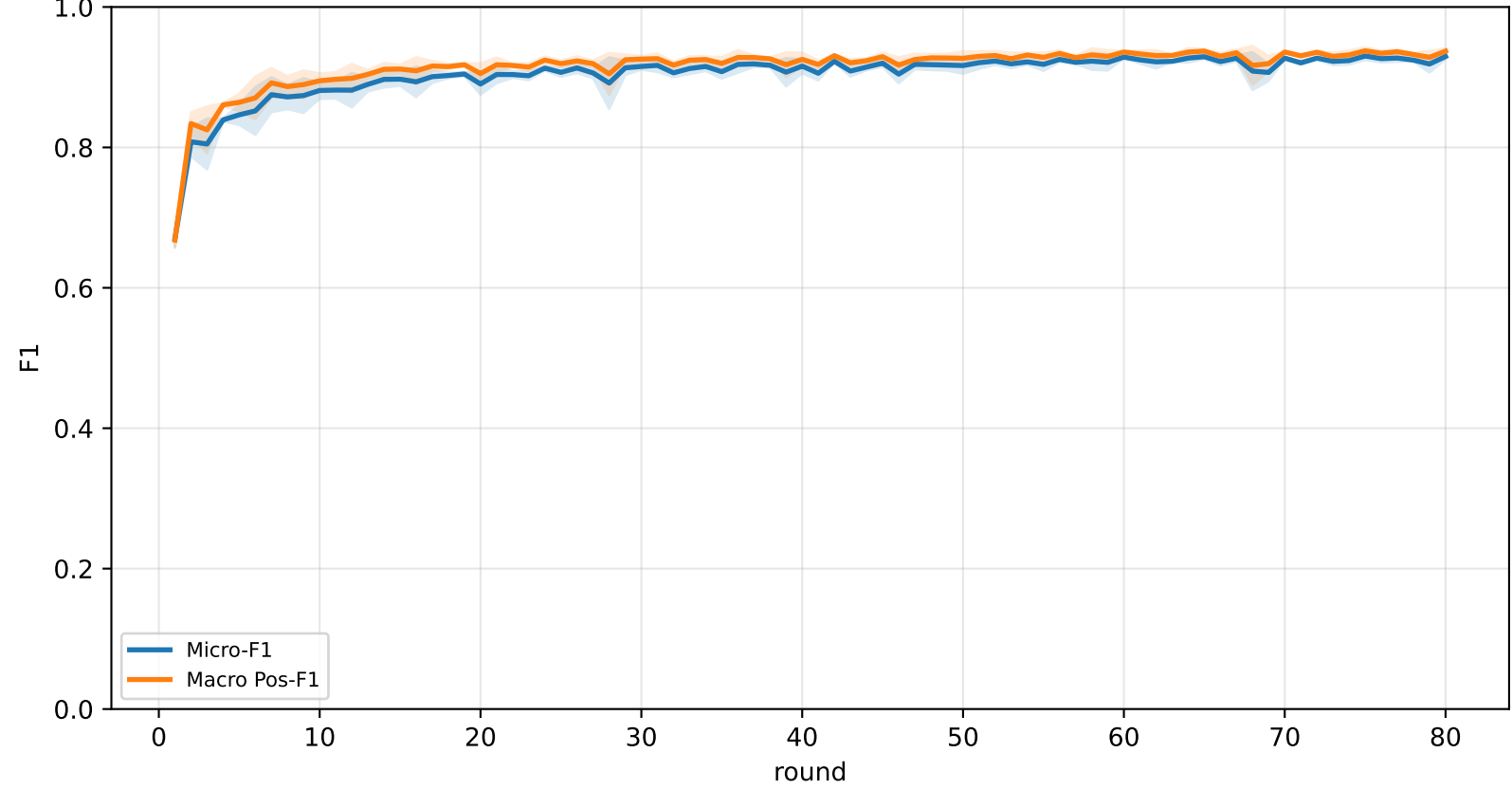
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.7% ± 0.2%	92.7% ± 0.2%	98.9% ± 0.1%	97.9% ± 0.1%	94.7% ± 1.2%	90.1% ± 0.6%	81.9% ± 0.3%
FedProx	93.0% ± 0.2%	93.7% ± 0.2%	98.9% ± 0.3%	97.9% ± 0.4%	95.7% ± 0.5%	92.6% ± 0.5%	83.6% ± 0.9%

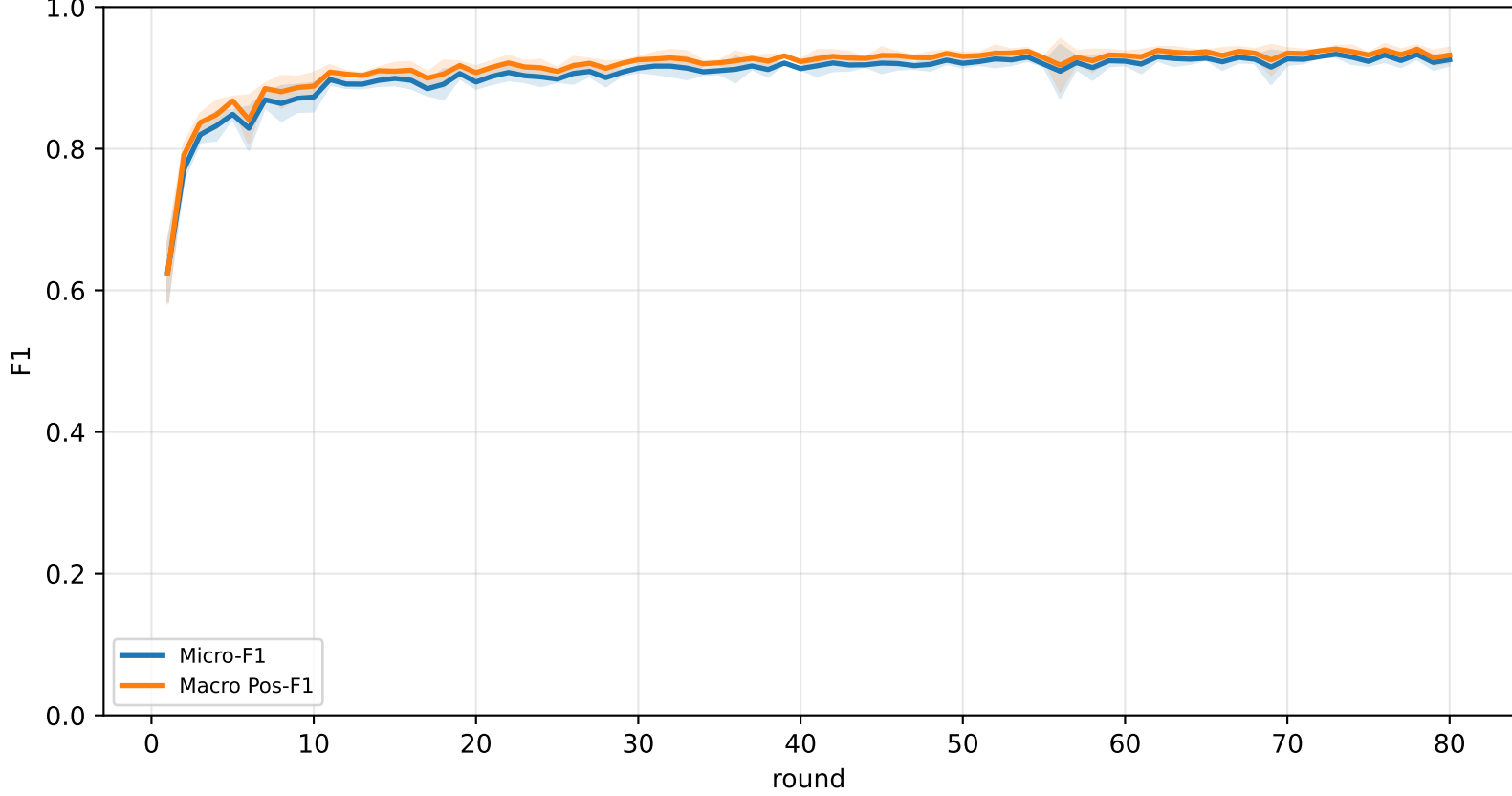
Validation loss comparison



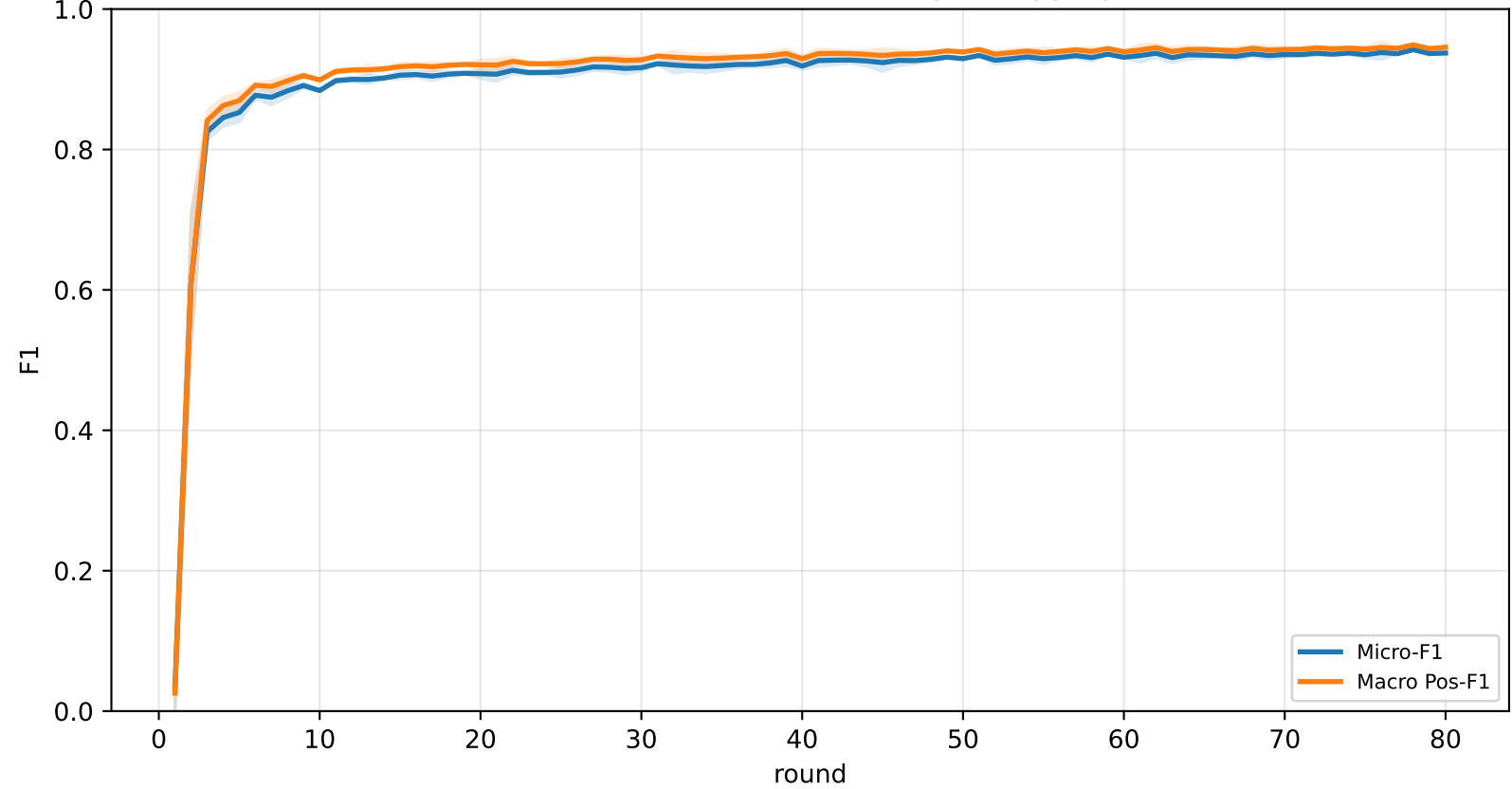
Pooled local centralized micro / macro F1



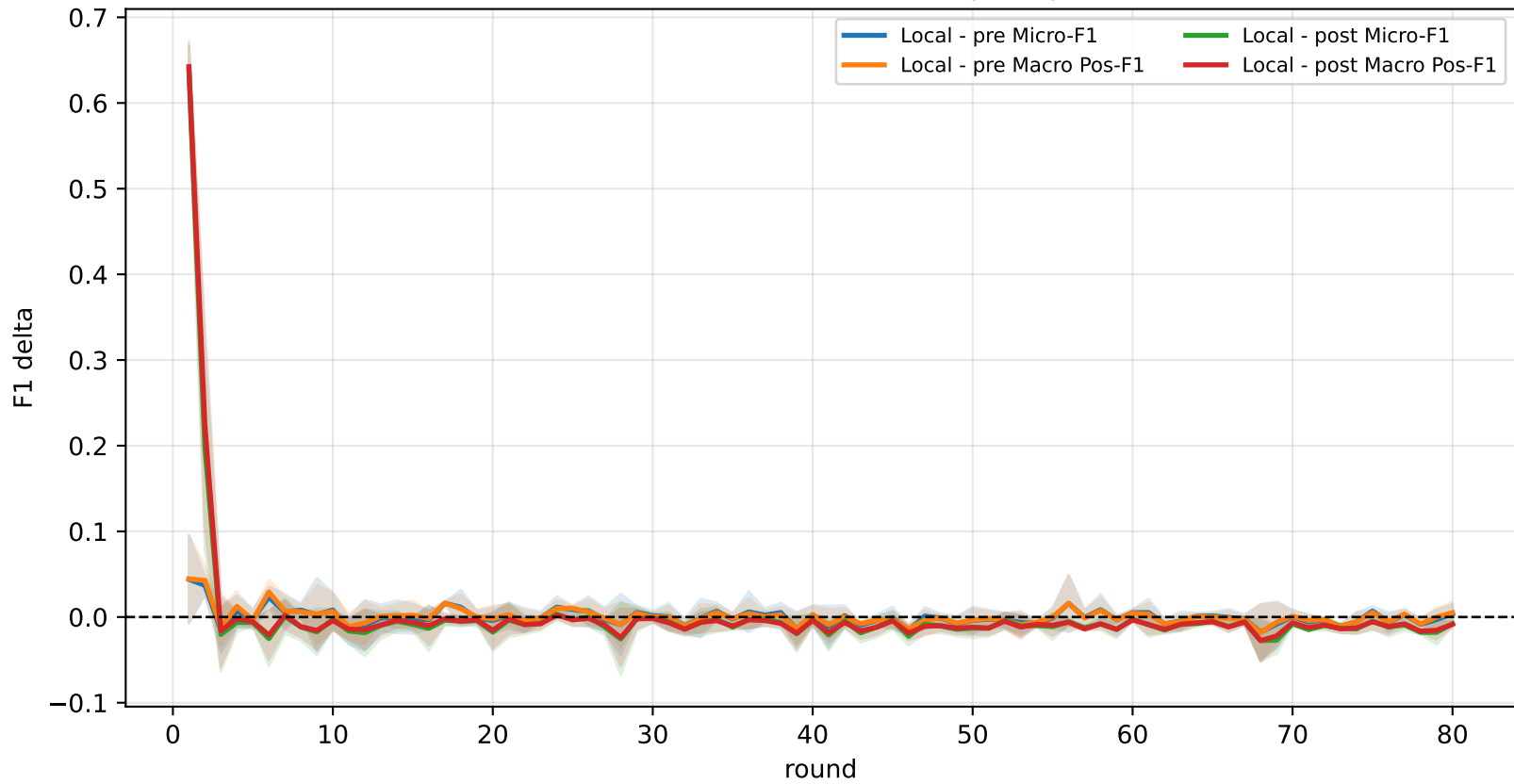
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)

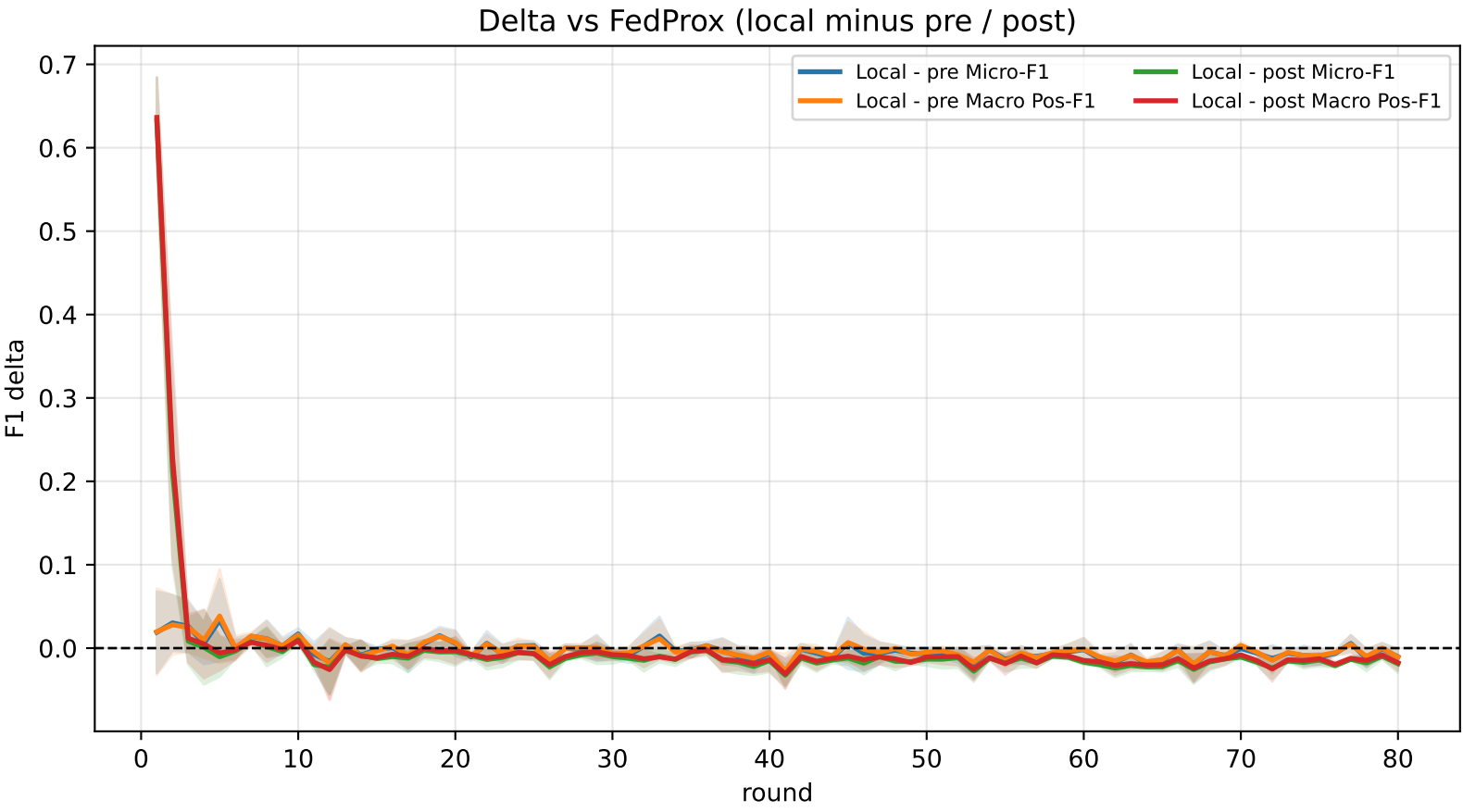
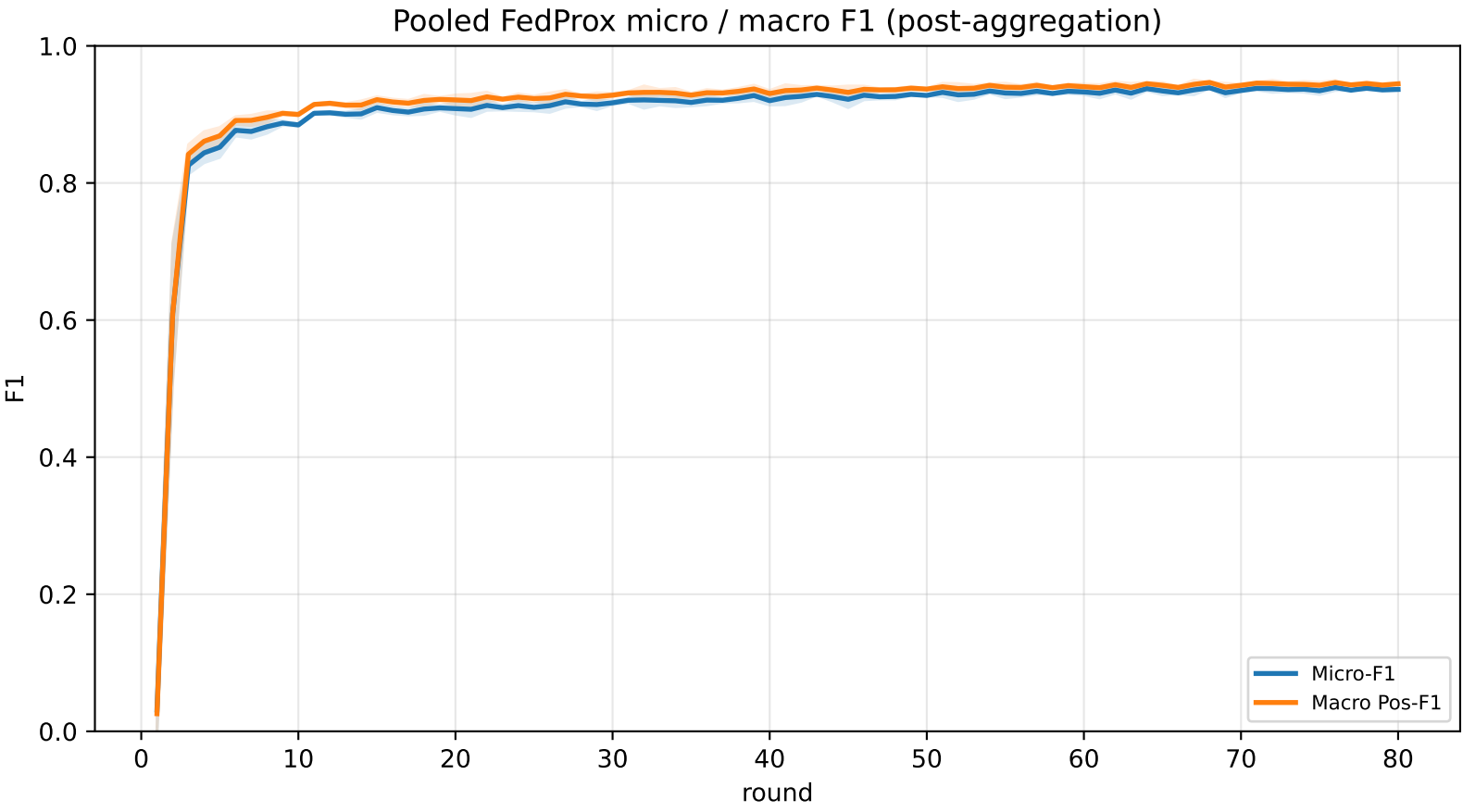
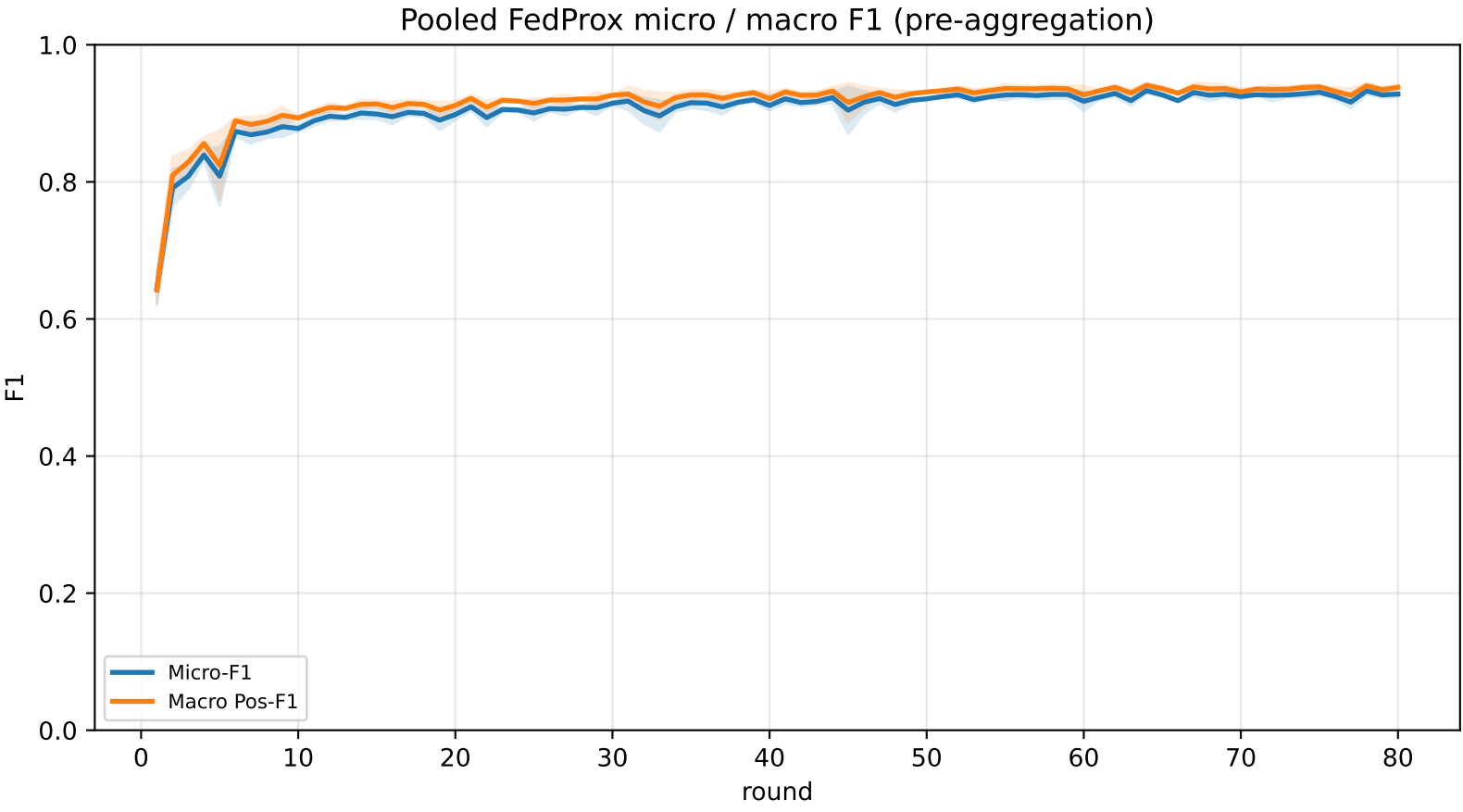
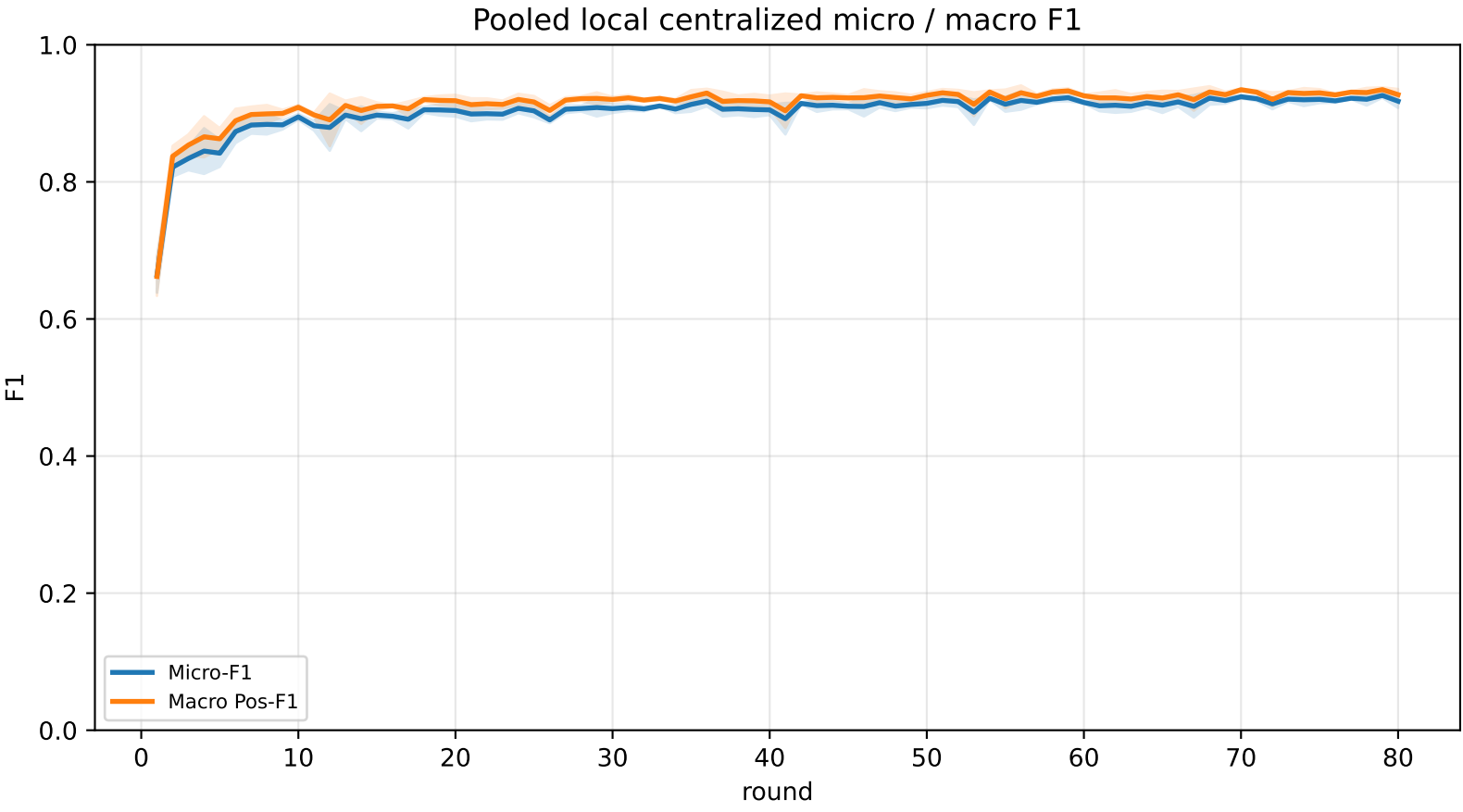
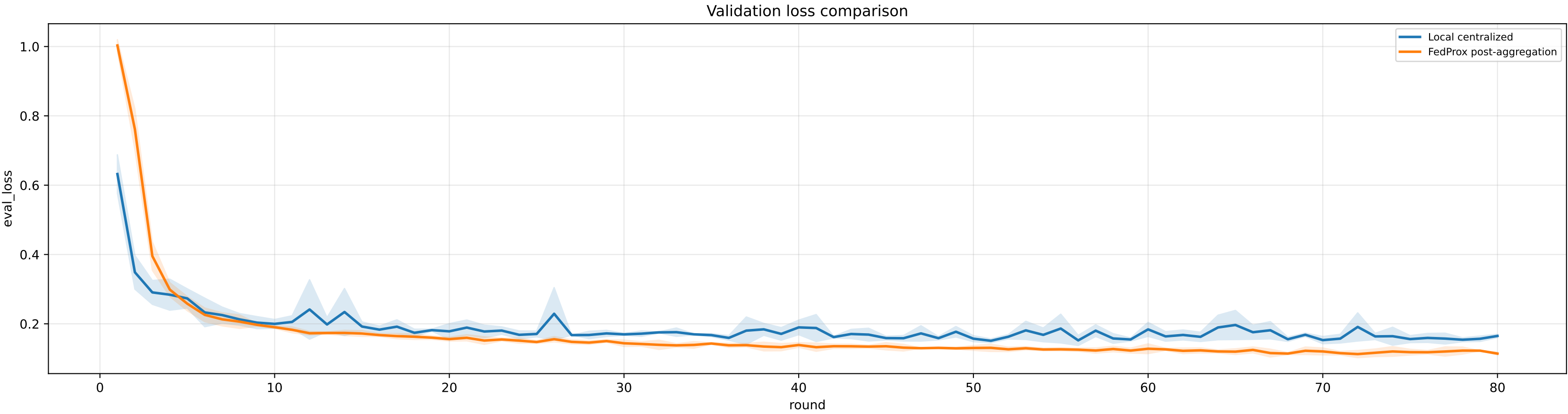


Delta vs FedProx (local minus pre / post)



family=specialized_cycle4 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 | family_centroid_jsd_mean=0.028

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.5% ± 0.4%	92.6% ± 0.2%	98.9% ± 0.4%	98.2% ± 0.1%	96.7% ± 0.5%	89.2% ± 0.2%	80.2% ± 1.5%
FedProx	93.0% ± 0.3%	93.8% ± 0.2%	99.1% ± 0.2%	98.2% ± 0.2%	96.0% ± 0.4%	92.1% ± 0.5%	83.7% ± 1.1%

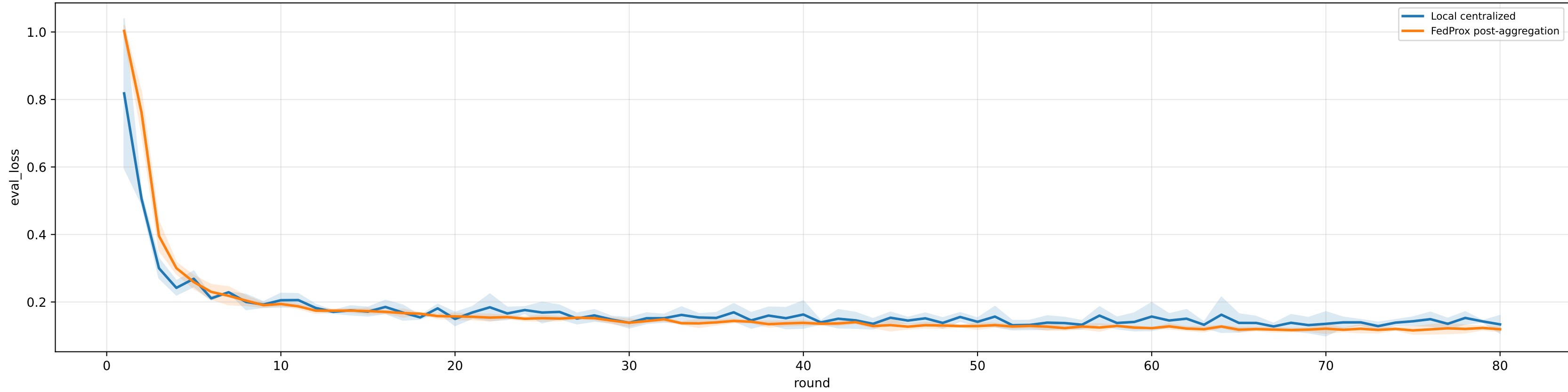


family=specialized_cycle5 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 13.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

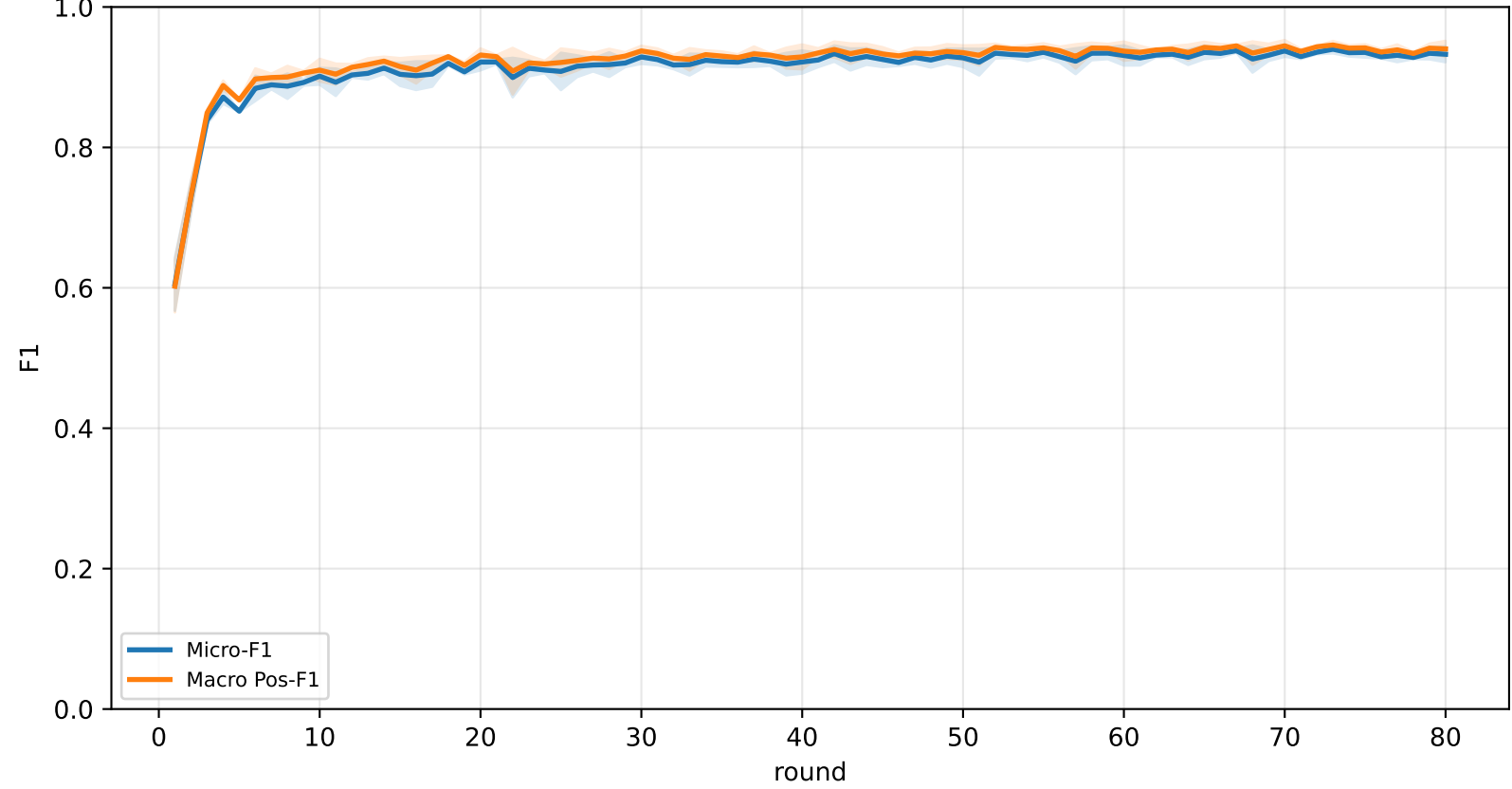
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.2% ± 0.7%	94.0% ± 0.6%	98.6% ± 0.5%	97.9% ± 0.1%	96.2% ± 0.5%	94.4% ± 0.3%	82.9% ± 2.4%
FedProx	93.2% ± 0.2%	94.0% ± 0.2%	98.8% ± 0.3%	98.7% ± 0.2%	96.0% ± 0.4%	92.5% ± 0.5%	84.0% ± 0.7%

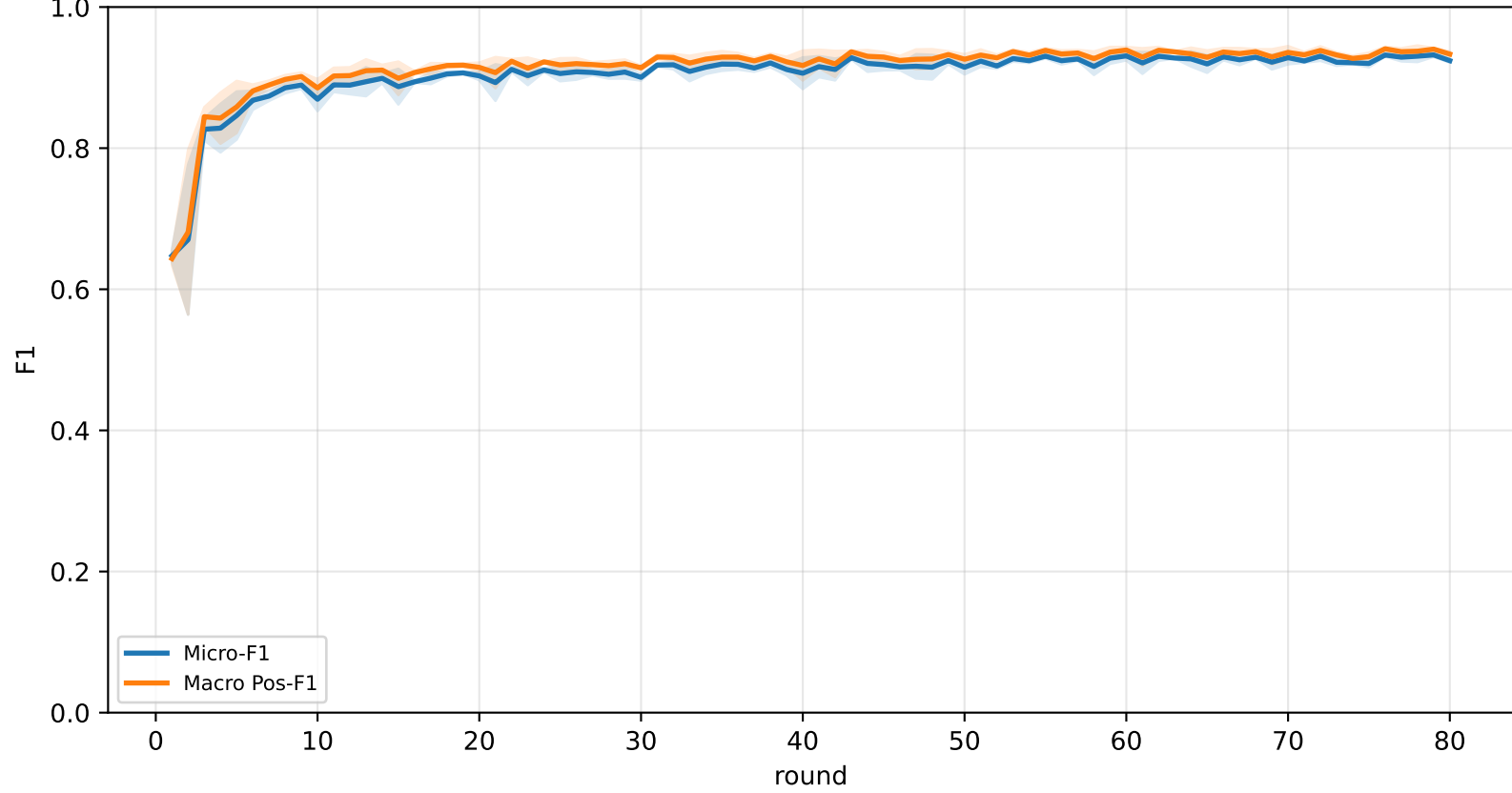
Validation loss comparison



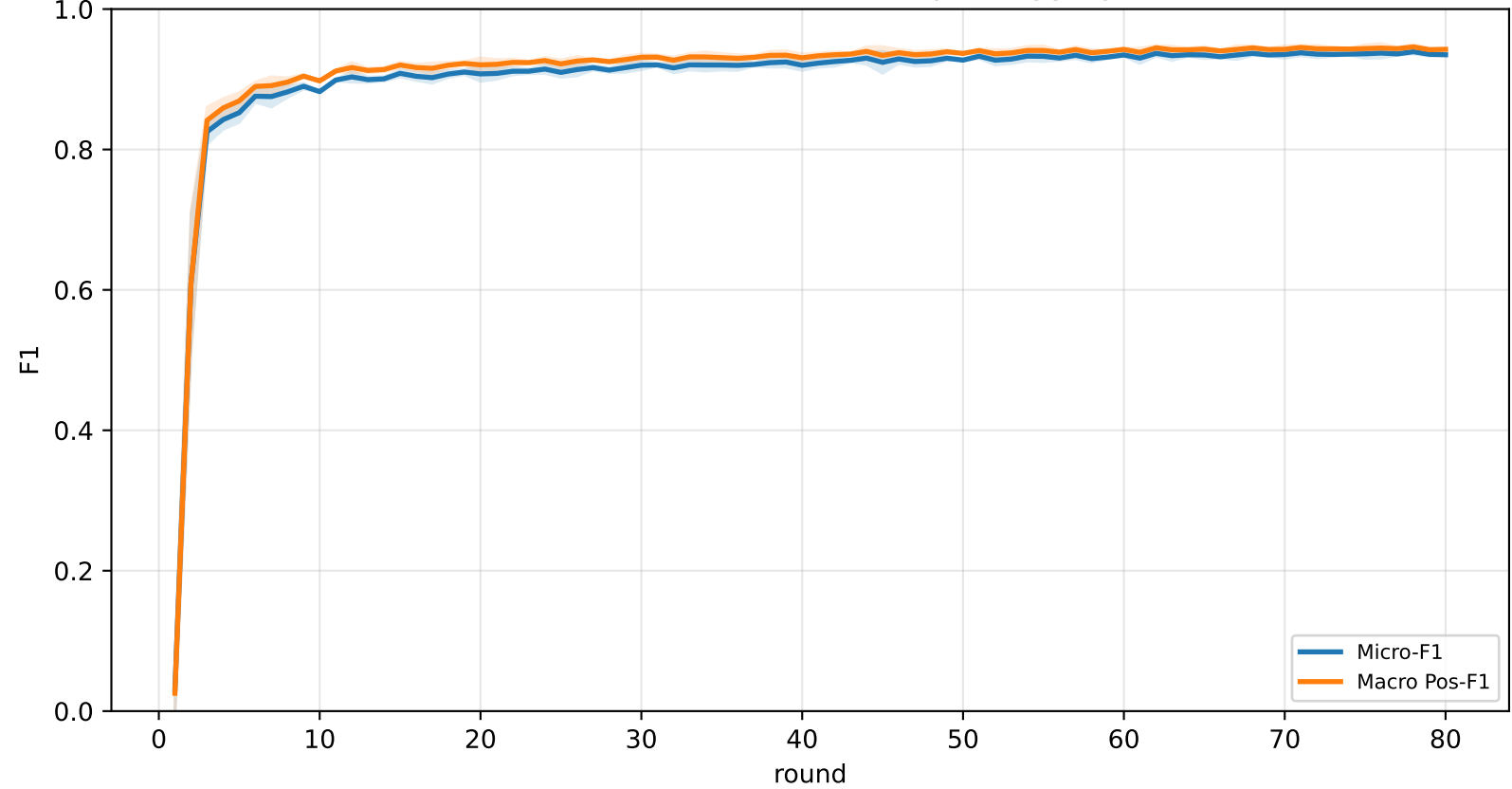
Pooled local centralized micro / macro F1



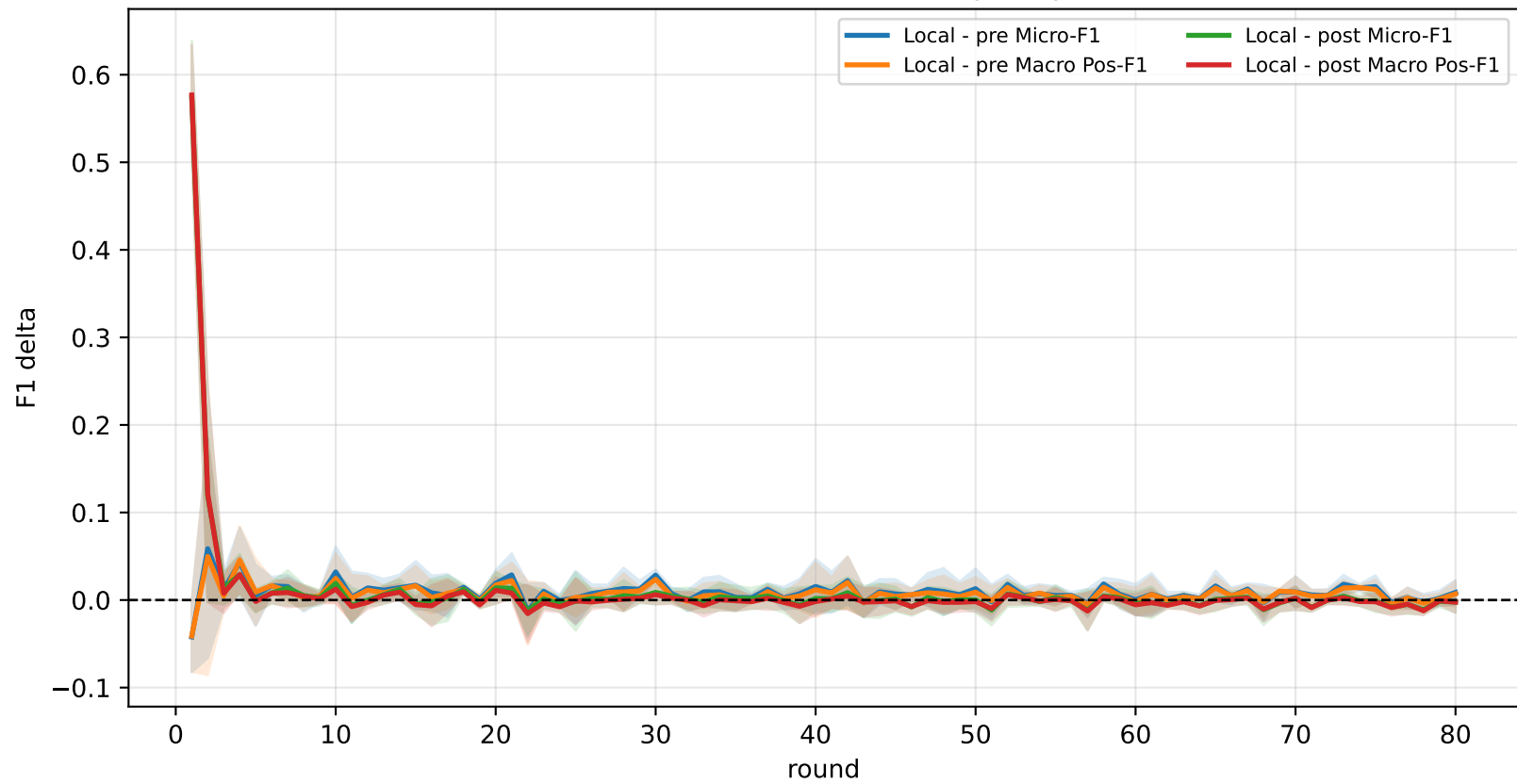
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

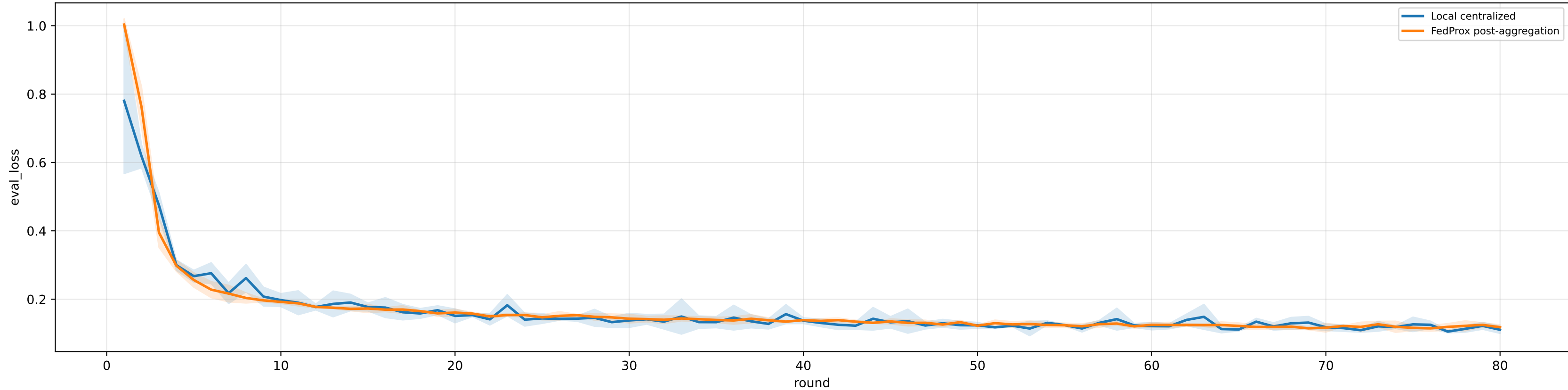


family=specialized_cycle6 | n_clients=1 | cycle2=visible 9.7%±0.0% / true 19.4% | cycle3=visible 11.8%±0.0% / true 23.6% | cycle4=visible 12.8%±0.0% / true 25.6% |
cycle5=visible 12.8%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.029 | task_jsd_max=0.030 |
family_centroid_jsd_mean=0.028

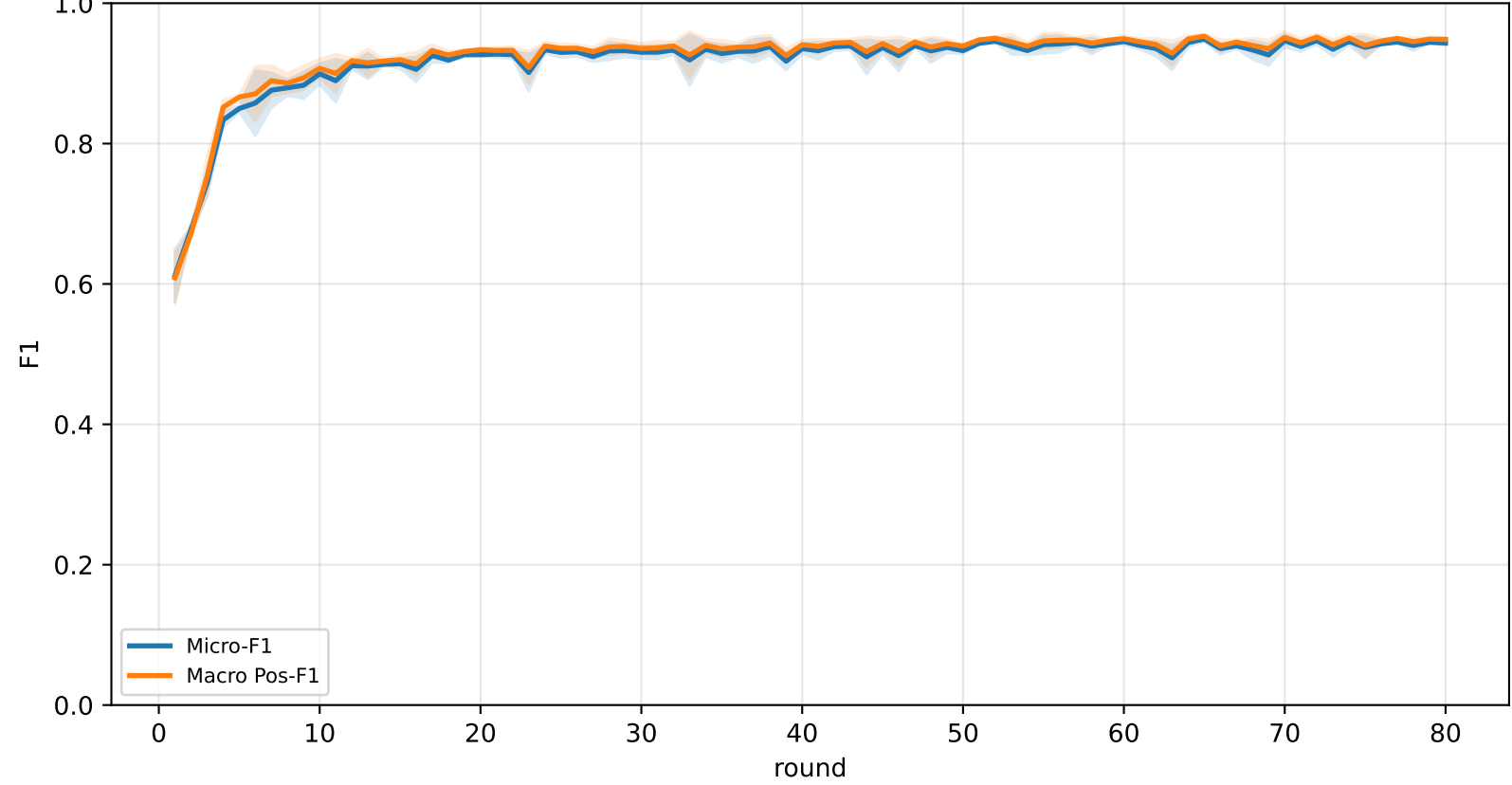
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.2% ± 0.5%	94.7% ± 0.4%	98.9% ± 0.3%	97.9% ± 0.3%	95.5% ± 0.4%	92.4% ± 1.0%	88.7% ± 1.2%
FedProx	93.2% ± 0.2%	93.9% ± 0.1%	98.9% ± 0.4%	98.2% ± 0.4%	95.9% ± 0.2%	92.4% ± 0.5%	84.2% ± 0.3%

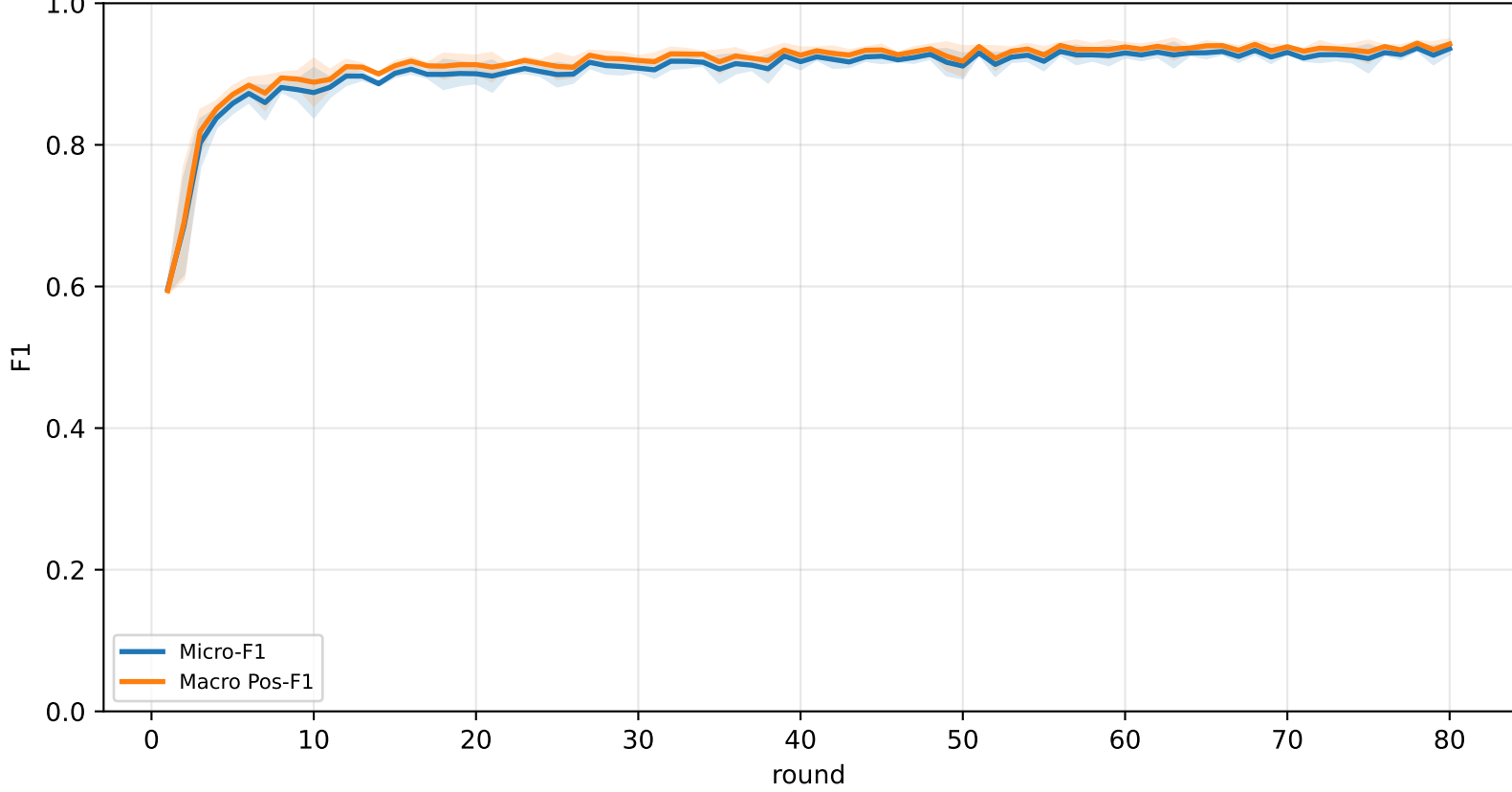
Validation loss comparison



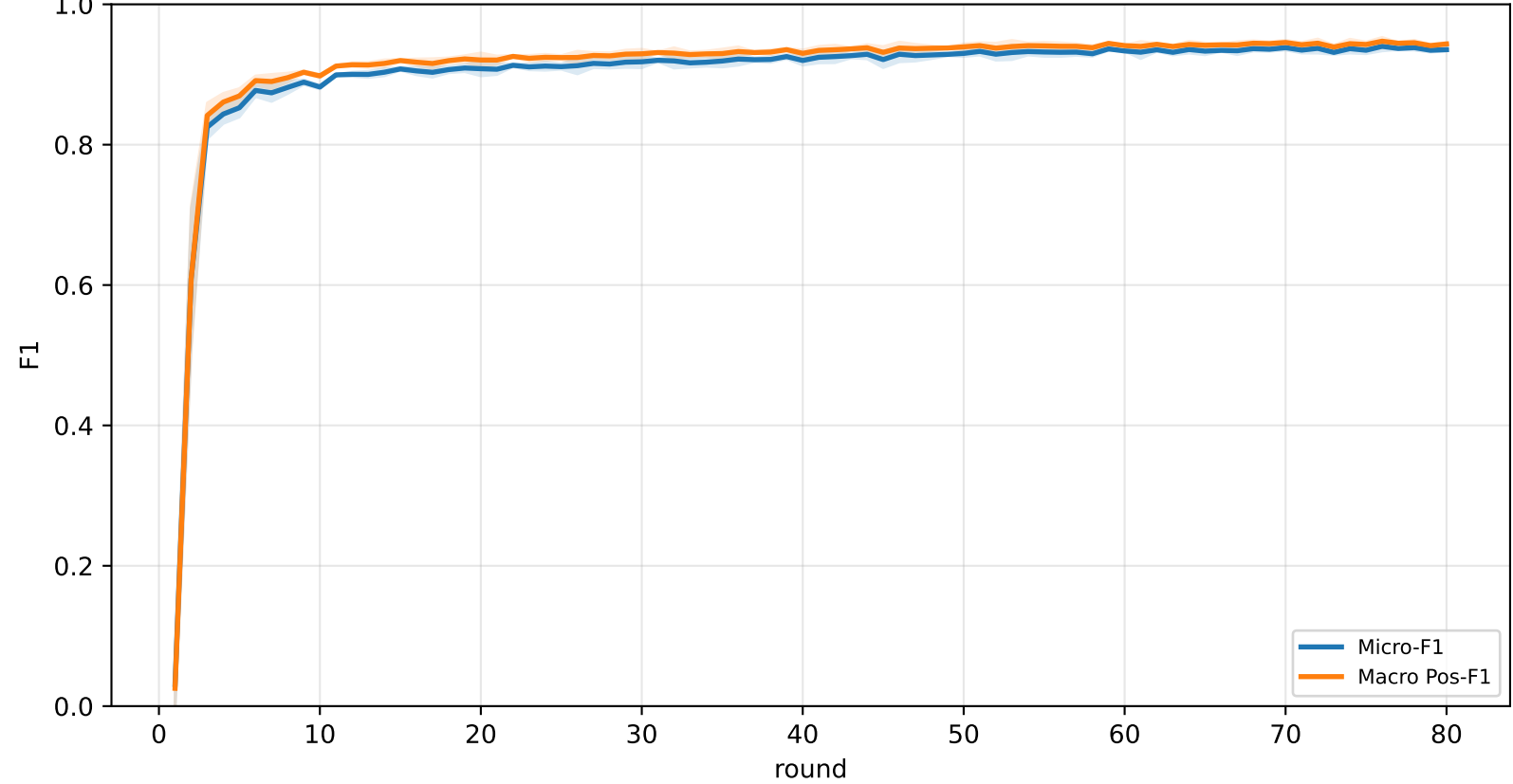
Pooled local centralized micro / macro F1



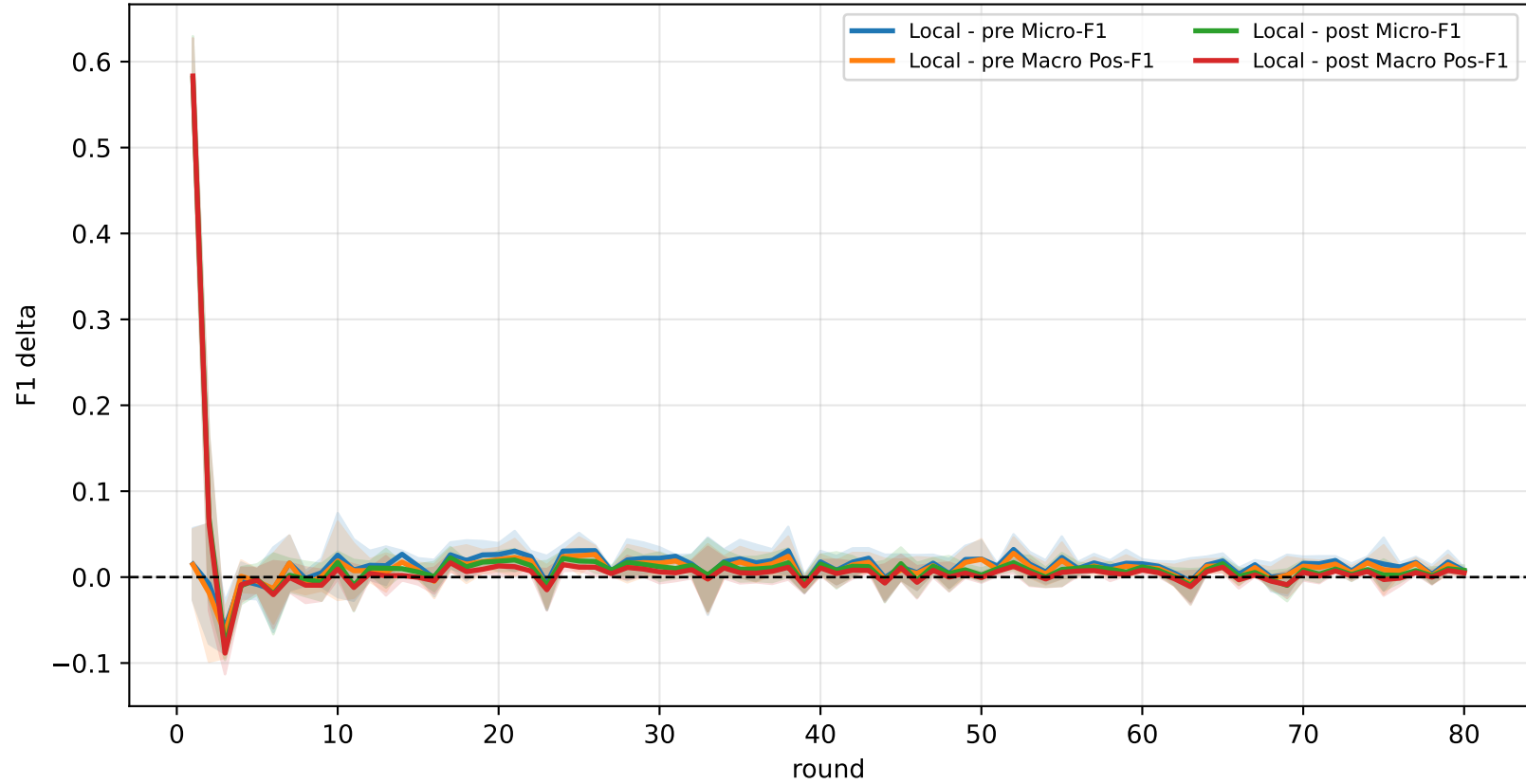
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)



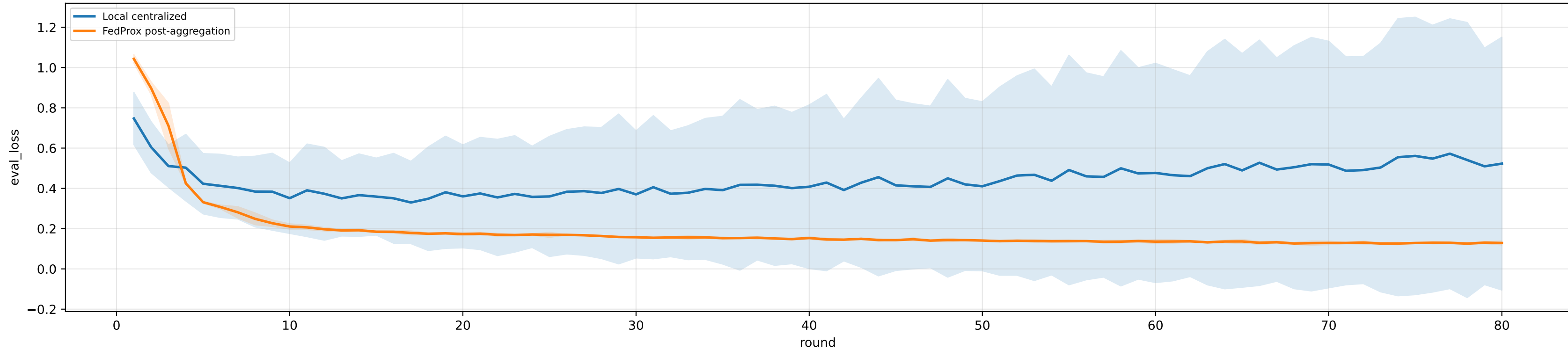
Controlled task specialization level: High | FedProx diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

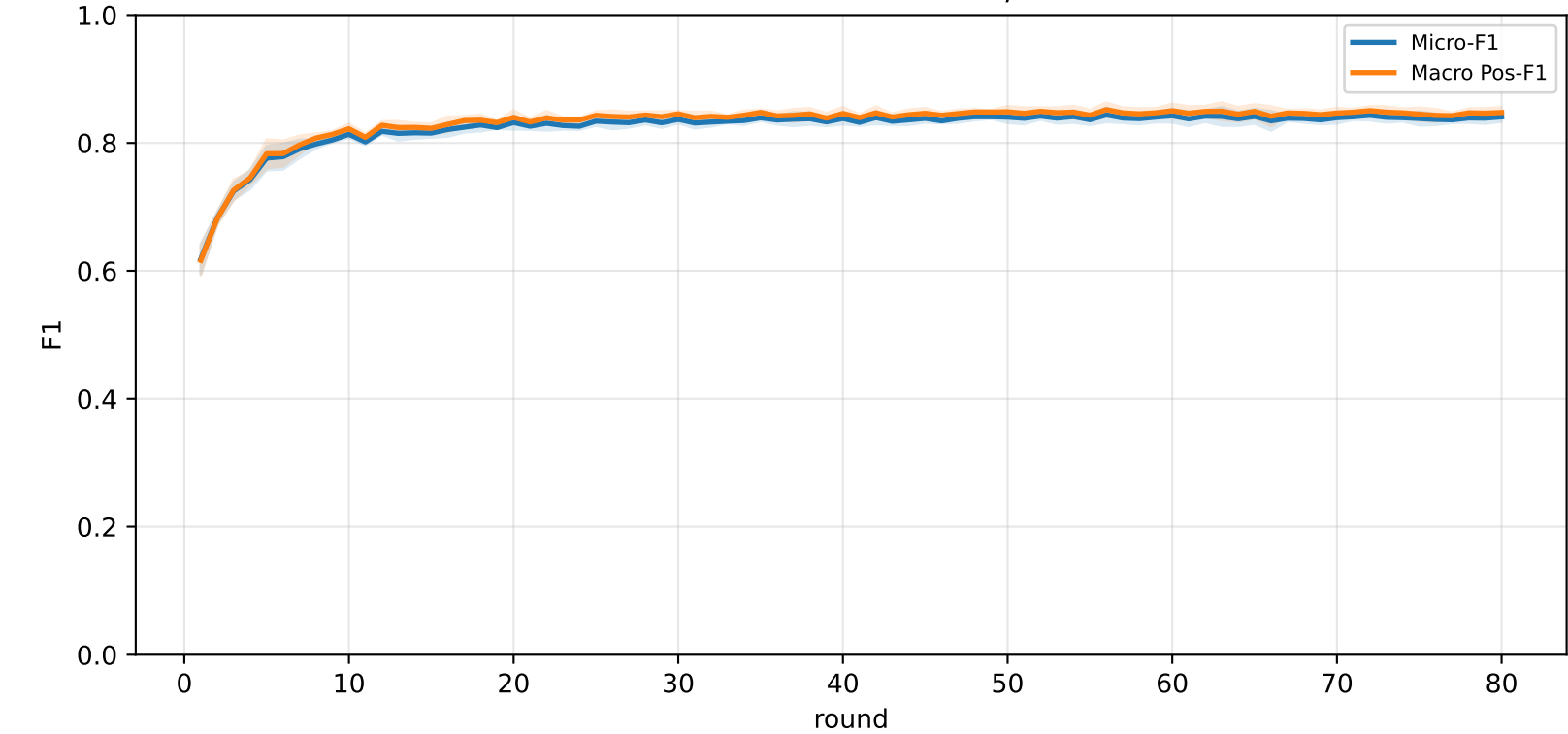
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.1% ± 0.3%	85.3% ± 0.3%	87.0% ± 1.6%	88.2% ± 0.8%	86.8% ± 0.5%	83.1% ± 1.1%	73.7% ± 0.3%
FedProx	92.8% ± 0.4%	93.6% ± 0.3%	98.9% ± 0.2%	97.9% ± 0.3%	95.3% ± 0.7%	92.4% ± 1.9%	83.7% ± 0.8%

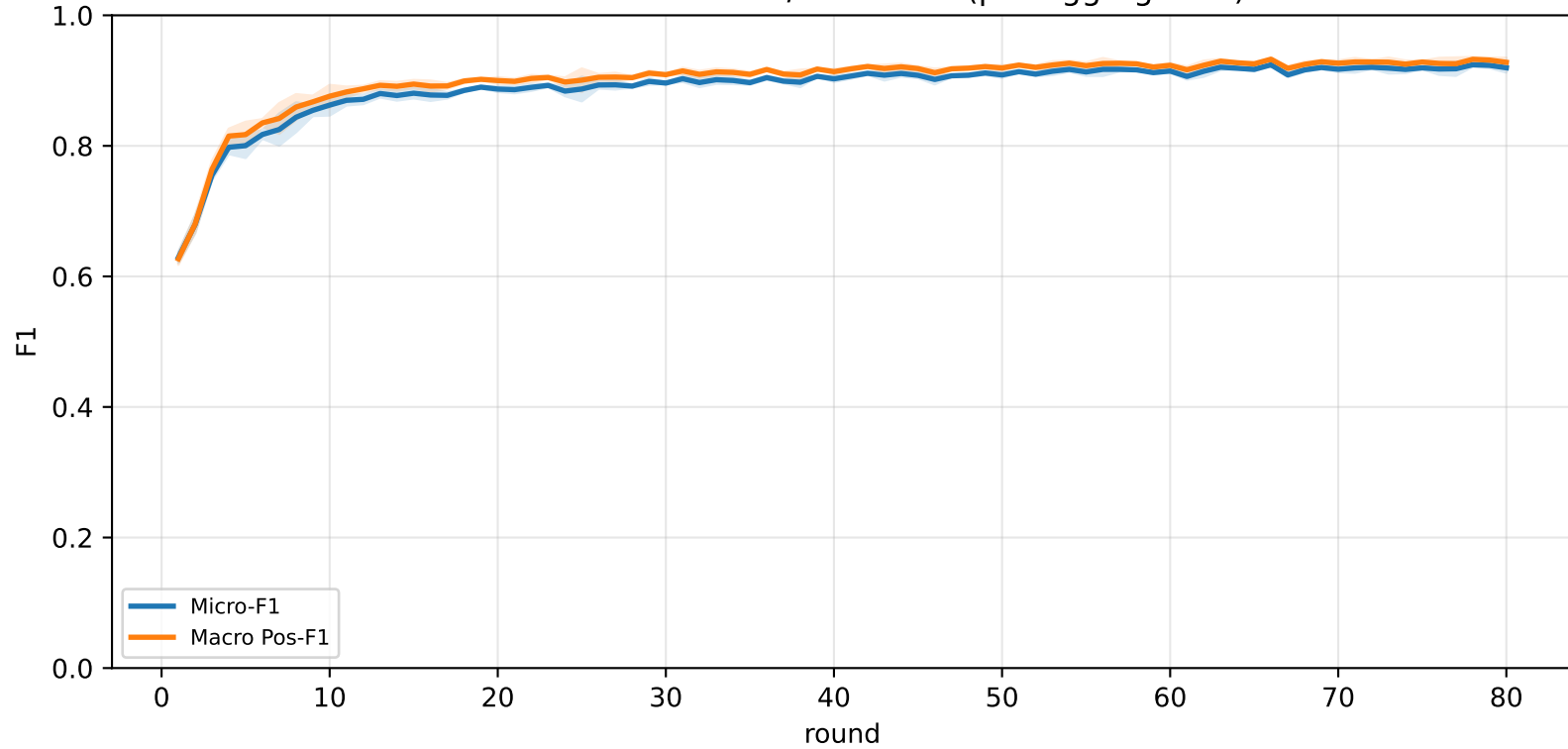
Validation loss comparison



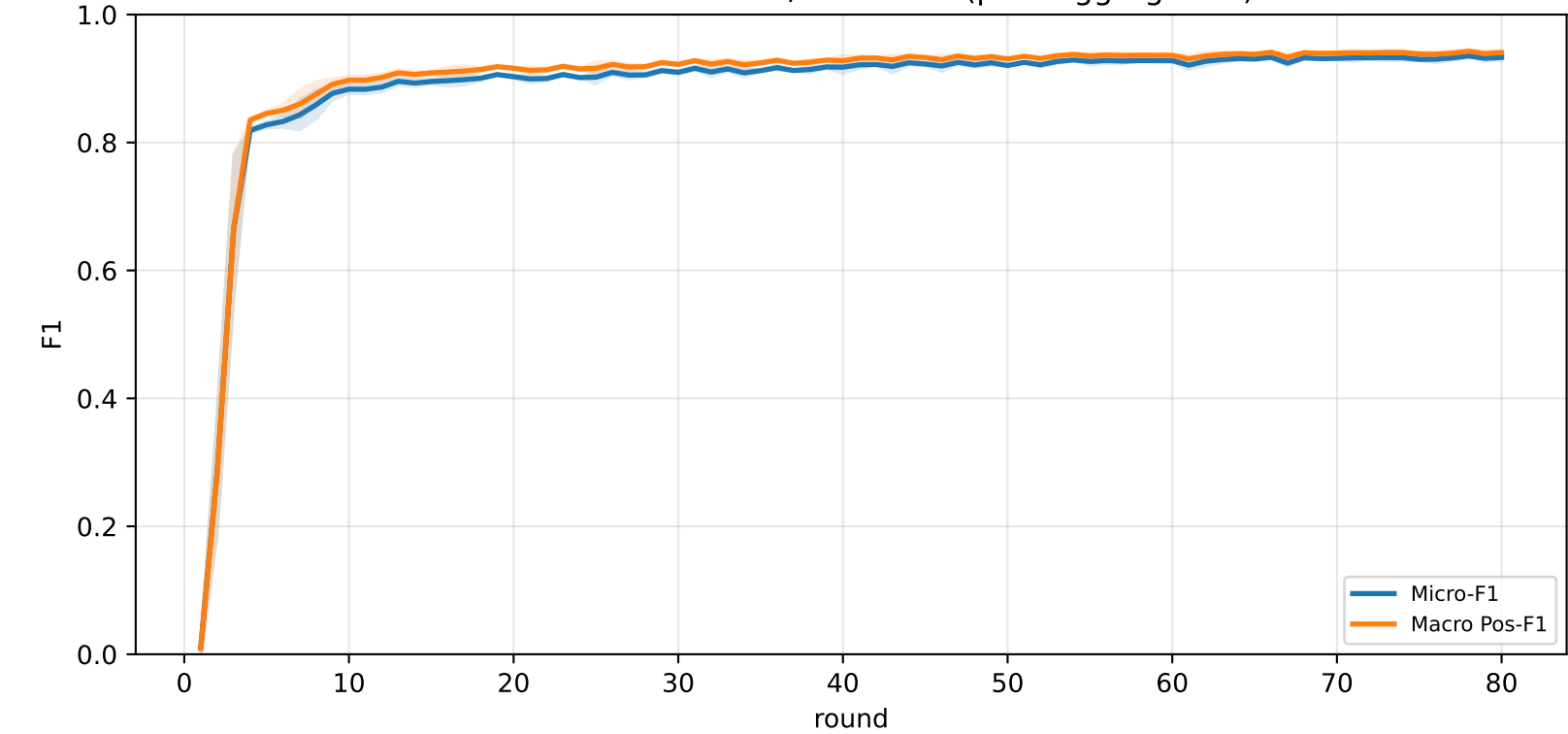
Pooled local centralized micro / macro F1



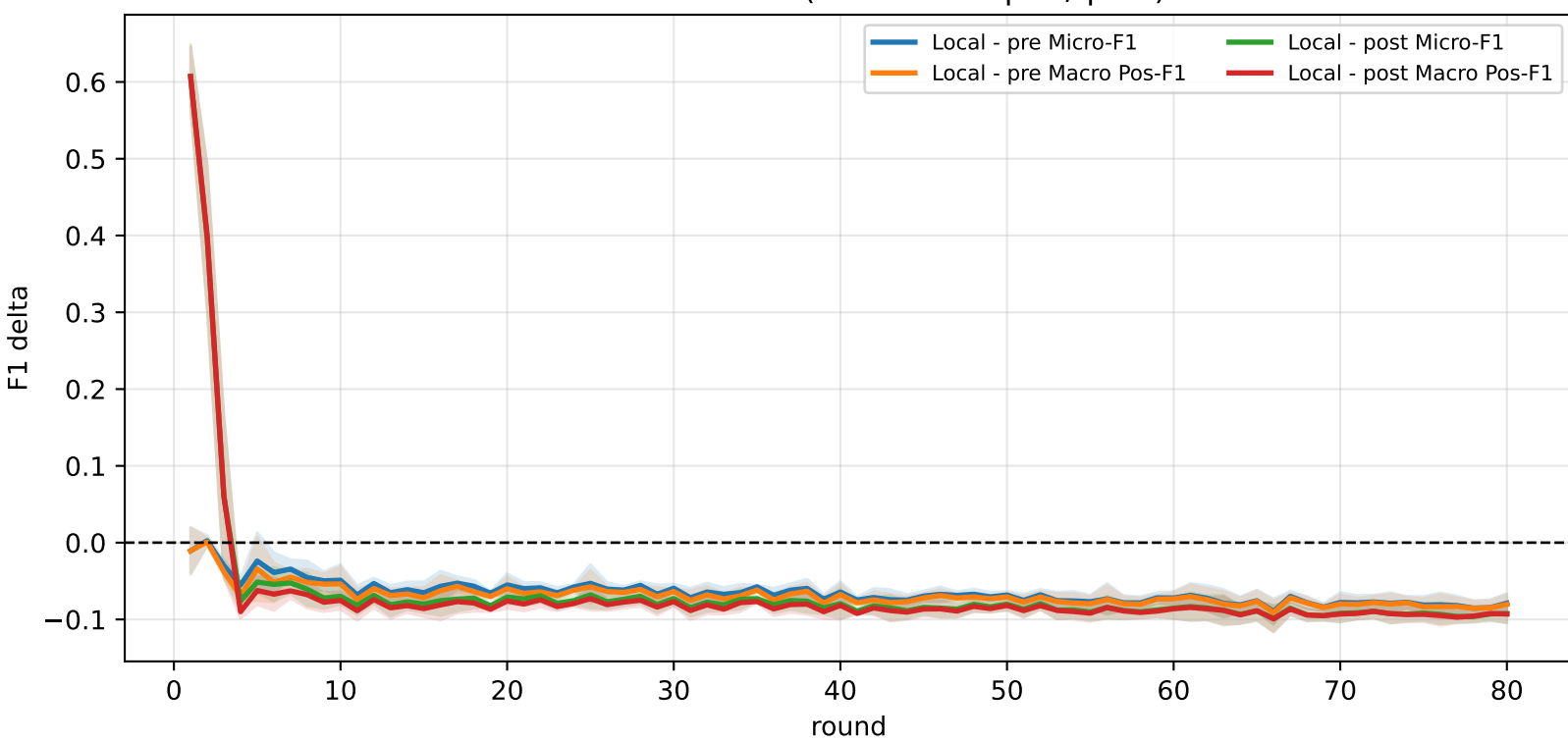
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)

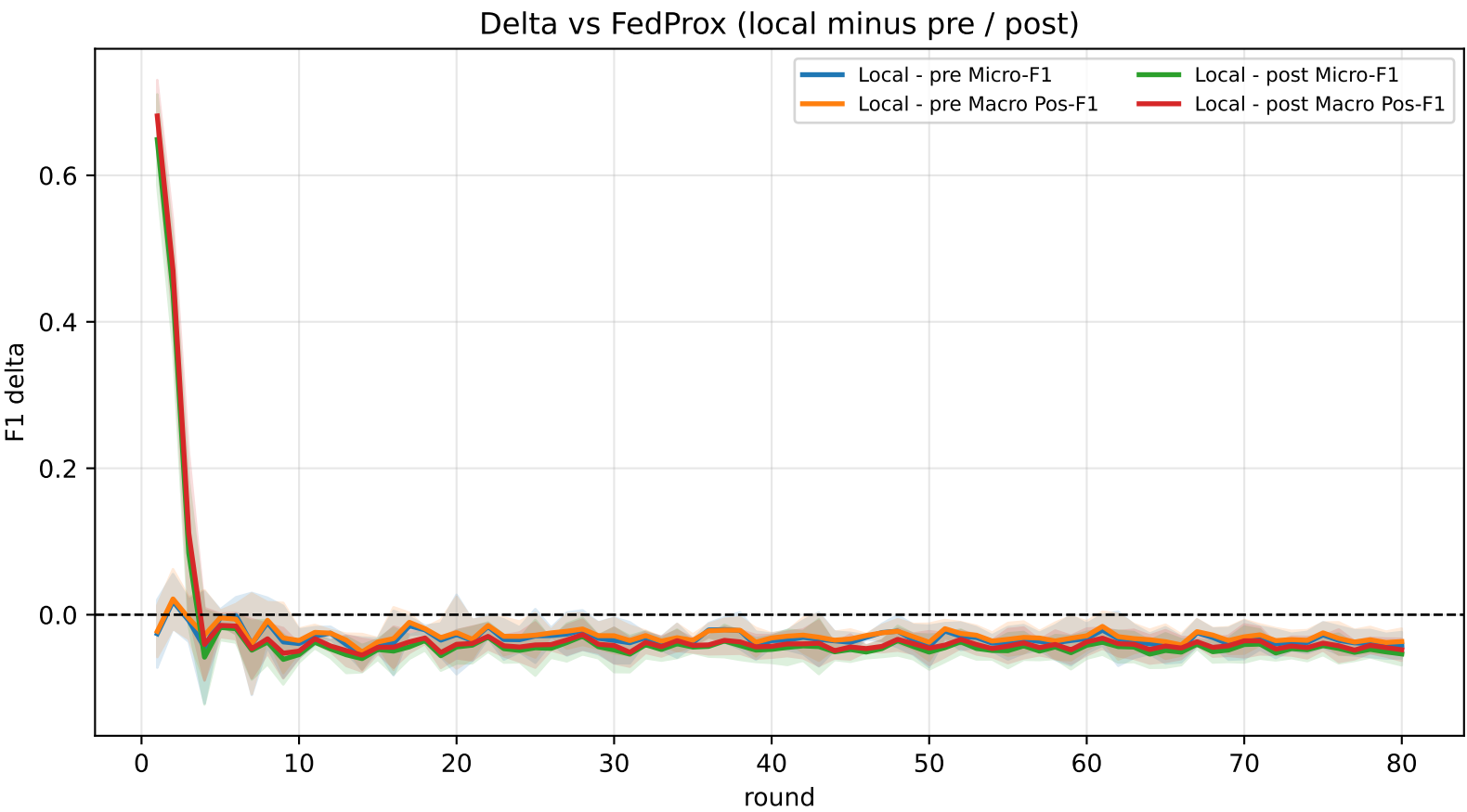
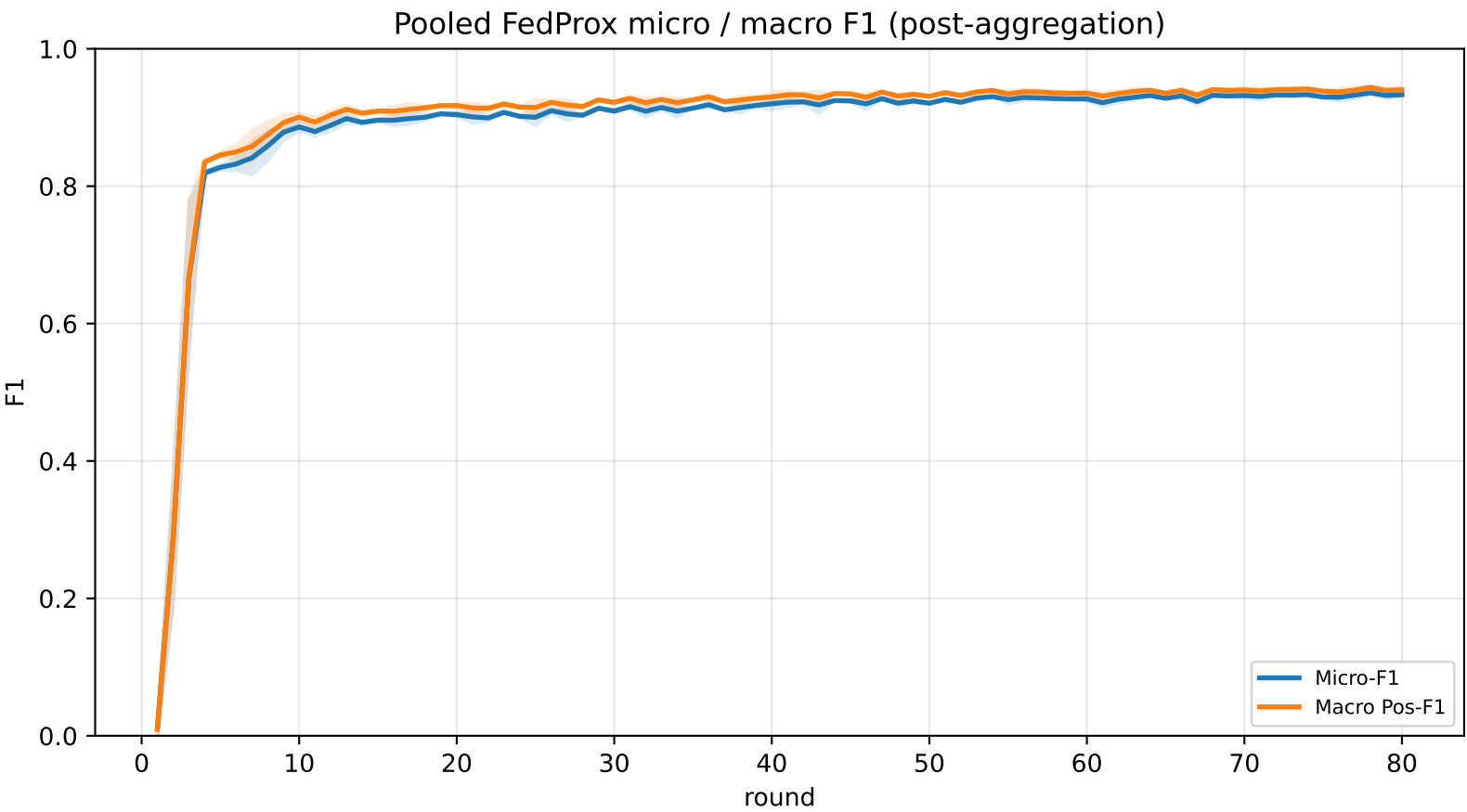
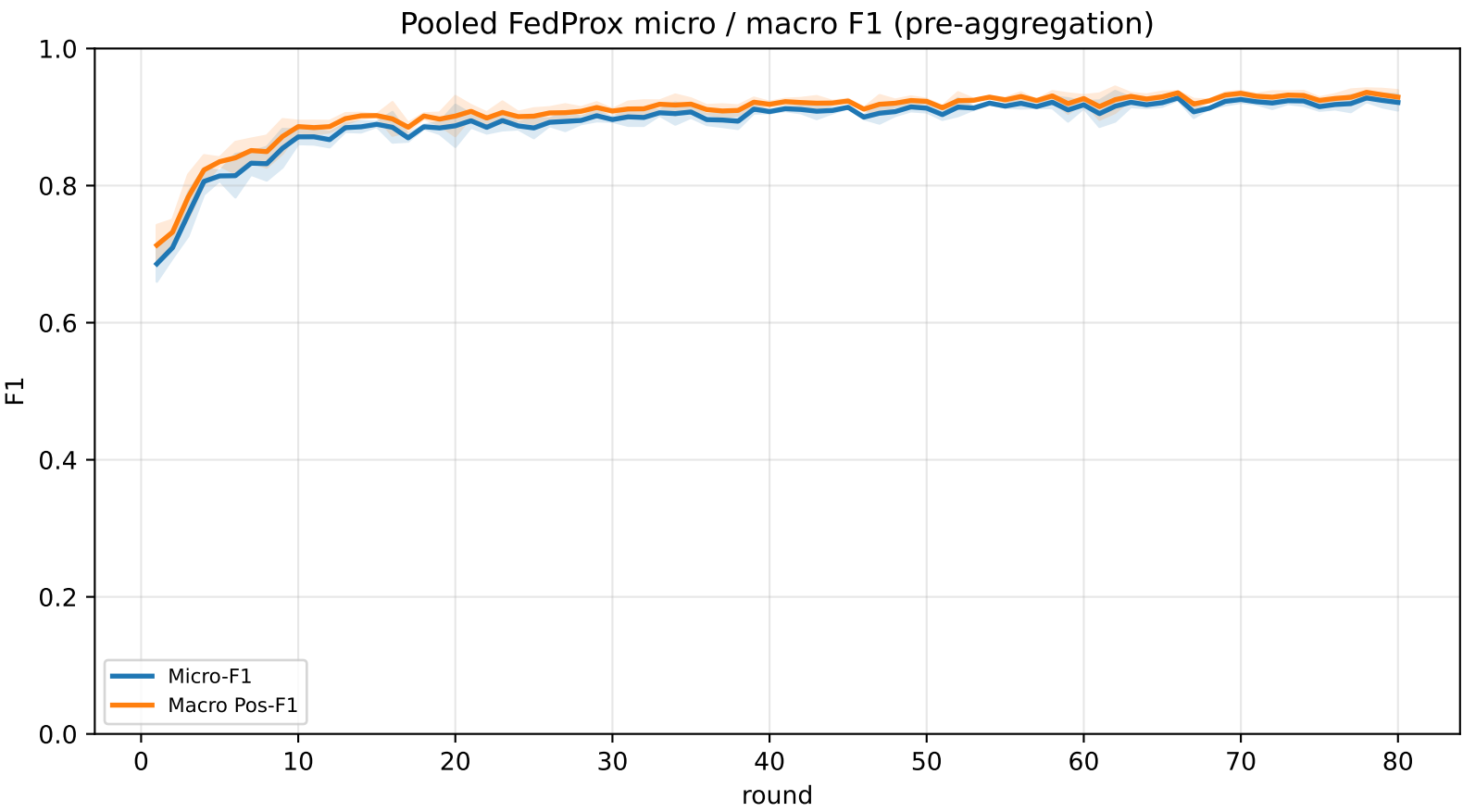
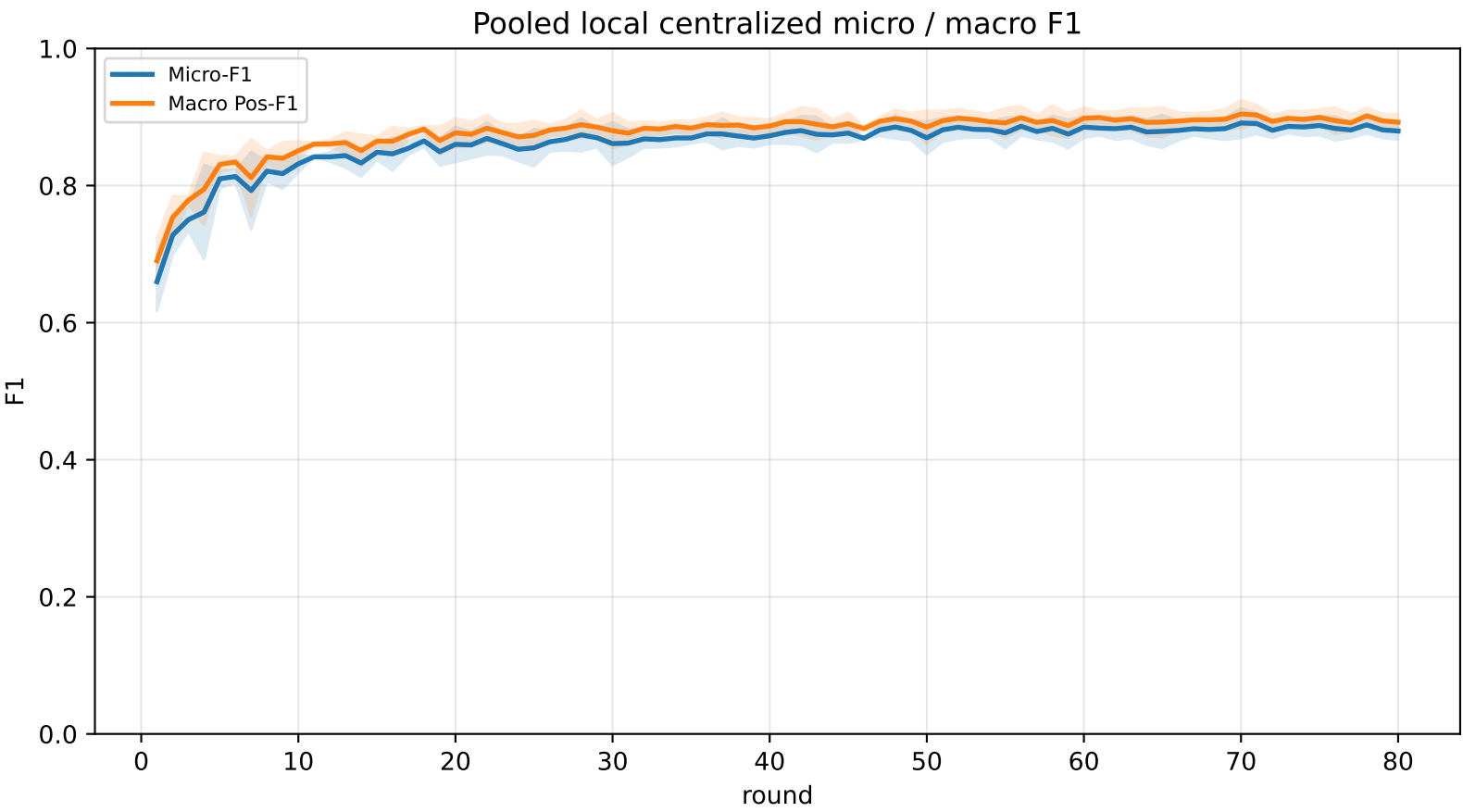
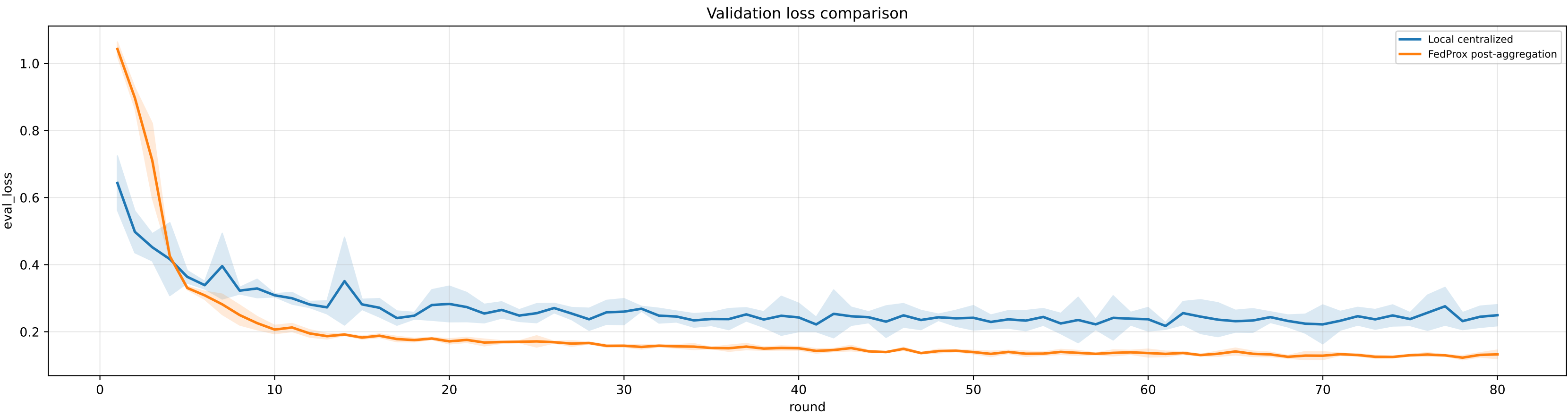


Delta vs FedProx (local minus pre / post)



family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.3% ± 1.5%	89.8% ± 1.2%	99.0% ± 0.2%	97.2% ± 0.4%	93.2% ± 0.6%	83.8% ± 5.3%	75.6% ± 1.0%
FedProx	93.0% ± 0.4%	93.8% ± 0.3%	99.1% ± 0.4%	98.0% ± 0.5%	95.4% ± 0.6%	92.5% ± 2.1%	83.9% ± 1.0%

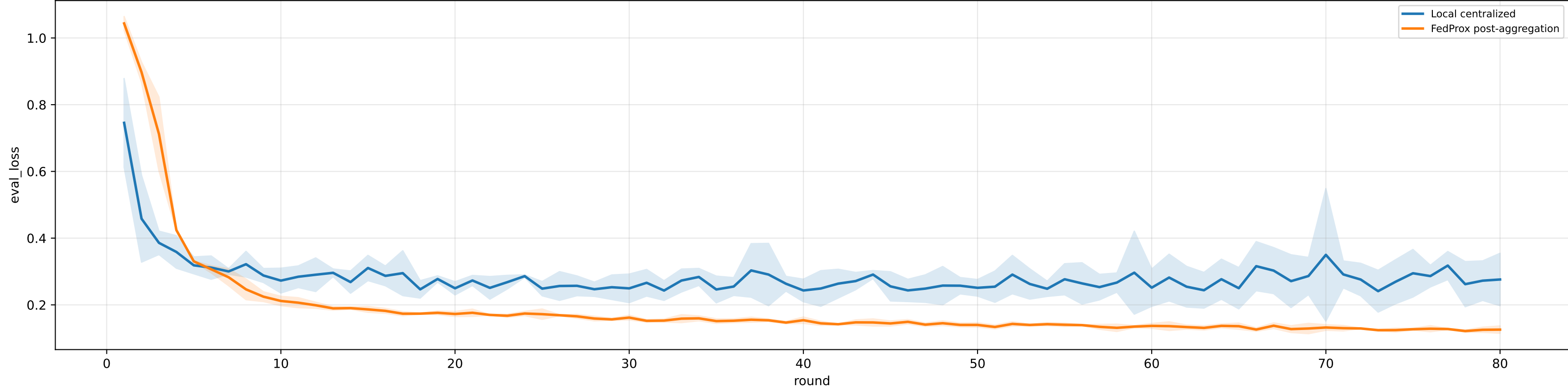


family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

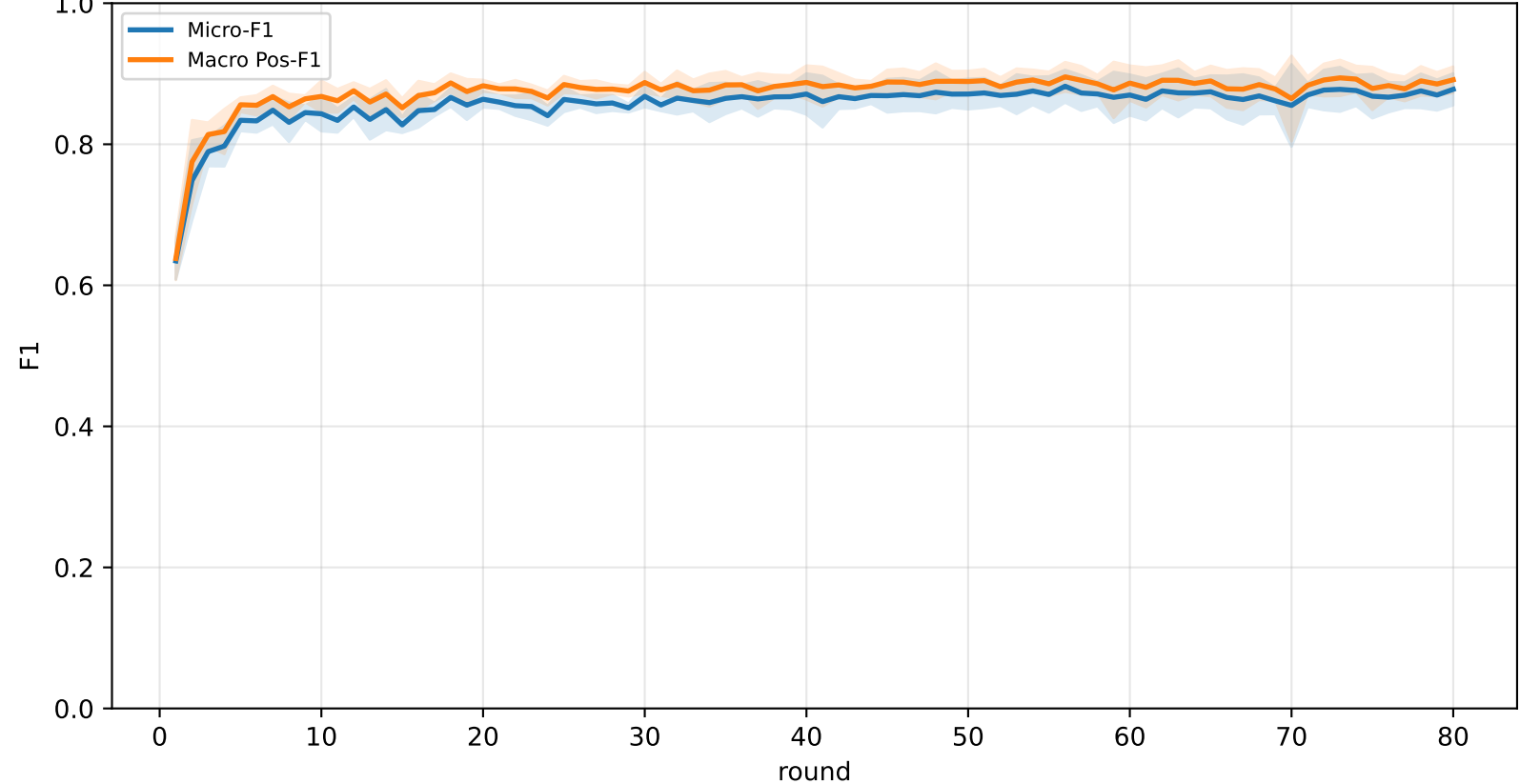
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.2% ± 2.5%	89.2% ± 1.7%	98.4% ± 0.6%	98.4% ± 0.3%	94.1% ± 0.6%	82.5% ± 4.4%	72.3% ± 4.5%
FedProx	92.6% ± 0.3%	93.4% ± 0.3%	98.8% ± 0.3%	97.5% ± 0.2%	95.1% ± 1.1%	92.3% ± 1.7%	83.6% ± 1.1%

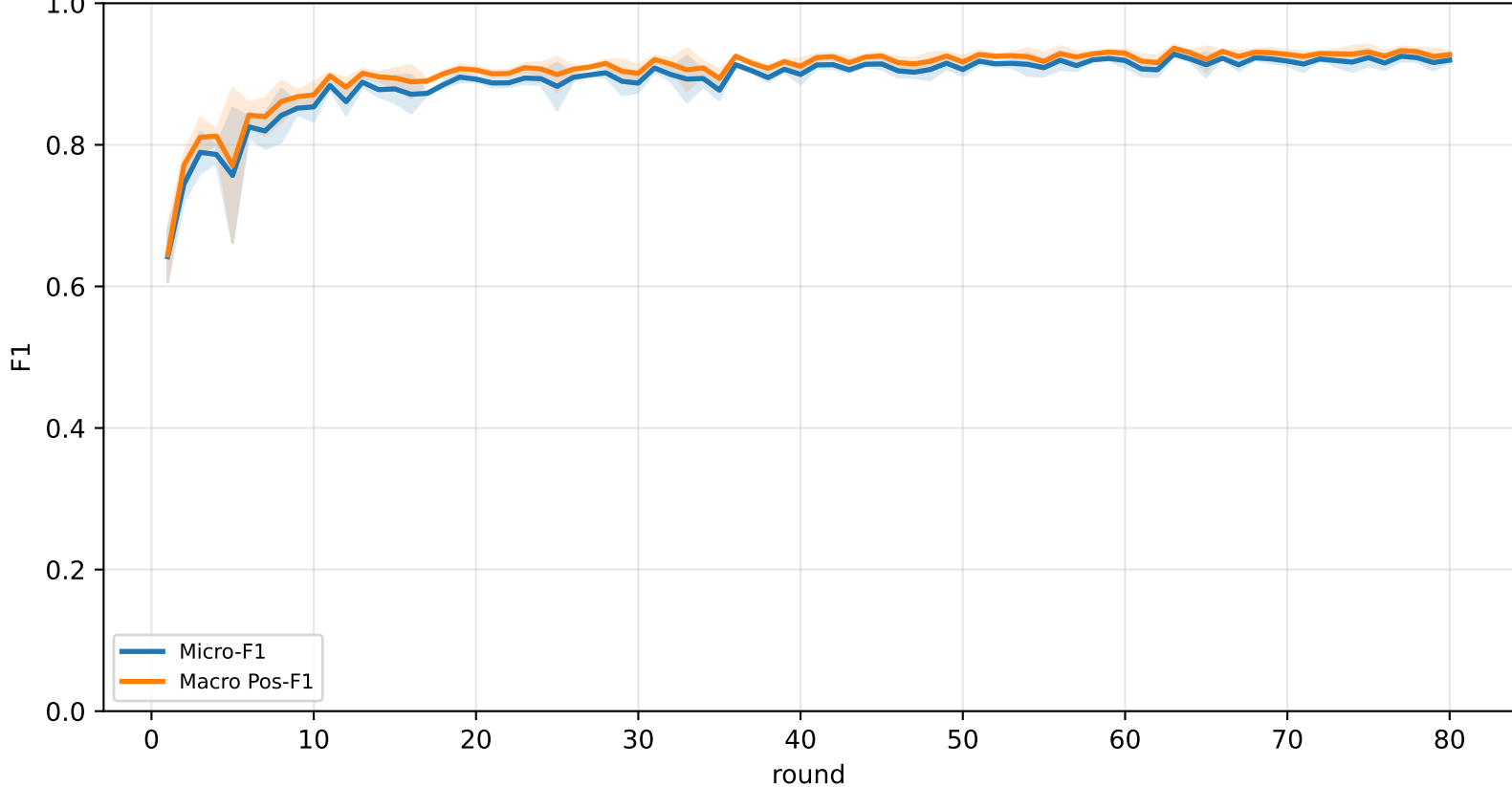
Validation loss comparison



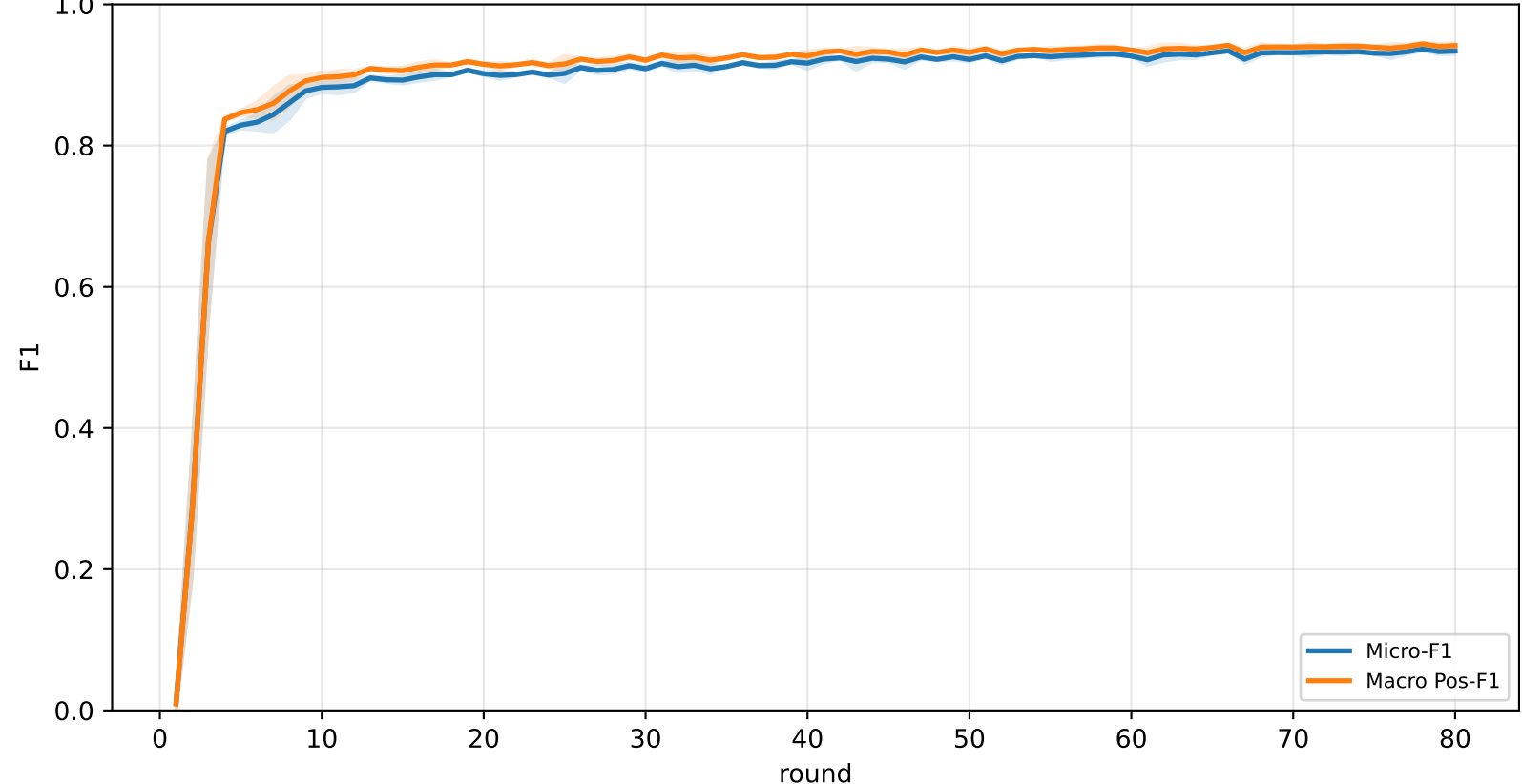
Pooled local centralized micro / macro F1



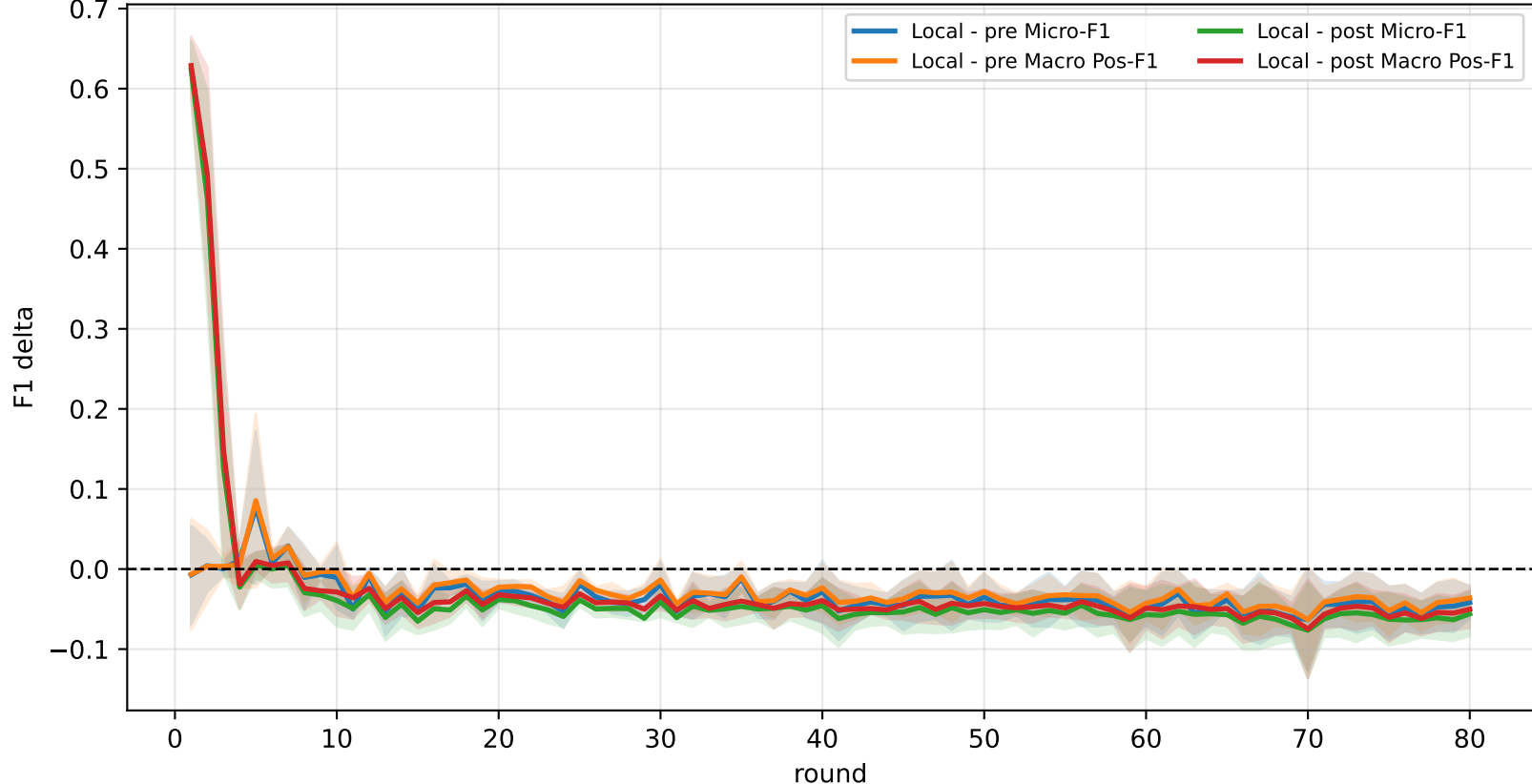
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)

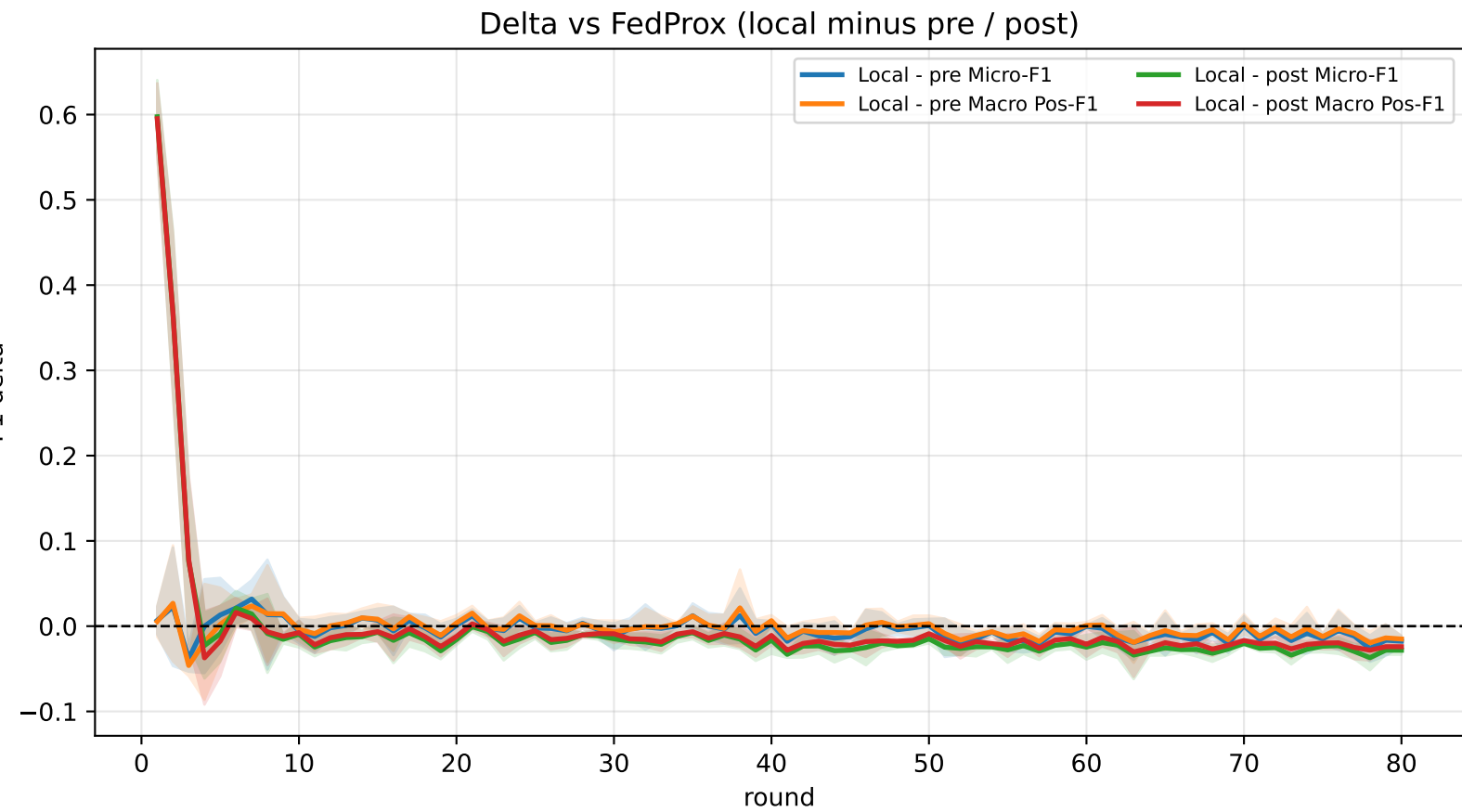
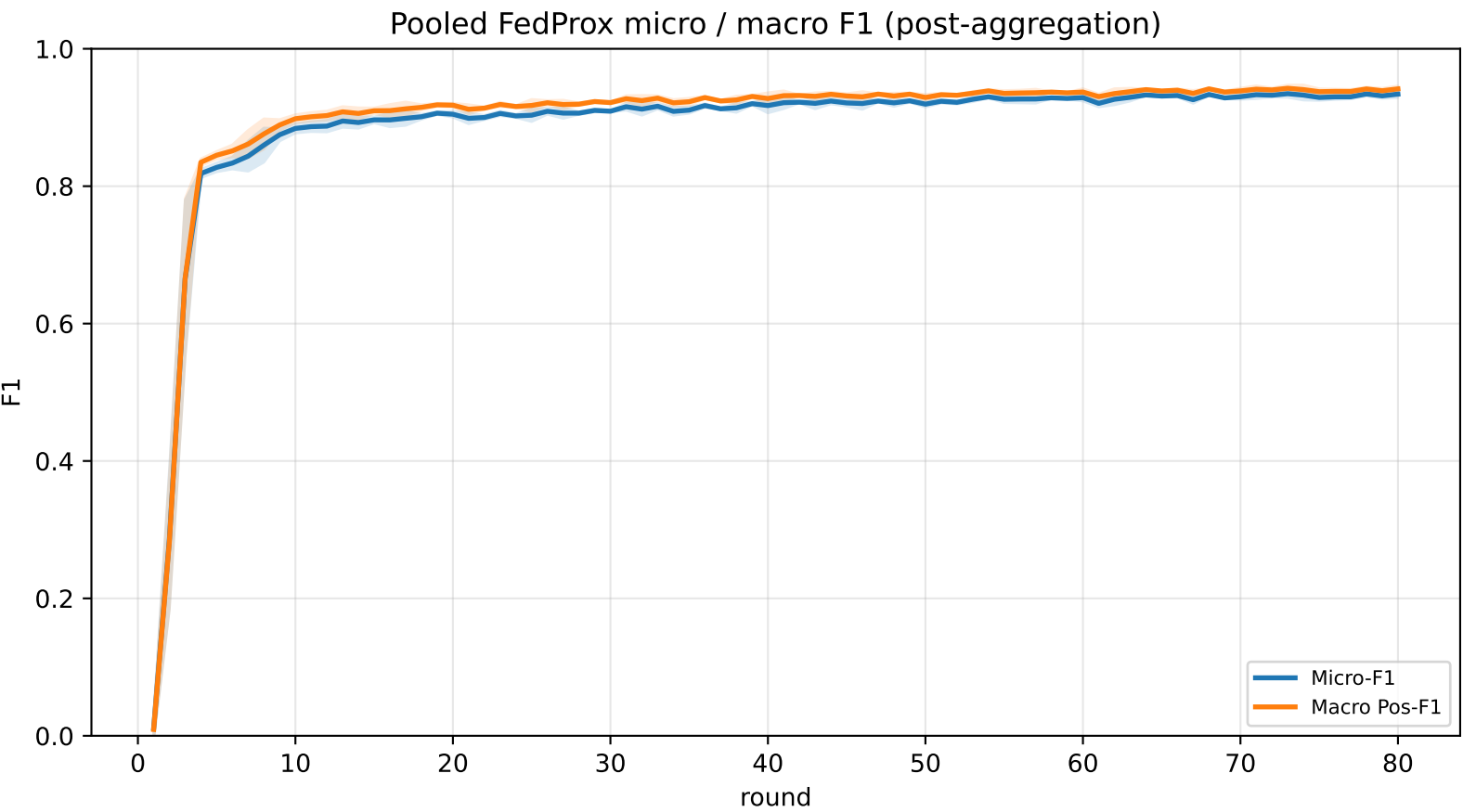
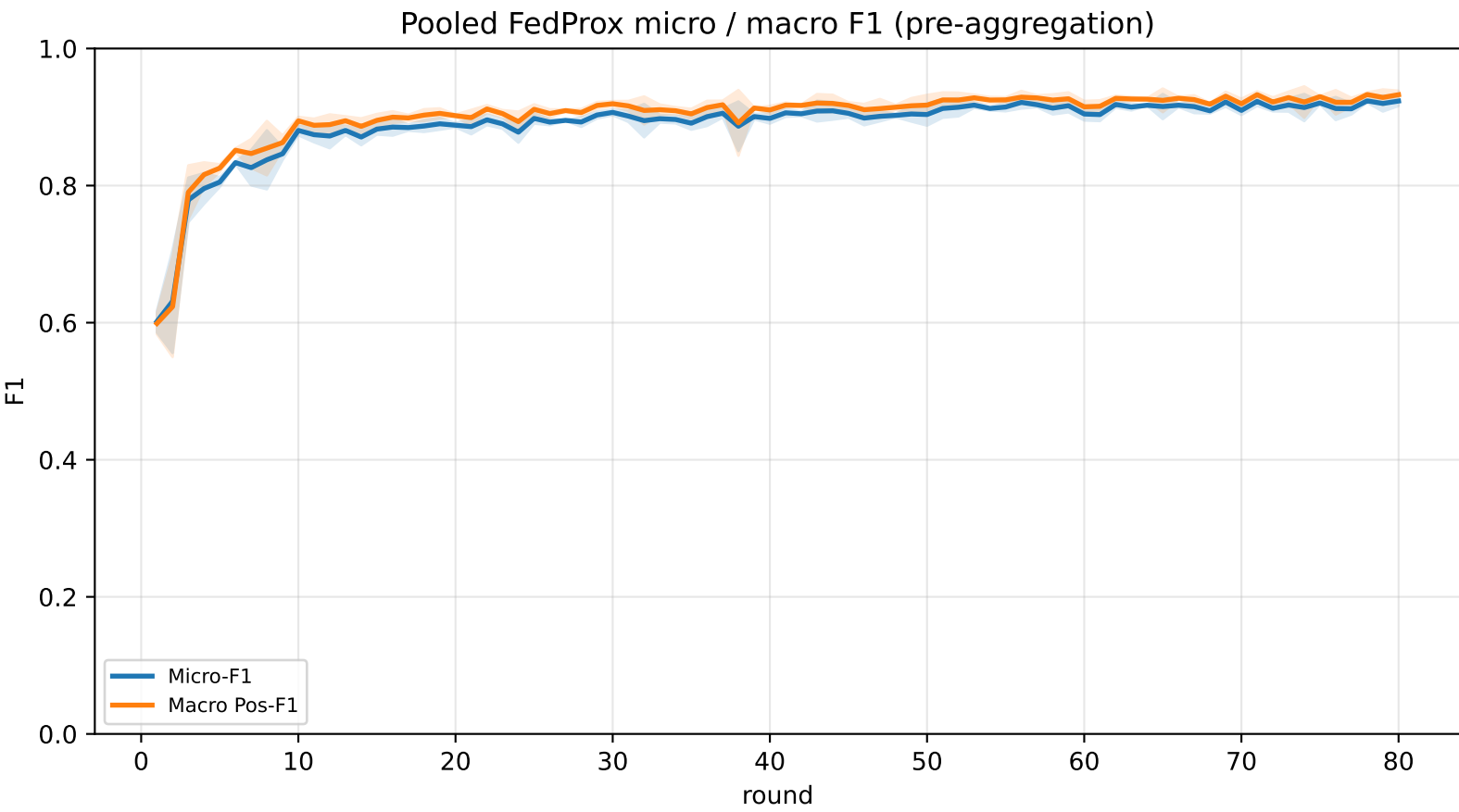
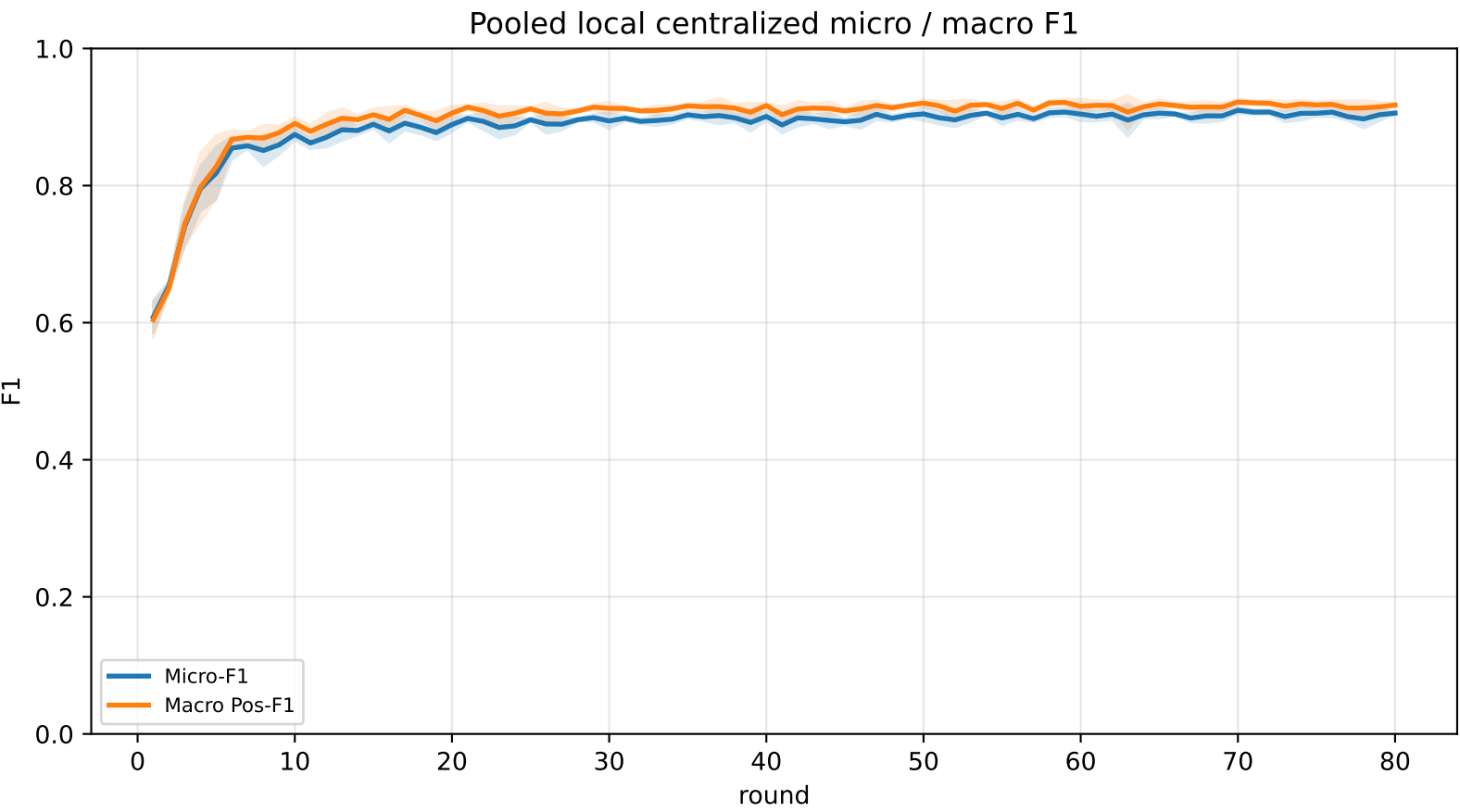
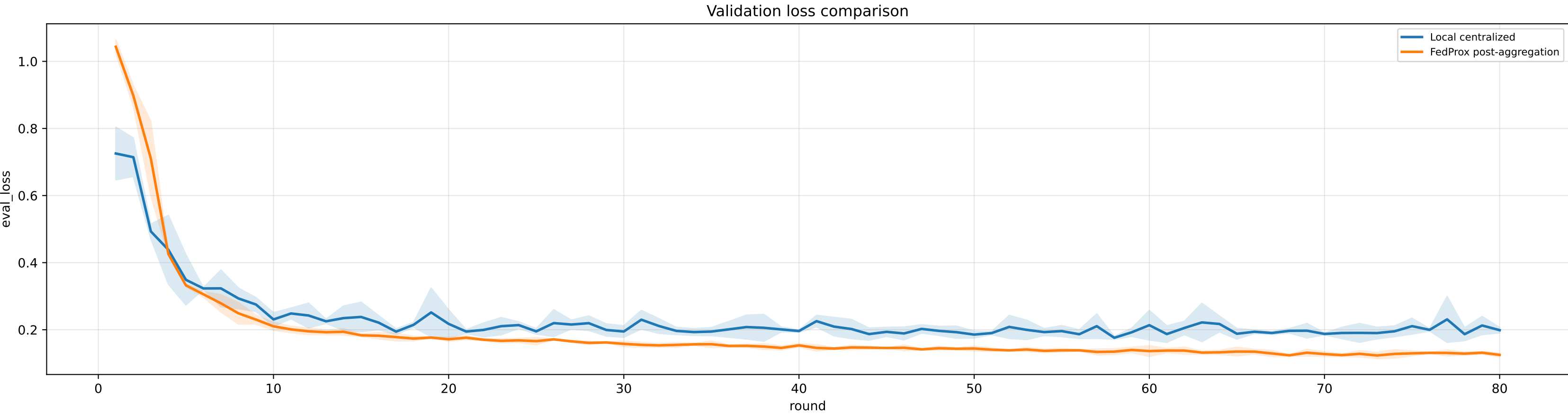


Delta vs FedProx (local minus pre / post)



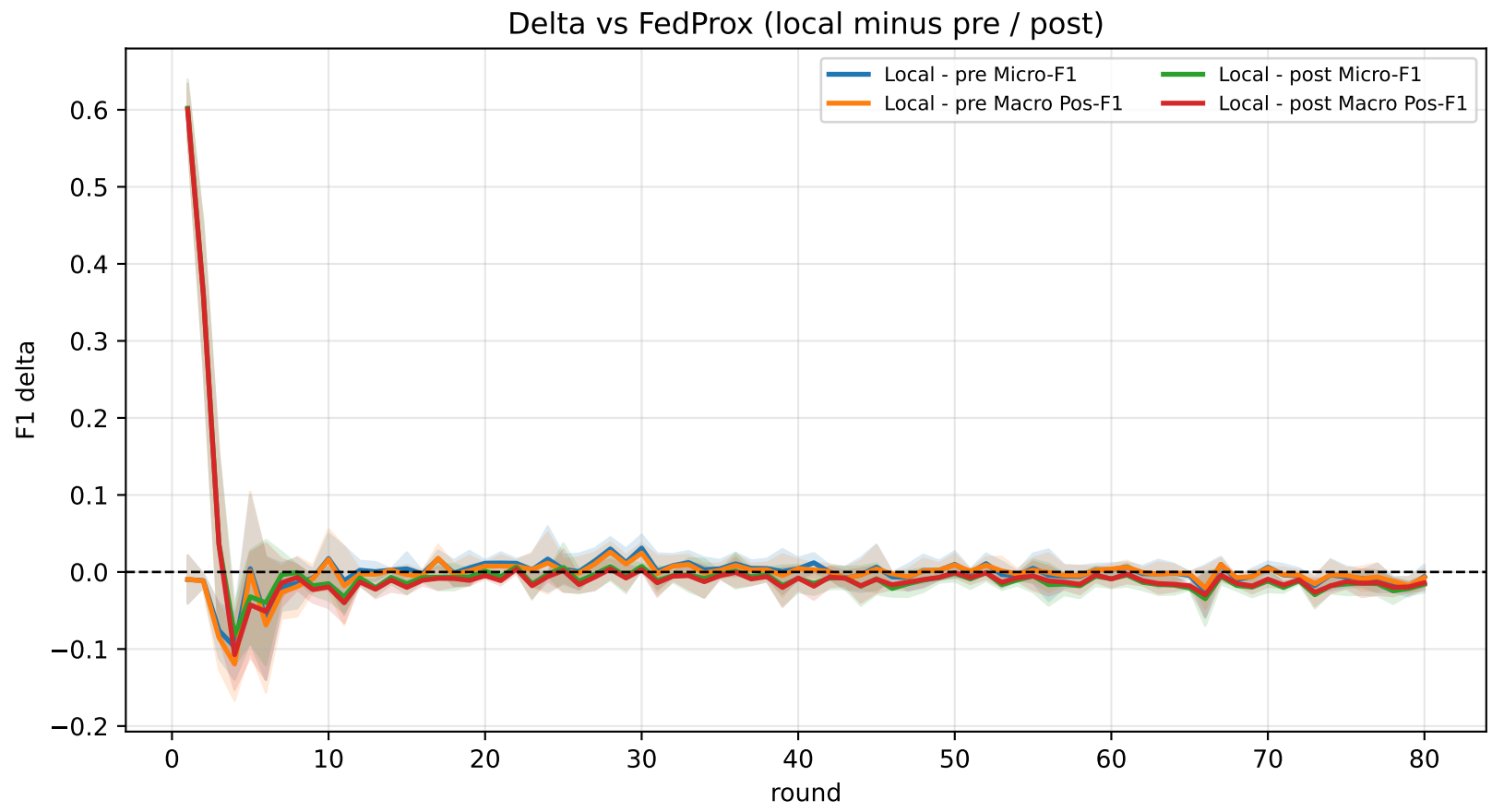
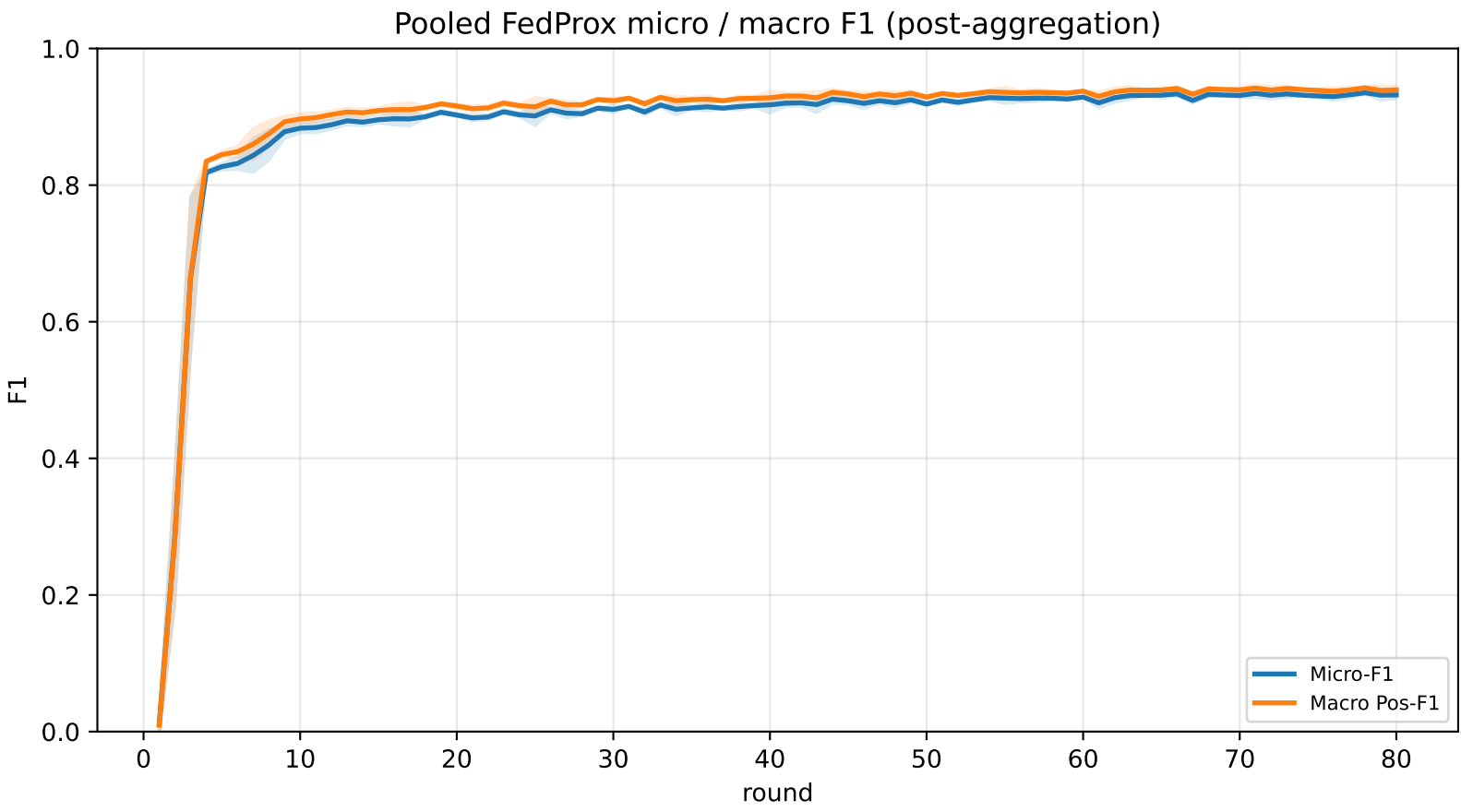
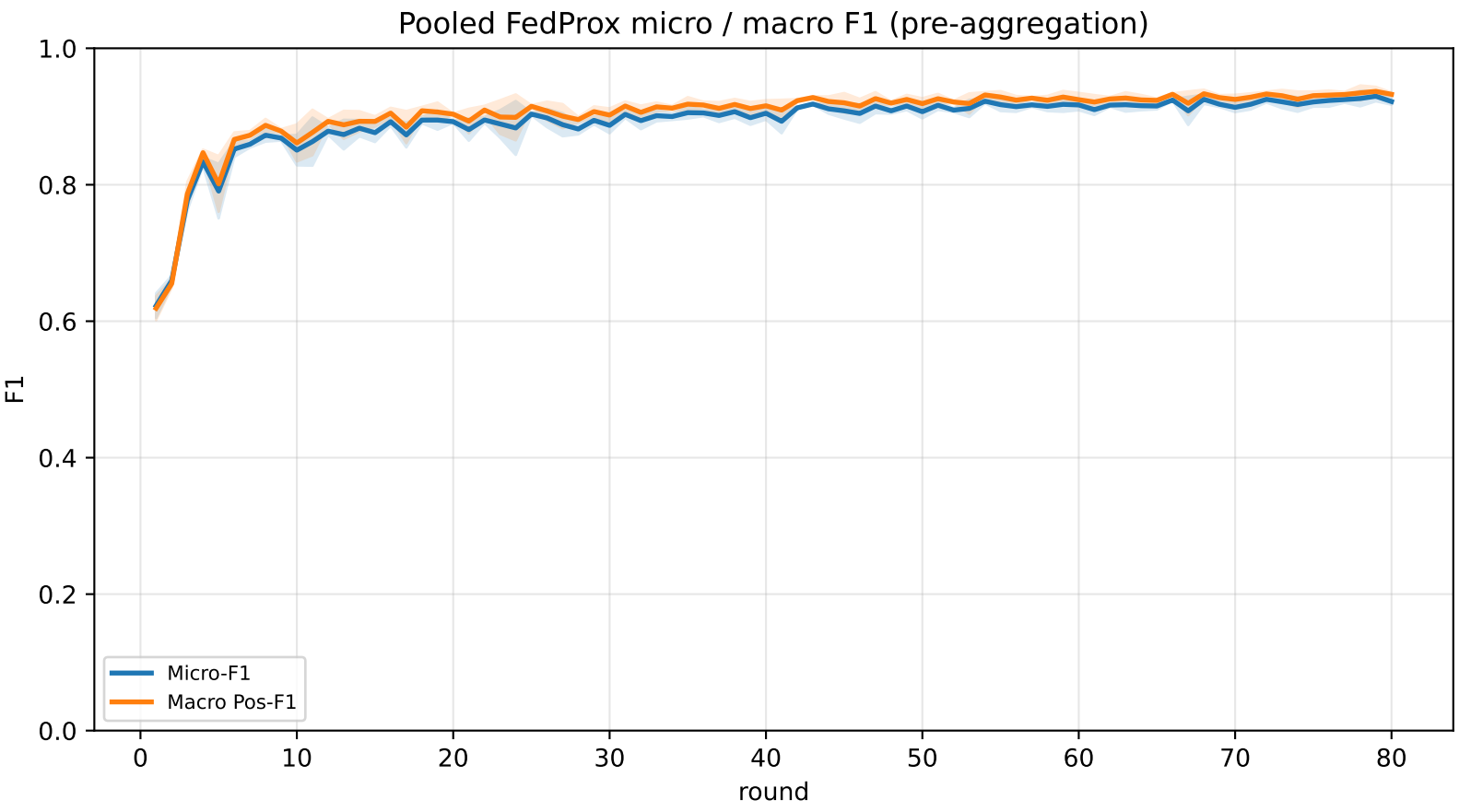
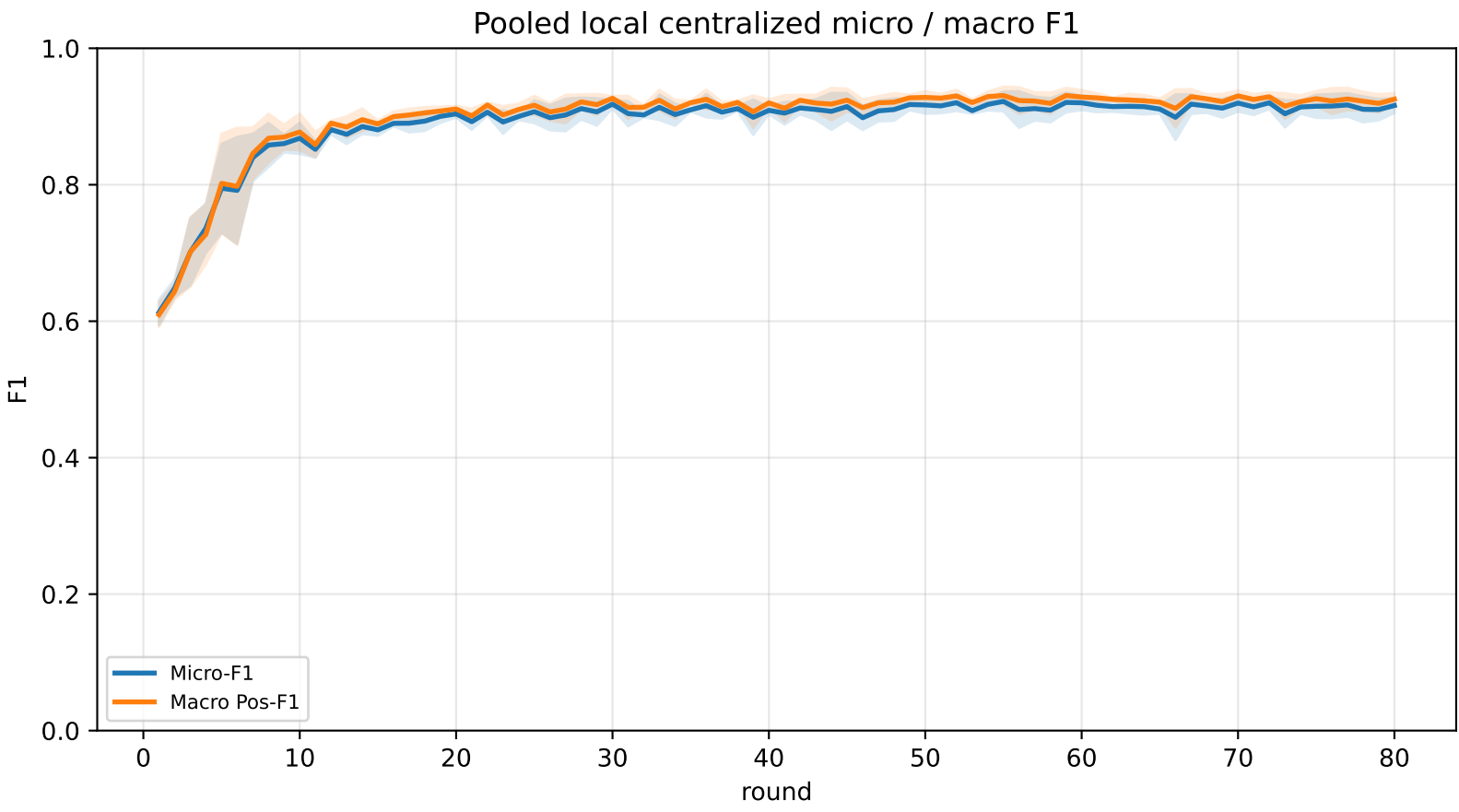
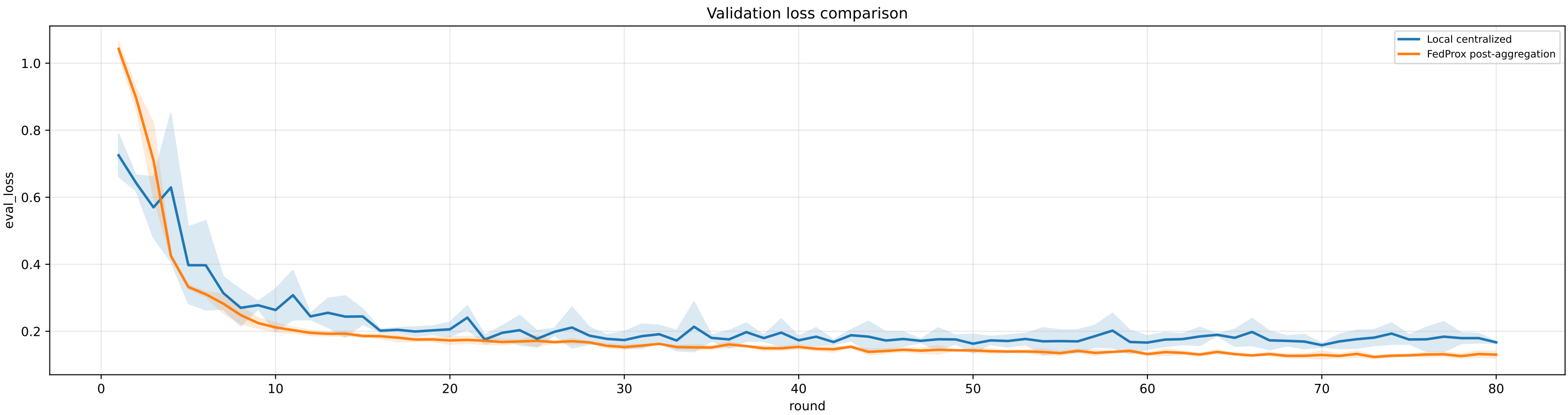
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.8% ± 0.2%	91.4% ± 0.1%	99.0% ± 0.3%	97.8% ± 0.2%	97.0% ± 0.3%	89.3% ± 0.6%	74.0% ± 0.3%
FedProx	92.7% ± 0.4%	93.6% ± 0.3%	99.0% ± 0.2%	97.6% ± 0.2%	95.3% ± 0.5%	92.0% ± 2.0%	83.8% ± 0.5%



family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

Best logged test performance (FedProx vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.3% ± 1.0%	92.4% ± 0.7%	98.4% ± 0.5%	97.7% ± 0.5%	94.7% ± 1.2%	93.5% ± 0.9%	77.8% ± 3.3%
FedProx	92.9% ± 0.4%	93.8% ± 0.4%	98.8% ± 0.3%	98.2% ± 0.3%	95.4% ± 0.6%	92.5% ± 2.0%	83.8% ± 0.7%

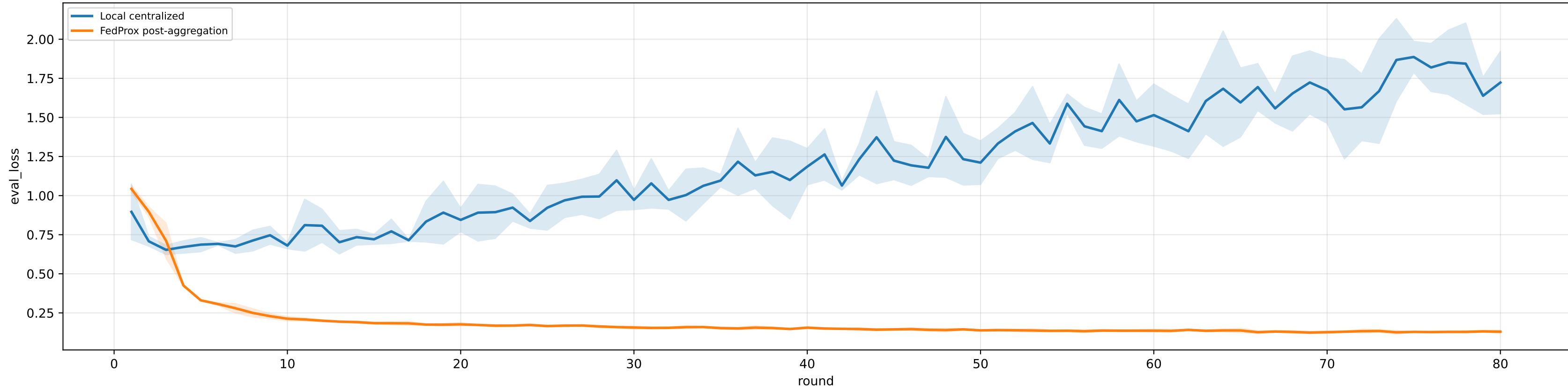


family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

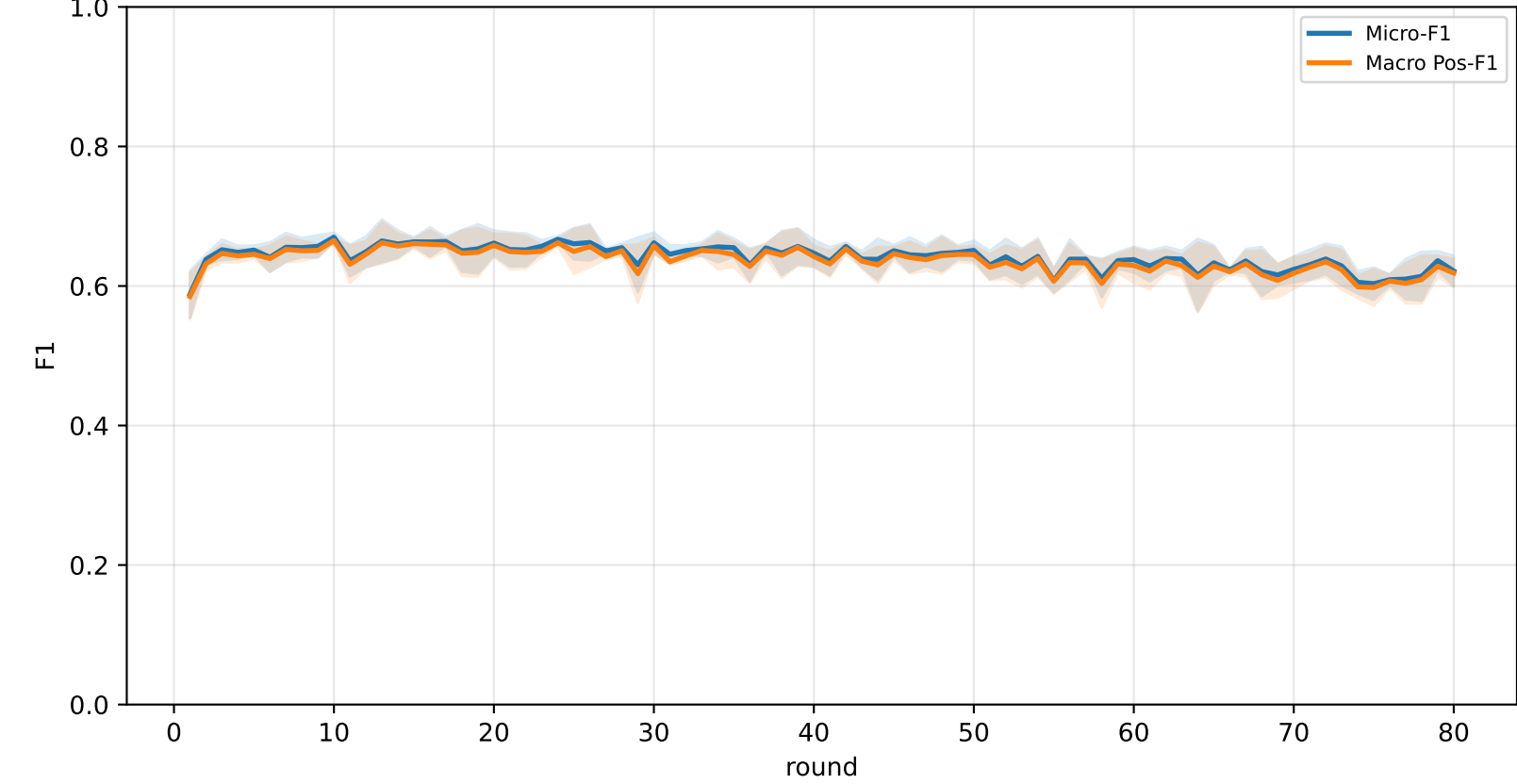
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.1% ± 1.1%	63.5% ± 1.0%	53.6% ± 1.8%	60.5% ± 0.7%	63.8% ± 0.9%	69.7% ± 0.8%	70.1% ± 1.9%
FedProx	92.8% ± 0.3%	93.7% ± 0.3%	98.8% ± 0.4%	98.0% ± 0.6%	95.3% ± 0.8%	92.4% ± 2.0%	83.6% ± 0.9%

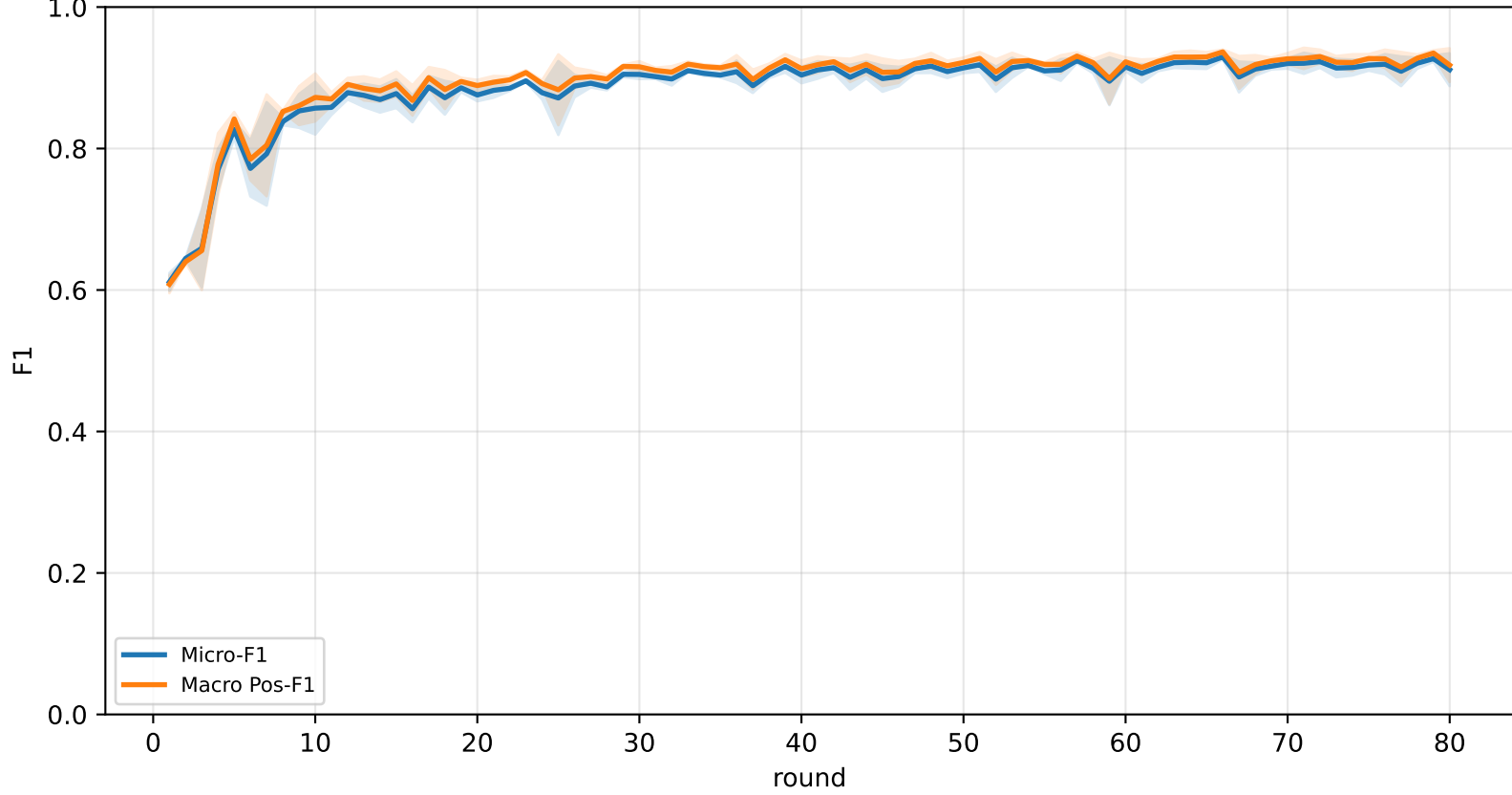
Validation loss comparison



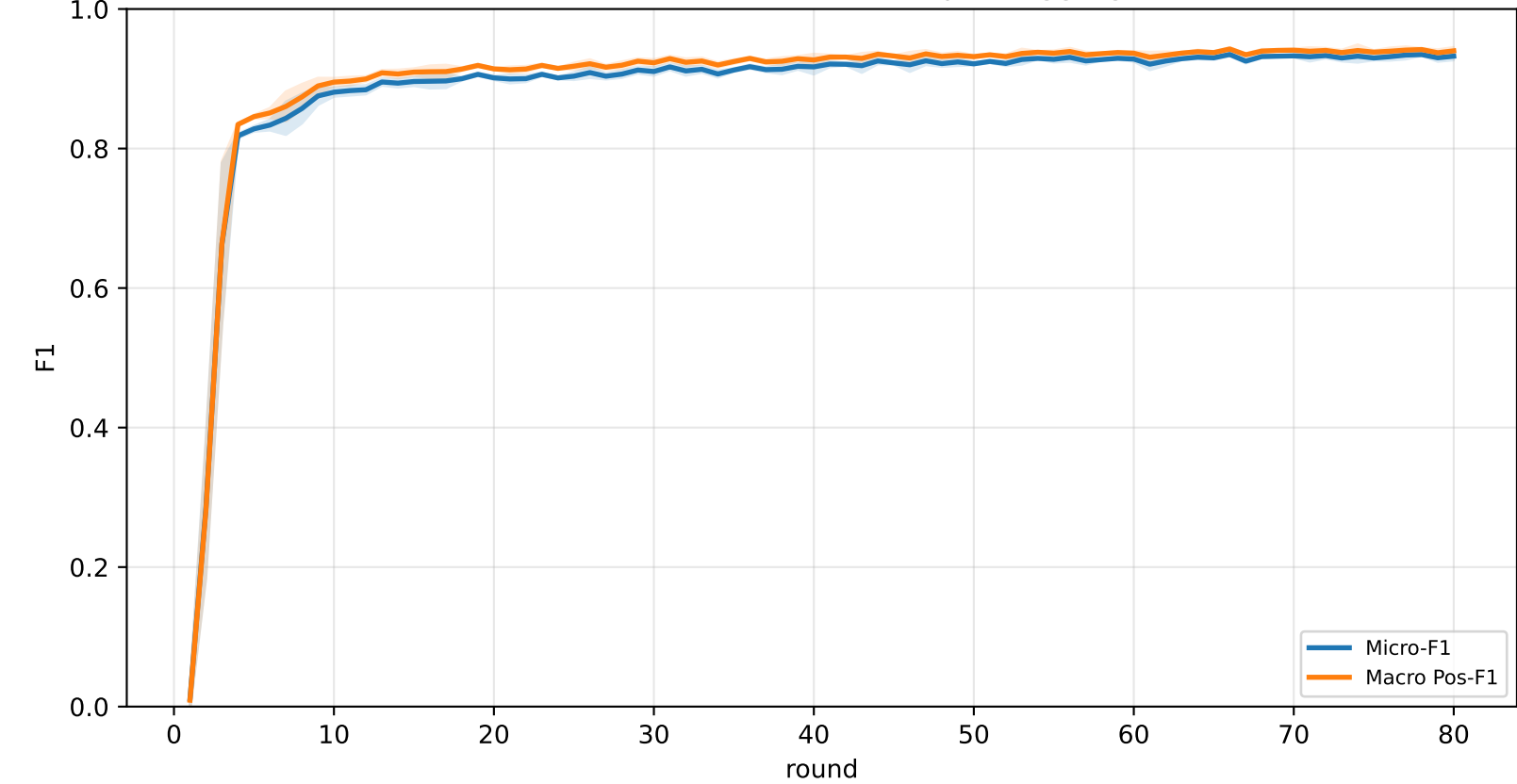
Pooled local centralized micro / macro F1



Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

